

Axxiom Elevator

Safety and Health Policies

Section 2: Specific Safety Policies

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General Safety Rules

(Reference: Policy AXX- 0010)

The safety rules in this section are general and by no means are an exhaustive list of the requirements by federal, state, or local regulations or codes. Employees and supervisors are responsible for abiding by these rules and all other regulations, codes, or safe practices that may apply to their work.

General Job Site Rules

1. Report all incidents and injuries, no matter how slight, immediately (or as soon as safe to do so) to your supervisor.
2. No horseplay or practical jokes are permitted. Running within the job is strictly prohibited, except in case of emergency.
3. No intoxicants of any kind are permitted on any site. Workers under the influence of intoxicants will immediately be removed.
4. No weapons of any type will be permitted on job sites, offices, or other Company property.

5. Workers must adhere to all site-specific rules, including the Emergency Action Plan for the worker's specific location.
6. The use of personal entertainment devices (radios, tape, CD Players, MP3 Players, cell phones, etc.) is prohibited on all job sites.
7. Cell phone usage is limited to work-related job duties or emergencies. If a worker must use a cell phone, they must ensure they are in a safe area, and are not walking around and talking on the phone.
8. Do not climb, swing, or slide down cables, ropes, guide rails, or pistons.
9. Never drop objects down the hoistway. Use proper methods to safely lower objects.
10. Observe and obey all safety and warning signage.
11. Replace guards and covers removed during maintenance prior to leaving jobs.
12. Always account for any electrical jumpers. Do not make homemade jumpers.
13. Never intentionally damage PPE or any other company equipment.
14. Before working on equipment, secure loose-fitting clothing and confine long hair.
15. Remove all conductive jewelry to include metal wedding bands, watches, necklaces, etc.
16. Keep work areas, ramps, runways, corridors, stairwells, offices, shops, and storage areas well lit. Notify the responsible party and your supervisor if the lighting is inadequate.

Housekeeping

A basic concept in any effective prevention endeavor is good housekeeping. Housekeeping greatly impacts the overall success of a safety program on a job site. Good housekeeping must be planned for at the beginning and followed through to the final clean-up.

1. Daily clean-up of each work area is required. Keep all scrap material, including lumber, rolling stock, and insulation, clear from work areas and passageways in and around buildings.
2. Keep all aisles, stairways, emergency exits, and fire extinguishers clear of material storage and debris.
3. Place all refuse and waste materials in the recognized waste containers for disposal.
4. Containers used for oily rags, flammables, or hazardous wastes must be covered and separated from regular refuse and waste.
5. All stacks or piles of material must be stable, with proper support as necessary.
6. Rolling stock must be chocked or effectively secured from displacement.
7. Do not allow extension cords, air hoses, welding leads, and similar equipment to create a tripping hazard.
8. Immediately clean up spills that present a slip hazard.
9. Immediately contain hazardous material spills and report them to your supervisor. Clean up and dispose of hazardous material spills in accordance with local, state, and federal regulations.
10. Clear the areas in front of and surrounding electrical installations (such as switch gear, temporary and permanent power panels, and transformers). A minimum of 3 feet of space is required in front of these units.

Working with Non-Company Employees

1. Requesting assistance from customers or other non-approved persons to complete work performed in the machine room or hoist way/truss of escalators and other equipment maintained by the Company is not allowed, unless specifically approved by the supervisor.

Communicating Safety and Incident Information

1. Management shall issue the Safety Policy to every employee. Each individual employee has a responsibility to read, understand and adhere to the rules and procedures in the Safety Policy. Direct any questions about the Safety Policy to your immediate Supervisor. Formal safety training will be provided, and workshop techniques will be included in on-the-job training. New employees will receive training prior to beginning work. Supervision will also provide training on an as needed basis. Toolbox safety meetings will be conducted on a weekly basis. The Safety Committee will hold quarterly meetings to review previous accidents, new policies, and procedures and to ensure progress toward established safety goals.

Safety Documentation

1. All mechanics and helpers will attend/conduct a toolbox meeting weekly. The meeting will be documented, and documentation turned into the supervisor. Mechanics working in a team shall hold a weekly toolbox talk safety meeting and document topics discussed. Management will provide Toolbox Safety Talk forms.

Elevator Specific Public Safety Procedures

1. All field employees shall immediately shut down any elevator operating in an unsafe manner. No employee will ever put an elevator back into service until unsafe operation is abated. The following defines unsafe equipment, and the procedures employees must follow when such a problem arises. Knowingly placing an elevator or escalator back in service in an unsafe condition will result in disciplinary action up to and including termination. Supervision and management will be held just as responsible as the field personnel if the unsafe condition was known.
2. Examples of Unsafe Equipment (not all inclusive):
 - Inoperable photo-eyes, safety edges, detector edges or door open buttons.
 - Erratic door closing.

- Excessive door closing speed.
- Excessive door closing torque.
- No door open pause time.
- Hoistway or car doors will not accept gib.
- Improper car leveling.
- Cab ceiling fixtures hanging precariously.
- Significant loss of oil in the hydraulic system.
- Safety circuits not functioning properly,
- Safety devices are not functioning properly.
- Unsafe or unreliable hoist equipment such as but not limited to:
 - Hoist ropes.
 - Machines
 - Sheaves
 - Shackles
 - Exposed wiring
- Water damage, vandalism, etc., that may affect the safe operation of the equipment.
- Any other condition that could compromise the safety of passengers or elevator personnel.

Procedures for Unsafe Equipment

1. Any employee who observes equipment operating unsafely must take the equipment **OUT OF SERVICE** immediately. **DO NOT RETURN UNSAFE EQUIPMENT TO SERVICE** until the unsafe condition is eliminated.
2. If the field employee cannot make necessary repairs to ensure safe operation, the employee must shut off the main line disconnect to cut all power to the elevator and lockout/ tagout/tryout.
3. Contact a supervisor for further instruction. The Supervisor will take steps to facilitate repairs.
4. Communicate with the customer and inform them of the unsafe condition of the equipment.
5. Fill out a time ticket stating that the equipment has been turned off, and why. Have the Customer sign the ticket or, if not available, leave a copy in a conspicuous place.
6. In the event that the unsafe equipment can be corrected, field personnel must ensure they take extreme care.

Gasoline / Propane and Other Fuel Storage

Gasoline Storage Requirements

- Store only in approved, labeled, and sealed containers (e.g., UL/FM-listed safety cans with flame arrestors).
- Limit storage to 5 gallons per container.

- Maximum indoor storage in general-purpose buildings should not exceed 25 gallons unless stored in an approved flammable liquid storage cabinet.
- Store in a cool, well-ventilated, and secure area, away from ignition sources.
- Keep containers at least 50 feet away from ignition sources and combustible materials
- No smoking or open flames within 25 feet of storage areas.

Propane Storage Requirements

- Store propane cylinders upright, with valve protection caps in place.
- Do not store propane indoors unless it's in a designated and ventilated propane cage or room compliant with NFPA 58.
- Outdoor storage must be in a secure, ventilated, and shaded area, at least:
 - 10 feet from buildings, exits, and ignition sources (for ≤500 lbs)
 - 25 feet from property lines or public areas (larger quantities)
- Never store propane cylinders near stairwells, doors, or in vehicle interiors.
- Cylinders must be inspected for damage or leaks before storage.

Oxygen and Acetylene Cylinder Storage

General Storage Requirements

- All cylinders must be stored upright and secured with chains, straps, or racks to prevent tipping.
- Cylinders must have valve protection caps in place when not in use.
- Store in dry, well-ventilated, and designated cylinder storage areas away from heat sources, electrical panels, or exit routes.
- Storage areas must be marked with "No Smoking" signs.

Separation Requirements (OSHA 1910.253(b)(4)(iii))

- Oxygen cylinders must be stored at least 20 feet from flammable gases (such as acetylene), OR
- Be separated by a noncombustible barrier at least 5 feet high with a fire-resistance rating of at least 30 minutes.

Acetylene Cylinder Storage

- Store valve end up, even when empty.
- Never store acetylene cylinders in areas exceeding 125°F.
- Avoid storage near open flames, welding operations, or energized electrical circuits.
- Never store lying down — the acetone inside may leak or impair performance.

Oxygen Cylinder Storage

- Keep cylinders clean and free of oil, grease, or other contaminants — oxygen accelerates combustion.
- Do not allow oxygen cylinders to come into contact with fuel gas lines or equipment.
- Never use oxygen to blow dust or for ventilation purposes.

Empty vs. Full Cylinder Management

- Mark cylinders as "Full," "In Use," or "Empty" using tags or color coding.
- Store empty cylinders separately from full ones.
- Close valves completely and replace valve caps on all stored cylinders.

Transportation

- Use approved carts with proper restraints for moving cylinders.
- Never drag, roll, or drop a cylinder.
- Valve caps must be in place during transport.

Inspection and Maintenance

Cylinders and storage areas must be visually inspected periodically for:

- Leaks or damage
- Proper labeling
- Secure storage

Damaged cylinders must be tagged and removed from service.

Emergency Procedures

In the event of a leak or fire:

- Evacuate the area immediately.
- Notify the supervisor and call 911.
- Do not attempt to move leaking cylinders without PPE and training.
- Fire extinguishers (Class B and C) must be available and accessible within 50 feet.

Job Hazard Analysis

(Reference: Policy AXX- 0011)

A Job Hazard Analysis (JHA) is a technique that focuses on job tasks to identify hazards before they occur. It focuses on the relationship between the worker, the task, the tools, and the environment. Supervisors can use the findings of a JHA to eliminate and prevent hazards in his/her workplace and as a tool for training new employees in the steps required to perform their jobs safely.

Development

1. A JHA will be required in the following situations:
 - When leadership and/or the project supervision believes the process, task, or procedure indicates a potential for injury and/or property damage.
 - When the project specifications require such development or
 - As a condition of approval for supervisors that do not meet the safety pre-qualification requirements. JHAs may be required to be turned in and reviewed before work is started.
 - When personnel on site determine there is a need for any reason.
 1. The Job Hazard Analysis worksheet may be used to complete the JHA. To complete this form, the following steps should be followed:
 1. List the key steps in the sequence in which they occur.
 2. Determine the hazard(s) for each step.
 3. Apply specific and effective safety measures to eliminate or control hazard(s).
 4. Be specific in the designations of protective devices and equipment.
 5. Include sufficient detail to preclude confusion and misunderstanding.
 6. Make changes as needed; JHAs are living documents.
 1. A JHA may be required initially before the commencement of work, weekly, or daily after that.
 2. All recognized hazards will be removed or mitigated before work is started. When this is not possible, work will be postponed until it is deemed safe by the Company.
 3. Hazards can be classified and then prioritized by severity, probability, and frequency of exposure.
 4. Risk assessments covering various scopes of work shall be completed and reviewed to ensure implementations are effective.

Training

Prior to the work, personnel involved with the operation will be instructed on the hazards involved and methods required to eliminate or control those hazards using the JHA.

Hazard Communication

(Reference: Policy AXX- 0012)

This program provides general safety guidelines and instructions for collecting, using, and storing chemicals at the various potential client facilities, job sites and Company facilities by employees and supervisors. It is a matter of Company policy to provide our employees with information about hazardous chemicals on the work site through our Hazard Communication Program, which includes a chemical inventory, container labeling, Safety Data Sheets (SDS), and employee information and training. The Hazard Communication program is stored electronically. Employees can request a copy of the written Hazard Communication Program

and the Company SDS binder at any time. Supervisors will be required to maintain the SDS list electronically and be made available upon request. A binder will be stored at the office containing program information. For the most up-to-date lists or information, please contact the Safety Manager.

For more information, the entire Hazard Communication OSHA standard can be accessed by clicking this link: [OSHA 1910.1200 Hazard Communication Standard](#)

Safety Data Sheets

1. Safety Data Sheets (SDS) are central to the Hazard Communication Program. The role of SDS is to provide detailed information on each hazardous chemical, its physical and chemical characteristics, and recommendations for appropriate protective measures. The Company shall have a safety data sheet in the workplace for each hazardous chemical used. The safety data sheet shall ensure that it is in English and includes at least the following section numbers and headings, and associated information under each heading:

Section 1, Identification

Section 2, Hazard(s) identification

Section 3, Composition/information on ingredients

Section 4, First-aid measures

Section 5, Fire-fighting measures

Section 6, Accidental release measures

Section 7, Handling and Storage

Section 8, Exposure controls/personal protection

Section 10, Stability and reactivity

Section 11, Toxicological information

Section 12, Ecological information

Section 13, Disposal considerations

Section 14, Transport information

Section 15, Regulatory information and

Section 16, Other information, including date of preparation or last revision.

1. SDS' are used to:

- Review hazards prior to use or, ideally, prior to purchase of a product.
- Provide emergency information in case of a spill, fire, or employee exposure involving the chemical.
- Provide information to employees about chemical hazards and methods of protection.
- Provide information necessary for environmental record keeping and reporting.

1. Hazard Communication Action Items Regarding SDS:

- Designate a person responsible for obtaining and maintaining SDS.
- For new chemicals: The employee or supervisor that requests a new chemical to be brought in should request the SDS for the chemical before that chemical is allowed on the premises or job site. The SDS can arrive with the shipment of chemicals.
- Obtain a SDS for each chemical used or stored in the workplace (using our Chemical Inventory List). They are available from the respective manufacturer, distributor, or supplier. If you do not receive one automatically, you should request one. If you receive one that is inadequate, for example, one with blank spaces, you should request an appropriately completed one. Contact the Safety Manager if you encounter difficulties. (A common practice is to request an SDS at the time a specific order is placed by noting on the purchase order "Shipment will not be accepted unless accompanied by a Safety Data Sheet.")
- SDS sheets will be sent to the Safety Manager as new chemicals are approved for use.
- Keep this binder and SDS readily accessible to all employees. This means that an office or filing cabinet that is sometimes locked is not appropriate.
- Upon written request, we will provide copies of specific SDS to former employees and/or their representatives within 15 days. They have the right to that information.
- Maintain and keep current the SDS file for all chemicals involved in the operation of the facilities and jobsites. Obtain updated SDS. Generally, SDS should be no more than 3-5 years old.
- A separate inactive SDS file must be kept for chemicals no longer used in the facility or on jobsites. These SDS sheets are to be kept for 30 years.

Outside Chemicals

Outside chemicals (i.e., chemicals brought from home or from another organization) are not permitted in Company facilities nor on Company jobsites. The purpose for this is to ensure that all chemicals are properly vetted and documented on the Chemical Inventory List. If there is a chemical needed, please notify the Safety Manager to properly scrutinize and add the chemical to the Hazard Communication Program.

Chemical Inventory List

1. The Hazard Communication Policy requires a list of hazardous chemicals in the workplace be complied with as part of the written Hazard Communication Program. The best way to prepare a comprehensive list is to survey the facility, technician trucks, and technician toolboxes. The Hazard Communication Policy covers chemicals in all forms – liquids, solids, gases, vapors, fumes, and mists.
2. While surveying, identify chemicals in containers, including transfer lines and piping, but also think about less obvious chemicals generated during work operations. For example, welding fumes are a source of chemical exposure.
3. To simplify record-keeping requirements, limit the list of on-site chemicals only to those that will be used. Dispose of expired or obsolete chemicals properly. Do not allow vendors to bring chemicals onsite unless they are necessary, and they are accompanied by an SDS. Add these chemicals to the Chemical Inventory List and include them in employee training.
4. Hazard Communication Action Items Regarding Developing a Chemical Inventory List:
 - Designate a person responsible for obtaining and maintaining the Chemical Inventory List.
 - Make a list of all chemicals in the workplace and that are used by Company employees on jobsites. Use the form in Appendix B to develop the Chemical Inventory List. The following information can be obtained from the label and/or SDS:
 - Product/Trade Name
 - Manufacturer Name
 - Place the copies of the completed list in the Hazard Communication binder and in the SDS binder(s).
 - Conduct regular inventories.
 - Update the Chemical Inventory List regularly when new chemicals are brought onsite.

Labeling

All containers (whether in the shop, office, or on a jobsite) of hazardous chemicals must be visibly labeled, tagged, or marked with the identity of the material and appropriate hazard warnings. If chemicals are transferred into other containers, we must ensure that these secondary containers are labeled as well, as outlined in the Secondary Containers section below.

If a chemical product container arrives without adequate labeling (e.g., labels or part of labels are missing or unreadable) or if there is some uncertainty about the contents of the container, contact the supplier to have them provide a label or take shipment away and replace the containers with adequately labeled ones.

1. Secondary Containers

Portable Containers

As often as possible the original chemical containers should be used. Sometimes, however, a chemical must be transferred to another, smaller container so that it can be used. These portable secondary containers must also be labeled with certain information as indicated below. This is important because the contents must be readily identified in case of overexposure, to prevent improper mixtures, and to ensure proper use and disposal.

Transfer containers should have a pour spout and a cover to prevent spills. Use dedicated labeled containers for transferring liquids.

Secondary container labels must have the following information:

- Product identifier
- Hazard warnings appropriate for employee protection. The hazard warning is a brief statement of the hazardous effects of the chemical (e.g., “flammable”, “causes lung damage”). Words, pictures, symbols, or a combination can be used to provide at least general information regarding the hazards of the chemical and appropriate protection. See Appendix E for more information.

Bulk Containers

Sometimes a chemical is delivered in large quantities or a chemical that needs to be stored in a large secondary container such as a tank. Whenever possible, try to use the label provided by the chemical manufacturer or supplier. If this is not available, label the container as indicated below. Signs, placards, or other written materials may be used instead of affixing labels to individual stationary containers if the alternative method identifies the container to which it is applicable and conveys the information as indicated below.

Chemical tanks and other container labels must have the following information:

- Product identifier
- Hazard warnings appropriate for employee protection. The hazard warning is a brief statement of the hazardous effects of the chemical (e.g., “flammable”, “causes lung damage”). Words, pictures, symbols, or a combination can be used to provide at least general information regarding the hazards of the chemical and appropriate protection. See Appendix F for more information.

Information to include on the secondary container label can be found on the original product label and on the product’s SDS.

Appropriate labels are:

- Clearly visible
- Labels obtained from suppliers.
- Photocopies of supplier labels
- Neatly printed labels constructed by employees that contain the required information.

1. Hazard Communication Action Items Regarding Labeling:

- Designate a responsible person for ensuring that the labels are maintained as required.
- Ensure that all containers, including secondary containers, are labeled and that all labels are visible and legible.
- Each container (primary container as it comes from the manufacturer or supplier) of hazardous chemicals must be labeled with the following information:
 - Identity of the hazardous chemical(s)
 - Appropriate hazard warnings
 - Name and address of the chemical manufacturer, importer, or other responsible party.
- Secondary container labels must have the following information:
 - Identity of the hazardous chemical(s)
 - Appropriate hazard warnings. The hazard warning is a brief statement of the hazardous effects of the chemical (e.g., “flammable”, “causes lung damage”).
- Words, pictures, symbols, or a combination can be used to provide at least general information regarding the hazard. See Appendix E.

Non-Routine Task / Product Trial

A “non-routine task” is one which the employee does not normally perform and for which he/she has not been previously trained and will be exposed to a hazardous chemical(s). An example of a non-routine task is periodic cleaning or maintenance of a piece of equipment, like cleaning a storage tank. An example of a product trial is when a chemical is brought in on a trial basis, to troubleshoot a problem, for instance, or as a part of testing out a piece of equipment that is being considered for purchase. A Job Hazard Analysis should also be utilized to ensure there are no mixing or air quality issues while using a new chemical.

Before employees start a non-routine task, obtain an SDS, review the hazards, determine proper employee protective measures, and conduct training with employees who will be using the product(s). Use the Non-Routine Task/Product Trial Form when hazardous chemicals are being used for non-routine tasks or during product trials for hazardous chemical(s). Refer to Appendix C.

Contractor Notification

The Company must notify outside contractors working at our facility (shop, warehouse, or jobsite) of the hazardous materials they may encounter in their work (for example racking installers working near where hydraulic oil is kept). A responsible person should be designated to provide this notification. Use the Contractor Notification Form in Appendix D to assist in and document this notification.

Training

Information and training are a critical part of the Hazard Communication Program. Through effective training, workers will learn to read and understand information from SDS and labels

and understand the risks of exposure to chemicals in the workplace as well as ways to protect themselves.

1. Each employee who may be exposed to hazardous chemicals when working must be provided information and trained:
 - Prior to initial assignment to work with a hazardous chemical
 - Annually
 - Whenever the hazard changes. Examples of a change in hazard might be a new work assignment, a change in a process, or use of a new chemical or new chemical formulation.

Training must be conducted in a language that is understood by the employee.

It is The Company's policy that all employees who may come in contact with chemicals must receive general training as outlined below. Employees who are expected to work more directly with chemicals must receive specific training about the chemicals they work with.

1. The following table outlines the type of Hazard Communication training to be received according to job type:

Job Type

General Hazard Communication Training

Specific Hazard Communication Training

Office Employees

X

Field Employees

X

Sales Employees

X

Shop Employees

X

Document all training by keeping sign-in sheets, course agendas, a list of training materials used, etc. Make sure the documentation includes the names of instructors, and if applicable, interpreters.

1. Training Materials

- Hazard Communications Safety Binder
- OSHA Hazard Communication Standard (Appendix A)
- SDS Binder(s) with Chemical Inventory List(s) (Appendix B)
- Hazard Communications training PowerPoints (Both General and Specific)
- Copies of GHS SDS and Labeling Quick Guide; refer to Appendix E
- SDS Application Download and Use Instructions (Appendix F)

General Hazard Communication Training for all Employees

The general training for employees who may be exposed to chemicals should include the following:

- 1 Explanation of the Hazard Communication Standard, including an explanation of the labeling system and SDS and how employees can obtain and use the appropriate hazard information.
- 2 Review the areas and operations within the facility where hazardous chemicals are stored, used, or might be encountered and, very generally, what hazards are involved.
- 3 Provide the location and availability of the Hazard Communication Safety Binder, the Chemical Inventory List(s), and SDS Binder(s), which are to be always kept accessible.
- 4 Review the location and operation of related emergency equipment (e.g., spill kits) and emergency eyewash station.

1. Specific Hazard Communication Training

In addition to the general training outlined above, specific training is required for all individuals who are expected to work with or come in contact with hazardous chemicals.

In addition to Company required training, some field personnel may also need to receive hazard communication and safety orientation training required by customers.

The following information is to be included in the specific training:

- 1 The labeling procedure used for secondary containers. (Refer to the Hazard Communication Policy – Labeling Section)
- 2 Review the specific chemicals employees will be working with; including their common chemical names, and where they are located in the facility. This will be performed for bulk chemicals used such as hydraulic oil. Employees are directed to the SDS or manufacturer's label for small, seldom used items. For the larger, more common use chemicals, the following detailed information (obtained from the respective SDS) must be covered (Refer to the SDS binder(s)):

1. The physical nature of the chemical (how to recognize it).
2. The specific hazards involved (fire and explosion, reactivity, and specific health hazards).

3. The health effects and likely symptoms of overexposure and the routes of exposure (skin contact, breathing, etc.).
4. First aid procedures.
5. Proper procedures in case of leaks or spills.
6. Proper disposal procedures, if appropriate.
7. Methods and observations that may be used to detect the presence or release of a hazardous chemical in the work area (for example, visual appearance or odor of the chemicals).
8. The measures employees can take to protect themselves from hazards, such as appropriate work practices, emergency procedures, and personal protective equipment to be used.
9. Hazard Communication Action Items Regarding Employee Information and Training:
 - Designate a person responsible for conducting the training. If needed identify and use an interpreter for bilingual training sessions.
 - Gather material that will be needed for this training including a computer and items listed above.
 - Develop a schedule to train employees at the appropriate intervals: new employees, annually, and upon change of hazard.
 - Use the training material listed in this policy to properly train your employees. Include information from SDS obtained from customers (if necessary).
 - Keep records of all training activities.

Recordkeeping

1. Training records must be maintained for a minimum of three years. The most recent record of training must always be kept on file, even if older than three years. Training records should be maintained in safety files, not individual personnel files.
2. SDS are updated occasionally with new information. The old SDS for a product can be discarded as long as the chemical formulation has not changed. If the product has changed formula, stamp or write an archive date on the old SDS and file it in an inactive or archived SDS file.
3. SDS on chemicals taken out of use must also be filed in an inactive SDS file (archive). Note the date the chemical was taken out of use and the date the SDS was archived.
4. When the Chemical Inventory List is changed or updated, the old list needs to be filed in a chronological file.
5. Keep the following records for thirty (30) years:
 - Archived SDS
 - Chemical Inventory List
 1. Hazard Communication Action Items Regarding Recordkeeping:
 - Designate a person responsible for record keeping.
 - Set up a Hazard Communication Program Records file. It is suggested that you keep it along with other OSHA records, logs, or files. Keep the following in it:
 - Current SDS

- Archived SDS
- Chemical Inventory List – Old and current
- Hazard Communication Training Records: rosters, training agendas, a list of training materials used.

Asbestos

(Reference: Policy AXX- 0013)

This program provides general safety guidelines and awareness to asbestos. It is a matter of Company policy to provide our employees with information about awareness, possible health effects, identification and training of asbestos and asbestos containing materials. The Asbestos program is stored electronically. Employees can request a copy of the written Asbestos Program and all other training materials at any time. A binder will also be stored at the office containing program information. For the most up-to-date lists or information, please contact the Safety Manager.

For more information, the entire Asbestos OSHA standard can be accessed by clicking this link: [OSHA 1926.1101 Asbestos Standard](#) or here: [OSHA 1910.1001 Asbestos Standard](#)

Definitions

Below are some definitions used in the Asbestos standard:

Asbestos-containing material (ACM): means any material containing more than 1% asbestos.

Employee exposure: means exposure to airborne asbestos that would occur if the employee were not using respiratory protective equipment.

Fiber: means a particulate form of asbestos 5 micrometers or longer, with a length-to-diameter ratio of at least 3 to 1.

Presumed asbestos containing material (PACM): means thermal system insulation and surfacing material found in buildings constructed no later than 1980.

Permissible Exposure Limit (PEL): Asbestos is an airborne concentration in excess of 0.1 f/cc as an eight (8) hour time-weighted average.

Surfacing Material (SM): Asbestos containing material that is sprayed, troweled-on, or otherwise applied to surfaces such as acoustical plaster on ceilings and fireproofing materials on structural members.

Thermal System Insulations (TSI): Asbestos containing material applied to pipes, fittings, boilers, tanks, ducts, or other structural components to prevent heat loss or gain.

Time-Weighted Average (TWA): The eight (8) hour TWA is the average concentration of a chemical in the air over an eight-hour exposure period.

What is Asbestos?

Asbestos is a naturally occurring fibrous silicate mineral. It consists of six types, all composed of long and thin fibrous crystals. Each fiber is made up of many microscopic “fibrils” that can be released into the atmosphere through abrasion and other processes. It has been used in products such as insulation for pipes, floor tiles, building materials, and vehicle brakes and clutches. Despite its adverse effects on human health, asbestos was commonly used as a building material until its dangers were more widely recognized in the 1970s. Many older buildings still contain asbestos, and its consequences can take decades to manifest due to a latency period of around 20 years.

1. Main Types of Asbestos

Identifying asbestos visually can be challenging, but here are some key points to consider:

Color and Texture:

- **Chrysotile:** The most common type of asbestos, it appears white and has a curly, layered structure. Chrysotile is often found in roofs, ceilings, walls, floor tiles, and car parts.
- **Amosite:** Brown in color, amosite poses a higher cancer risk than chrysotile. It's commonly used in insulation for pipes, ducts, and appliances.
- **Crocidolite:** Blue in color, crocidolite is less common and has fine, needle-like fibers.

Form and Appearance:

- Asbestos is usually found in clumps or groups, not as individual fibers.
- It may appear as crumbling material, especially if disturbed.

Testing

- You cannot visually confirm asbestos just by looking at it.
- If you suspect a material contains asbestos, treat it as if it does and contact your supervisor.
- Testing by a licensed asbestos lab is necessary for accurate identification.

Asbestos fibers are often microscopic and can remain airborne for days, making inhalation a serious risk. If you encounter any suspicious material during modernizations or repairs, contact your supervisor promptly.

Asbestos Health Hazards

Lung Cancer:

- Asbestos exposure combined with tobacco smoking significantly increases the risk of developing lung cancer.

Mesothelioma:

- A rare cancer affecting the membrane lining the lungs, chest cavity, or abdominal cavity.
- Signs may not appear until 30 to 40 years after exposure.

Other Cancers:

- Asbestos exposure also raises the risk of laryngeal, ovarian, pharyngeal, stomach, and colorectal cancers.

Working Near Asbestos

Employees will not perform any construction, maintenance, or repairs in areas where there is a potential exposure to a PACM/ACM, or in regulated areas in the normal scope of their work. If any employee is asked to enter such areas, they need to contact their supervisor before doing so. Company employees are not trained to abate asbestos. Employees will not, under any circumstances, intentionally disturb an ACM or PACM that will create an exposure above the PEL. If an accidental disturbance occurs that may exceed the PEL, the employee will immediately vacate the site until the asbestos can be removed and contact their supervisor.

1. Asbestos Investigation Guidelines

In the event a PACM is identified in an area where Company employees are performing work, please follow the steps set out below to ensure the health and safety of our employees.

Identification of PACM

An employee who identifies a PACM should notify his/her supervisor immediately and suspend any work in the area.

Investigation of PACM and Notification to Building Owner

A qualified supervisor will inspect the area in question to determine if potential exposure exists.

When potential exposure is confirmed, the supervisor will meet with an authorized Customer/building owner representative to request documentation that the questionable material is or is not an ACM. The documentation shall include verification that the area has been tested in accordance with all OSHA guidelines and that the levels, if any, are below the PEL.

If the building Owner provides test results, e-mail them to the Safety Manager for review.

If test results reveal that there are no ACMs, or levels lower than the PEL per OSHA guidelines, the Safety Manager will notify the supervisor that the area is clear, and that work can continue.

NOTE: If the material is ACM and exposure levels are lower than the PEL, the supervisor or owner must post signs and notify employees not to disturb the area containing asbestos.

If the building owner is unable to produce documentation that the area is free of asbestos, the supervisor shall notify the building Owner that testing, and air monitoring is required.

NOTE: If test results reveal ACM, air monitoring must be performed. If monitoring reveals levels above PEL, work in the area will not proceed until ACM is abated. Have the building Owner sign the PACM Evaluation Form.

Sample Removal

If building Owner fails to provide testing documentation or refuses to test the PACM in accordance with regulations, then a proper consultant shall be called to perform sampling at the Owner's cost. Test results shall be sent to the Safety Manager for review.

Test Results

If test results reveal no exposure, or levels lower than those permissible under OSHA guidelines, the Safety Manager will notify the supervisor that the job is cleared, and that work can continue.

If test results reveal levels above the permissible levels per OSHA guidelines, the Safety Manager will notify the supervisor and building owner that service will be suspended until the asbestos is abated and the area is safe for our employees.

When the job site is cleared and/or levels are below the permissible guidelines, the supervisor shall make the test results available in the office and at the job site.

Training

1. All field employees will review asbestos awareness routinely. Awareness training will include:
 - The Company procedures for handling sites with PACM or ACM.
 - Ways to recognize asbestos.
 - Adverse health effects of asbestos exposure.
 - Operations that could result in asbestos exposure and the importance of protective controls to minimize exposure.
 - The appropriate work practices for performing jobs around asbestos.
 - Medical surveillance program requirements.
 - The contents of the asbestos Policy.

Sampling

If a supervisor determines the need to sample material to determine the presence of asbestos (that will not be covered by the building owner), notify the Safety Manager to make necessary arrangements with a licensed consultant.

Monitoring

A qualified person shall monitor air quality when necessary. Air samples will be monitored in accordance with OSHA regulations for PELs.

Recordkeeping

1. **Objective Data Records** - Where the Company uses objective data to demonstrate that products made from or containing asbestos cannot release fibers in concentrations at or above the PEL or STEL, they will keep an accurate record for as long as it is relied on.

Monitoring Records - All records of all employee exposure monitoring will be maintained for at least 30 years and made available to the affected employees, former employees, their designated representative, and/or OSHA's Assistant Secretary when requested.

Medical Surveillance Records - All records will be maintained for the duration of the employee's employment plus 30 years and made available to the subject employee or anyone having written consent of that employee and or/OSHA's Assistant Secretary.

Training Records – Training records will be kept for three (3) years for field employees.

Other Recordkeeping - All other records shall be maintained for one (1) year beyond the last date of employment for each employee.

Hand and Power Tools

(Reference: Policy AXX-0014)

The Company and employees share the responsibility for the safe condition of tools and equipment, but employees have the responsibility for properly using and maintaining tools.

Employees are required to use Company-approved tools and are required to inspect all tools and equipment before each use.

Employees must wear appropriate Personal Protective Equipment (e.g., safety goggles, gloves, etc.), due to hazards that may be encountered while using portable power and hand tools.

When working around flammable substances, sparks produced by iron and steel hand tools can ignite. Where this hazard exists, spark-resistant tools made from brass, plastic, aluminum, or wood should be used.

If any tool is defective, employees must not use the tool and shall contact their Supervisor for a replacement. Apply a "DEFECTIVE - DO NOT USE" tag to the tool.

Hazard Recognition

In the process of removing or avoiding hazards, employees must recognize hazards associated with different tools and safety precautions necessary to prevent those hazards. At times, a Job Hazard Analysis may be used to think about the possible hazards with using a particular tool in a specific area.

General Safety Precautions

Employees who use hand and power tools, and who are exposed to the hazards of falling, flying, abrasive, and splashing objects, or exposed to harmful dusts, fumes, mists, vapors, or gases shall be provided with PPE necessary to protect them from the hazard.

Following five basic safety rules can prevent hazards involved in the use of power tools:

- 1 Keep all tools in good condition with regular maintenance.
- 2 Use the right tool for the job.
- 3 Examine each tool for damage before use.
- 4 Operate according to Manufacturer instructions.
- 5 Provide and use appropriate PPE.

Employees and supervisors have a responsibility to work together to establish safe working procedures. If a hazardous situation is encountered, it should be immediately brought to the attention of a supervisor.

Hand Tools

Employees shall not use unsafe hand tools. Tools must be used for their intended purpose. An example of improper use is using a screwdriver as a chisel, which may cause the tip of the screwdriver to break and fly.

Do not use wrenches, including adjustable, pipe end, and socket wrenches when jaws are sprung to the point that slippage occurs.

Keep wooden tool handles free from splinters or cracks. Keep tool handles tight. Do not tape tool handles.

1. Knife Safety

Use these rules when utilizing knives in the workplace:

Cut away from your body

- This is the first rule of knife safety and prevents the blade from slipping or sticking.

Use a stable surface

- Don't leave knives on worktops where they can be pushed off.

Wear protective equipment

- This includes cut-resistant gloves on your knife hand and safety glasses to protect your eyes.

Keep knives sharp

- Check for dullness, chips, or roughness, and replace blades when necessary.

Store knives safely

- Keep them in a knife rack or drawer with the handles facing out, or in a container labeled "knives only". Don't leave knives loose or near the edge of a counter, and don't store them with the cutting edge exposed.

Maintain knives

- Clean knives immediately after use, and throw away broken, dull, or rusty blades in a puncture-resistant container.

Be careful when carrying knives

- If you need to carry an unprotected knife, carry it with the blade pointed down and close to your side.

Power Tool Precautions

Power tools can be hazardous when used improperly. There are several types of power tools, based on the power source they use: electric, pneumatic, liquid fuel, hydraulic, and powder-actuated.

Power tool users should observe the following general precautions:

1. Read the tool manufacturer owner's manual.
2. Never carry a tool by the cord or hose.
3. Never yank the cord or the hose to disconnect it from the receptacle.
4. Keep cords and hoses away from heat, oil, and sharp edges.
5. Disconnect tools when not in use, before servicing, and when changing accessories such as blades, bits, and cutters.
6. Keep observers at a safe distance away from the work area.
7. Secure work with clamps or a vise, freeing both hands to operate the tools.
8. Avoid accidental starting. The worker should not hold a finger on the switch button while carrying a plugged-in tool.
9. Tools should be maintained with care. They should be kept sharp and clean for the best performance.
10. Follow instructions in the Operator's Manual for lubricating and changing accessories.
11. Be sure to keep good footing and maintain a good balance.
12. The proper apparel should be worn. Loose clothing, long hair, or jewelry can become caught in moving parts.
13. All portable electric tools that are damaged shall be removed from use and tagged "Defective".

Guards

When power operated tools are designed to accommodate guards, they shall be equipped with such guards when in use. **DO NOT REMOVE** safety guards when using a tool. One or more types of guarding may be provided to protect the operator and other employees from hazards such as those created by:

- Point of Operation.
- In-running nip points.
- Rotating parts
- Flying chips and sparks

Portable circular saws, having a blade diameter greater than two (2) inches, shall be equipped with a guard above and below the base plate or shoe. When the tool is withdrawn from the work, the lower guard shall automatically and instantly return to a covering position.

Safety Switches

All power tools that are equipped with a constant pressure switch or control that will shut off the power when the pressure is released, shall remain unaltered.

All hand-held powered drills, tappers, fasteners, drivers, horizontal, vertical, and angle grinders with wheels greater than two (2) inches shall be equipped with a constant pressure switch.

Power Tools

1. Electric Tools

The possibility of electrocution is a hazard associated with electric tools.

Among the hazards of electric-powered tools, are burns and slight shocks, which can lead to injuries or even heart failure. Under certain conditions, even a small amount of current can result in fibrillation of the heart and eventual death. A shock can also cause the user to fall off a ladder or other elevated work surface.

To protect employees from shock, tools shall have a three-wire cord with ground and be grounded, be double insulated, or be powered by a low-voltage isolation transformer. Three-wire cords contain two current carrying conductors and a grounding conductor. One end of the grounding conductor connects to the tool's metal housing. The other end is grounded through a prong on the plug.

Do not remove the third prong from the plug. All electrical tools should have double insulation.

1. General Practices When Working with Electric Tools

- Operate electric tools within their design limitations.
- Gloves are required during use of electric tools.
- Eye and/or face protection as required.
- When not in use, store tools in a dry place.
- Do not use electric tools in damp or wet locations.
- Work areas shall be well lit.
- Protect all electric tools with a GFCI device when plugging into temporary wiring.

1. Powered Abrasive Wheel Tools

Powered abrasive grinding, cutting, polishing, and wire buffing wheels create special safety problems because they may generate airborne fragments.

Before an abrasive wheel is mounted, inspect it closely and sound or ring-test it to make sure that it is free from cracks or defects. To test, tap wheels gently with a non-metallic instrument. Do not use it if they sound cracked or dead because they could fly apart in operation. An undamaged wheel will give a clear metallic tone or "ring."

Before use, ensure that the abrasive wheel is compatible with the tool.

To prevent the wheel from cracking, the user should ensure it fits freely on the spindle. Tighten the spindle to hold the wheel in place without distorting the flange. Follow the manufacturer recommendations. Take care to assure that the spindle wheel will NOT exceed the abrasive wheel specifications.

Due to the possibility of a wheel disintegrating (exploding) during start-up, employees should never stand directly in front of the wheel as it accelerates to full operating speed.

Portable grinding tools shall be equipped with safety guards to protect workers, not only from the moving wheel surface, but also from flying fragments. In Addition:

1. Always use eye protection.
 2. Turn off the power when not in use.
 3. NEVER clamp a hand-held grinder in a vise.
1. Pneumatic Tools

Pneumatic tools are powered by compressed air and include chippers, drills, hammers, and sanders.

Pneumatic tools present several safety hazards. The most common danger is getting hit by one of the tool's attachments or by a fastener.

1. Eye protection is required, and face protection is recommended for employees working with pneumatic tools.
 2. When working with noisy tools, such as jackhammers, proper hearing protection is required.
 3. When using pneumatic tools, employees shall check to see that the tool is fastened securely to the hose to prevent disconnection. A short wire or positive locking device attaching the air hose to the tool will serve as an added safeguard.
 4. A safety clip or retainer shall be installed to prevent attachments, such as chisels on a chipping hammer, from being unintentionally shot from the barrel.
 5. Set screens to protect nearby workers from being struck by flying fragments around chippers, riveting guns, staplers, or air drills.
 6. Do not point compressed air guns toward anyone. Users should NEVER "dead-end" a tool against themselves or another person.
 7. If using pneumatic equipment to clean with, the PSI will be set at no more than 30 PSI and proper safety glasses and other PPE will be worn.
1. Powder-Actuated Tools

Powder-Actuated tools are prohibited.

1. Hydraulic Power Tools

The manufacturer's recommended safe operating pressure for hoses, valves, pipes, filters, and other fittings shall not be exceeded.

Jacks

The manufacturer rated capacity shall be clearly marked on all jacks and not exceeded.

All jacks: lever and ratchet jacks, screw jacks, and hydraulic jacks, shall have a device that stops them from jacking up too high.

Do not use a jack to support a lifted load. Once the load has been lifted, use a block to support the load.

Use wooden blocking under the base, if necessary, to make the jack level and secure. If the lift surface is metal, place a one (1) inch thick hardwood block, or equivalent, between it and the metal jack head to reduce the danger of slippage. To properly set up a jack, ensure the following:

- The base rests on a firm level surface.
- The jack is correctly centered.
- The jack head bears against a level surface.
- The lift force is applied evenly.

Proper maintenance of jacks is essential for safety. Inspect all jacks before each use and lubricate regularly. Jacks shall be inspected, not less than the following:

1. For constant and intermittent use, once every year.
2. For jacks subjected to abnormal load or shock, immediately before and thereafter.

Tag defective jacks and do not use until repaired. Hydraulic jacks exposed to freezing temperatures shall be filled with the manufacturer recommended liquid.

Ladder and Stairway Safety

(Reference: Policy AXX-0015)

The purpose of a ladder and stairway safety policy is to establish guidelines for the safe use of ladders and stairways. It ensures that employees, contractors, and visitors understand how to use ladders safely and avoid accidents related to slips, trips, and falls. This policy covers ladder types, proper usage, inspection, and maintenance requirements. This policy also covers proper stairway design in order to provide awareness to unsafe stairways.

Definitions

Single-Cleat Ladder - A ladder consisting of a pair of side rails, connected by cleats, rungs, or steps.

Double-Cleat Ladder - A ladder with a center rail to allow simultaneous two-way traffic, for employees ascending or descending.

Failure - Load refusal, breakage, or separation of components.

Fixed Ladder - A ladder that cannot be readily moved or carried, because it is an integral part of a building or structure.

Handrail - A rail used to provide employees with a handhold for support.

Job-Made Ladder - A ladder that is fabricated by employees, typically at the construction site. Not commercially manufactured.

Load Refusal - The point at which the structural members lose their ability to carry the load.

Point Of Access - All areas used by employees, for work related passage from one area or level to another.

Portable Ladder - A ladder that can be readily moved or carried.

Riser Height - The vertical distance, from the top of a tread or platform/landing to the top of the next higher tread or platform/landing.

Side-Step Fixed Ladder - A fixed ladder that requires a person to get off at the top to step to the side of the ladder side rails, to reach the landing.

Stair Rail System - A vertical barrier erected along the unprotected sides and edges of a stairway to prevent employees from falling to lower levels.

Temporary Service Stairway - A stairway where permanent treads and/or landings are to be filled in later.

Through Fixed Ladder - A fixed ladder that requires a person getting off at the top, to step between the side rails of the ladder, to reach the landing.

General Requirements

No employee is allowed to work with ladders until they have been properly trained. Before using any ladder, inspect it to ensure it meets the manufacturers and company requirements. Before climbing a ladder, inspect the setup to ensure that it is properly secured.

- A stairway or ladder shall be provided at all worker points of access, where there is a break in elevation of 19 inches or more, and no ramp, runway, embankment, or personnel hoist is provided.

- When there is only one point of access between levels, it shall be kept clear to permit free passage. If free passage becomes restricted, a second point of access shall be provided and used. When there are more than two points of access between

levels, at least one point of access shall be kept clear.

- All stairway and ladder fall protection systems required by these rules shall be installed and all duties required by the stairway and ladder rules shall be performed before employees begin work.
- Unsafe Customer Site Work Area Access: If the customer worksite doesn't allow safe access to elevator equipment and/or elevator workspaces the employee will notify the office immediately before proceeding and note conditions on the time ticket.

Stairways

The following general requirements apply to all stairways that Company employees may encounter at jobsites. If any of the following conditions are encountered or are out of specification, the employee is to find another way to access the area or call a supervisor for help.

- All stairway parts shall be free from dangerous conditions, such as protruding nails.
- Slippery conditions on stairways.
- DO NOT use stairways where the treads and/or landings have NOT been filled in with concrete or other material, unless the pans of the stairs and/or landings are temporarily filled in with wood or other material. Temporary treads shall be made of wood or other solid material and installed to the full width and depth of the stair.
- DO NOT use spiral stairways that are NOT part of a permanent structure.
- Stair rail systems and handrails shall be surfaced to prevent injuries, such as

punctures or lacerations, and to keep clothing from snagging.

- Handrails shall provide an adequate handhold for employees to grasp, to prevent falls.
- The ends of stair rail systems and handrails shall be constructed to prevent dangerous conditions such as rails protruding beyond the end posts of the system.
- Unprotected stairway landing sides and edges shall be constructed with standard 42-inch (+/- 3") guardrail systems.

1. Using Electric Stair Climber Machines

Electric stair climbers are an excellent way of moving heavy objects up or down stairs. However, there are some safety concerns one needs to know to use them:

- Know how to use the climber. It is too late to figure out how to use the machine when you are already moving an object. Ask for help or the manufacturer's instructions.
- Know your power source: If your climber is battery-operated, ensure you have enough power before use.

- Body Positioning: Many climbers have taller handles to aid the user in leveraging the unit and object being moved. Keep your posture as straight as possible and pay attention to the clearance at your feet.
- Losing balance: Plan your path and look for hazards before using the climber. If you happen to lose balance, or the equipment begins to slip, LET IT GO. There is an instinct to attempt to catch or move your body in an attempt to save the object. Most often this ends in serious injury or death, especially when on stairs.
- Never allow anyone to be below the object being moved for any reason, even if the object being moved is strapped.

Ladders

The following general requirements apply to all ladder usage and inspection:

- Check ladders for defects before use.
- Clear scrap and material away from the base and top of the ladder.
- Secure the base and top of the ladder to prevent accidental movement.
- Set the ladder on a firm, level surface. If the ladder must sit on soft surfaces or rough

soil, use mud sills.

- It is often tempting to borrow ladders from other trades or customers. However, this is against policy due to the possible hazards that may be known or unknown.
- Single-width job-built ladders are designed for one worker at a time. A double width

ladder can be used by two workers, providing they are on opposite sides.

- Ensure that rails on ladders extend at least three (3) feet above the landing.
- Set straight or extension ladders one foot out for every three (3) to four (4) feet up.
- Do not position ladders against flexible or moveable surfaces.
- Do not use a portable ladder (A-Frame) as an extension ladder.
- Before setting up ladders, **ALWAYS** check for overhead electrical or power lines.
- ALWAYS face the ladder when climbing up or down and while working from it.
- Maintain 3-point contact when climbing up or down. A 3-point contact is two hands

and one foot or two feet and one hand on the ladder at all times.

- Keep your center of gravity between the side rails. When climbing up or down, do

NOT carry tools or materials in your hands. Use a hoist rope instead.

- Keep boots clean from mud, grease, or any slippery materials that could cause loss of

footing.

- When working six (6) feet or more above the ground or floor, wear a safety

harness with the lanyard tied off to a proper structure.

- NEVER erect ladders on boxes, carts, tables, or other unstable surfaces.
- NEVER use ladders horizontally as scaffold planks, runways, or any other service

for which they are not designed.

- Stand no higher than the third rung from the top on any ladder. Maintain knee contact

Balance.

- DO NOT splice short ladders together to make a longer ladder.
- DO NOT use ladders for bracing.
- DO NOT jump off or onto the ladder or try to “walk” a ladder while on it.
- DO NOT set up ladders in doorways, passageways, driveways, or any other location

where they can be struck or knocked over. Use caution tape around work area to

keep other trades away from the area.

- NEVER rest a ladder on its rungs. Ladders must rest on their side rails. When

erecting long, awkward, or heavy ladders, get help to avoid injury from overexertion.

Wooden Ladder Requirements

Wood ladders are prohibited on the job.

Aluminum Ladders

Aluminum and other conductive ladders are prohibited on the job.

Portable Ladder Requirements

Non-self-supporting and self-supporting portable ladders shall support at least four times the maximum intended load; extra heavy-duty type 1A metal or plastic ladders shall sustain 3.3 times the maximum intended load. The ability of a self-supporting ladder to sustain loads shall be determined by applying the load to the ladder in a downward vertical direction. The ability of a non-self-supporting ladder to sustain loads shall be determined by applying the load in a downward vertical direction, when the ladder is placed at 75.5 degrees from horizontal.

- Based on the nominal length-of the ladder, each section of a multi-section ladder

shall overlap the adjacent section by at least the number of feet stated in the following table:

Nominal Ladder Length (ft)

Minimum Overlap (ft)

36 feet and under

3

37-48 feet

4

49-60 feet

5

- Extension ladders shall be equipped with positive stops, which will ensure the overlap specified in the table above.
- A metal spreader or locking device shall be a component of each stepladder to securely hold the front and back sections in the open position.

Fixed Ladders

The following general requirements apply to all fixed ladders that Company employees may encounter at jobsites. If any of the following conditions are encountered or are out of specification, the employee should find another way to access the area or call a supervisor for help.

- A fixed ladder shall support at least two loads of 250 lbs. each, concentrated between any two consecutive attachments. Fixed ladders also shall support added anticipated loads caused by ice buildup, winds, rigging, and impact loads resulting from the use of ladder safety devices.
- Individual rung/step ladders shall extend at least 42 inches above an access level or landing platform, either by the continuation of the rung spacing as horizontal grab bars or by providing vertical grab bars that shall have the same lateral spacing as the vertical legs of the ladder rails.
- The rungs of individual rung/step ladders shall be shaped to prevent slipping off the end of the rungs.
- The rungs and steps of fixed metal ladders manufactured after January 14, 1991, shall be corrugated, knurled, dimpled, coated with skid-resistant material, or treated to minimize slipping.

- The minimum perpendicular clearance between the centerline of fixed ladder rungs, cleats, and steps, and any obstruction on the climbing side of the ladder shall be 30 inches. If obstructions are unavoidable, clearance may be reduced to 24 inches, provided a deflection device is installed to guide workers around the obstruction.
- The step-across distance, from the nearest edge of the ladder to the nearest edge of equipment or structure, shall not be more than 12 inches, or less than two (2) and a half (1/2) inches. A landing platform shall be provided if the step-across distance exceeds 12 inches.
- Fixed ladders without cages or wells shall have at least a 15-inch clear width to the nearest permanent object, on each side of the centerline of the ladder.
- Fixed ladders shall be provided with cages, wells, ladder safety devices, or self-retracting lifelines, where the length of climb is less than 24 feet, but the top of the ladder is at a distance greater than 24 feet above lower levels.
- If the total length of a climb on a fixed ladder equals or exceeds 24 feet, the following requirements shall be met: fixed ladders shall be equipped with either (a) ladder safety devices; (b) self-retracting lifelines, and rest platforms at intervals not to exceed 150 feet; or (c) a cage or well, and multiple ladder sections, each ladder section not to exceed 50 feet in length. These ladder sections shall be offset from adjacent sections, and landing platforms shall be provided at maximum intervals of 50 feet.
- The side rails of through or side-step fixed ladders shall extend 42 inches above the top level or landing platform served by the ladder. For a parapet ladder, the access level shall be at the roof, if the parapet is cut to permit passage through it; if the parapet is continuous, the access level is the top of the parapet.
- Steps or rungs for through-fixed ladder extensions shall be omitted from the extension; and the extension of side rails shall be flared to provide between 24 inches and 30 inches of clearance between side rails.
- Fixed ladders shall be used at a pitch, no greater than 90 degrees from the horizontal, measured from the backside of the ladder.

Ladder Inspection and Maintenance

All ladders should have at least a cursory inspection before use. Portable ladders should be inspected carefully each day, or if it was previously used by someone else.

- Inspect ladders for structural rigidity.
- Inspect non-skid ladder feet for wear, imbedded material, and proper pivot action on swivel feet.
- Replace frayed or worn ropes on extension ladders, in accordance with the Manufacturer's guidelines.
- Check all ladders for grease, oil, caulking, imbedded stone, and metal, or other materials that could make them unsafe.
- Ensure all ladders have the legible manufacturer's stickers or tags intact and easy to read.

Ladders with structural defects, such as broken or missing rungs, cleats, or steps, broken or split rails, corroded components, or other faulty or defective components shall immediately be marked defective, or tagged with "DANGER - DO NOT USE" or similar language and removed from service until repaired. At no time should a ladder be "field modified" for any reason.

Axxiom Elevator will not repair ladders. If your ladder is defective for any reason, bring it in and turn it in to your supervisor. The supervisor will then issue you a replacement ladder as soon as possible.

Training Requirements

Each employee shall be trained in the proper use of ladders and stairways. Employees shall recognize hazards related to ladders and stairways and use proper procedures to minimize these hazards. Training shall include, but not be limited to:

- The nature of fall hazards in the work area.
- The correct procedures for erecting, maintaining, and disassembling the fall protection systems used on job sites.
- The proper construction, use, placement, and care in handling of all stairways and ladders.
- The maximum intended load-carrying capacities of ladders used.

Re-training should be provided for each employee, as necessary, so that the employee maintains the understanding and knowledge necessary for compliance with the Safety Policy.

Electrical Safety

(Reference: Policy AXX-0017)

Electricity is a serious workplace hazard, capable of causing both employee injury (shocks, electrocution, fires, and explosions) as well as serious property damage. By providing personnel with proper training in safe electrical work practices, Axxiom Elevator hopes to reduce the risk of such incidents.

General Requirements (Examination, Installation & Use of Equipment)

Inspect all electrical equipment to ensure it is free from recognized hazards that are likely to cause death or serious physical harm to employees. ASSUME ALL CIRCUITS ARE LIVE UNTIL VERIFIED!

Safety of equipment shall be determined by the following:

- Suitability of equipment, for an identified purpose, may be evidenced by instructions, labeling, or by certification for that identified purpose.
- Mechanical strength and durability. For parts designed to enclose and protect other equipment, this includes the adequacy of the protection provided.
- Electrical wire insulation (and it's condition). Does it have insulation, should it have insulation and is it in good condition.
- Heating effects under usage conditions.
- Arcing effects that are possible, tools that could cause arc effects, and how to protect existing situations and locations from inadvertent arcing or grounding.
- Classification by type, size, voltage, current capacity, and specific use.
- Other factors that contribute to the practical safeguarding of employees who use or are likely to come in contact with the equipment.

Approval

All electrical conductors and equipment used by employees shall be approved by a supervisor. Insulated tools shall meet the expectations of the company and shall be approved by the supervisor. Voltage testers and meters shall have a minimum 1000V tolerance.

Marking

All electrical equipment must be properly marked providing the voltage, current, wattage, or other ratings as necessary.

Working Space and Clearances

Sufficient access, adequate lighting, and working space shall be provided and maintained around all electrical equipment. If performing electrical work in a confined space that either does not have adequate lighting, or other unique circumstances, a Job Hazard Analysis must be completed.

Working Clearances

Working clearances for live parts, operating at 600 volts or less, and likely to require examination, adjustment, servicing, or maintenance while live may not be less than:

Minimum Clear Distance for Condition (2) (ft)

Nominal Voltage to Ground

(a)

(b)

(c)

3 (1)

3 (1)

3

151-600

3 (1)

3.5

4

1. The minimum clear distance may be 2 feet 6 inches for installations built prior to April 16, 1981.
2. Conditions (a), (b), and (c) are as follows:

(a). Exposed live parts on one side and no live or grounded parts on the other side of the working space, or exposed live parts on both sides effectively guarded by suitable wood or other insulating material. Insulated wire or insulated busbars operating at not over 300 volts are not considered live parts.

(b). Exposed live parts on one side and grounded parts on the other side.

(c). Exposed live parts on both sides of the workspace not guarded as provided in Condition (a) with the operator between.

The minimum headroom for working spaces near service equipment, switchboards, panel boards, or motor control centers shall be six (6) feet three (3) inches.

Guarding

Live parts of electric equipment, operating at 50 Volts or more, shall be guarded against accidental contact. Guarding live parts shall be accomplished by one of the following methods:

- Location in a cabinet, room, vault, or similar enclosure, accessible only to qualified persons. Use of permanent, substantial partitions, or screens to exclude unqualified persons.
- Location on a suitable balcony, gallery, or platform, elevated and arranged to exclude unqualified persons.
- An elevation of eight (8) feet or more above the floor.

Entrance to rooms and other guarded locations containing exposed live parts shall be marked with conspicuous warning signs forbidding unqualified persons to enter. Electric installations that are over 600 Volts and that are open to unqualified persons shall be made with metal-enclosed

equipment or enclosed in a vault or area controlled by a lock. In addition, equipment shall be marked with appropriate caution signs.

Over-current Protection

The following requirements apply to over-current protection of circuits rated 600 volts, nominal, or less:

- Protect conductors and equipment from over-current in accordance with their ability to safely conduct current and the conductors shall have sufficient current-carrying capacity to carry the load.
- Over-current devices shall not interrupt the continuity of the grounded conductor unless all conductors of the circuit are opened simultaneously, except for motor-running overload protection.
- Over-current devices shall be readily accessible and not located where they could create a safety hazard by being exposed to physical damage or located in the vicinity of easily ignitable material.
- Fuses and circuit breakers shall be located or shielded so that employees will not get burned or otherwise injured by their operation, e.g., arcing.

Grounding of Equipment Connected by Cord and Plug

Exposed noncurrent-carrying metal parts of cord and plug connected equipment that may become energized shall be grounded in the following situations:

- When in a hazardous (classified) location.
- When operated at or over 110 volts to ground, except for guarded motors and metal frames of electrically heated appliances, if the appliance frames are permanently and effectively insulated from ground
- When using any equipment listed below:
 - Hand-held motor-operated tools.
 - Cord-and plug-connected equipment, used in damp or wet locations, or by employees.
 - Standing on the ground or on metal floors.
 - Utilizing alternative power sources (ex. Generators).
 - Sump pumps.
 - Tools likely to be used in wet and/or conductive locations.
 - Portable hand lamps.
 - Fans

Tools likely to be used in wet and/or conductive locations need not be grounded, if supplied through an isolating transformer, with an ungrounded secondary of not over 50 Volts. Listed or labeled portable tools and appliances protected by a system of double insulation, or its

equivalent, do not need to be grounded. If such a system is employed, the equipment shall be distinctively marked to indicate that the tool or appliance uses a system of double insulation.

If a tool requires disposal for any reason, give it to your supervisor and get a replacement.

1. Safety Related Work Practices

Protection of Employees: Except for inspection, observation, and troubleshooting, employees will NOT work near any part of an energized circuit that the employee could accidentally contact in the course of work. When working on or near these circuits, the employee shall be protected against shock by de-energizing the circuit and grounding it or by guarding it effectively by insulation or other means.

1. Passageways and Open Spaces

Employees shall use barriers or other guarding to ensure that workspace for electrical equipment will NOT be used as a passageway during the time when energized parts of electrical equipment are exposed. Walkways and similar working spaces shall be kept clear of electric cords.

1. Flexible Cords, Extension Cords and Cables

Unless you are using original manufacturer equipment, use only No. 12 flexible cords or larger to provide electrical power. Flexible cords and cables shall be suitable for conditions of use and location. Flexible cords and cables shall only be used for:

- Pendants
- Wiring of fixtures
- Connection of portable lamps or appliances
- Elevator cables

Do not use flexible cords as a substitute for fixed wiring systems. Extension cord sets, used with portable electric tools and appliances shall be three-wire type and designed for hard or extra-hard usage. Flexible cords used with temporary or portable lights shall be designed for hard or extra-hard usage.

1. Identification, Splices and Terminations

Portable cord and plug connected equipment and flexible cord sets (extension cords) shall be visually inspected before use for external defects (such as loose parts, deformed, and missing pins, or damage to outer jacket or insulation), and for evidence of possible internal damage (such as pinched or crushed outer jacket).

- If a cord is damaged, either have the cord replaced or repaired by a "qualified" person.
- Cords smaller than No. 14 gauge CANNOT be repaired in the field and must be replaced.

- Flexible cords shall be used only in continuous lengths. Splicing of any cords is prohibited.
- Flexible cords shall be connected to devices and fittings so that strain relief is provided, which will prevent pull from being directly transmitted to joints or terminal screws.
- Flexible cords and cables shall be protected from accidental damage. Sharp corners and projections shall be avoided. When passing through doorways or other pinch points, flexible cords and cables shall be provided with protection to avoid damage.

Electrical Shock

An electrical shock is the physiological reaction that occurs when a person's body comes into contact with live electricity and becomes part of an electrical circuit. The shock can feel like a tickle, but it can also cause serious injury.

1. Electrical Shock Effects

An electric shock can cause a range of symptoms, depending on the severity of the shock, including:

- Nervous system effects: Pain, tingling, numbness, weakness, or difficulty moving a limb
- Central nervous system effects: Dazedness, amnesia, seizures, or respiratory arrest
- Other effects: Burns, muscle pain and contractions, headache, vision or hearing problems, abdominal pain, or chest pain
- Cardiac effects: Irregular heartbeat (arrhythmia) or cardiac arrest
- Breathing effects: Difficulty breathing or no breathing at all

Electric shocks can also cause long-term effects, such as permanent scars from burns, cataracts, or ongoing pain, tingling, numbness, and muscle weakness.

When a shock occurs, the victim may be dazed or may experience amnesia, seizure or respiratory arrest. Long-term damage to the nerves and the brain will depend on the extent of the injuries and may develop up to several months after the shock. This type of damage can also cause psychiatric disorders.

1. Severity of Electrical Shocks

The severity of an electrical shock depends on several factors, including:

Current

The amount of current passing through the body, measured in amperes, is the most important factor in determining the shock's intensity. A person can typically tolerate less than 10 milliamperes (mA) of current without losing control of their arm and hand muscles. However, a current as low as 0.25 mA can cause a tingling or buzzing sensation without causing injury.

Time

The length of time the current passes through the body is also important.

Path

The path the current takes through the body can also affect the severity of the shock. For example, shocks from light switches are usually mild, while contact with industrial power sources can have severe effects.

Type of current

Whether the current is alternating current (AC) or direct current (DC) can also affect the severity of the shock.

Body resistance

The amount of resistance in the body can also affect the severity of the shock.

Arc Flash

An arc flash is a release of light and heat that occurs when an electrical current leaves its intended path and travels through the air to another conductor or the ground. This is also known as an arc fault, which is a type of electrical explosion or discharge.

Three factors determine the severity of an arc flash injury: proximity of the worker to the hazard, temperature, and the time it takes for the circuit to break. Because of the violent nature of an arc flash exposure, employee injuries can be very serious.

The temperatures of an arc flash can reach up to 35,000°F (19,400°C) at the arc terminals, which is almost four times hotter than the sun's surface.

Arc Blast

An arc blast is a pressure wave that occurs at the same time as an arc flash, an electrical phenomenon that happens when electricity escapes from a system and travels through the air.

Arc flash and arc blasts can propel molten metal, equipment parts, and debris at speeds of up to 700 miles per hour (mph) and 300 meters per second. These speeds are fast enough to penetrate human bodies and standard personal protective equipment (PPE). Arc flashes can also produce extreme temperatures, pressures, and sounds:

Temperatures: Arc flashes can reach temperatures of over 30,000 degrees Fahrenheit.

Pressures: Arc flashes can generate pressures of hundreds to thousands of pounds per square foot (lb./ft²).

Sounds: Pressure waves from arc flashes can produce sounds that exceed 140–160 decibels (dB).

Injuries can include external burns (i.e. severe burns to the skin), internal burns and intoxication from inhaling hot gasses and vaporized metal, hearing damage, eye damage and blindness from the ultraviolet light of the flash as well as many other devastating injuries.

Arc Flash Hazards and Electrical Safe Work Practices

Axxiom Elevator follows recommendations set forth in the Arc-Flash Hazard Analysis Position Paper written by the National Elevator Industry, Inc. As recommended in NFPA 70E, NEII commissioned an Arc-Flash Hazard Analysis (in compliance with Institute of Electrical and Electronic Engineers, Inc. (IEEE) Standard 1584-2002 for procedures for calculating the incident energy of the arc-flash) by an independent consultant to determine at what level an arc flash hazard existed to employees who work on energized elevator equipment.

Based on the Arc-Flash Hazard Analysis, the arc-flash boundaries at the elevator / escalator controllers ranged from 3 in. to 16 in. from the exposed components and the incident energy calculated at 18 in. ranged from 0.06 cal/cm² to 0.95 cal/cm² which indicates that the arc-flash hazard to employees is primarily to the hands and arms.

The surest means of avoiding an arc-flash hazard is to LOCK-OUT and TAG-OUT the electrical service to a controller and VERIFY (TRY-OUT) that the equipment is in a zero-energy state. As stated in Section 7 of the Elevator Industry Field Employees' Safety Handbook, "unless it is not feasible, (i.e.: inspecting; troubleshooting; observing; etc.) employees shall not perform any work on equipment where there is a potential to come in contact with energized mechanical or electrical hazards until all sources of energy have been de-energized, grounded or guarded". If the equipment must remain energized to perform work, effective insulation and safe electrical working practices should be observed.

Described below are several work practices that may be used to reduce arc-flash hazards when working on energized equipment:

- **Guarding:** - Where possible, install temporary guarding to protect from inadvertent contact.
- **Fuses:** - Verify that the correct size, type and capacity are installed.
- **Personal Protective Equipment:** Use appropriate PPE to protect body parts within the range of 3-16 inches from components that are not otherwise guarded.

Examples of appropriate PPE are:

- Non-conductive eye protection

- Clean leather or fire-resistant gloves
- Natural fiber or fire-resistant rated long-sleeved shirts and pants, or fire-resistant rated long-sleeved coveralls
- **Metallic Articles:** Remove metal articles such as watches, chains, bracelets, earrings, belt buckles and key chains before troubleshooting.
- **Instruments:** - Use category III multi-meters and be familiar with use and limitations. Follow manufacturer's instructions and precautions. Use UL or CSA labeled scopes tested for 1000 V.
- **Lockout/Tag-out/Try-out:** When troubleshooting is complete and further work can be accomplished without the equipment being energized, follow the lockout/tag out/try-out procedures before commencing repairs or service work.
- **Special Conditions:** Troubleshooting in wet, hot, or cold conditions calls for extra caution. Hazards created by water, snow, or condensation in the work area can cause slips, falls, and accidental contact. Don't troubleshoot unless you can keep your shoe/ boot soles dry.
- **Main Line Disconnect:** - **DO NOT OPEN THE MAIN-LINE DISCONNECT SWITCH COVER or the 110 disconnect switch cover for any reason.** If power is not being supplied to the elevator controller (e.g. open main-line fuses, etc.), advise the building owner to correct the condition and notify your supervisor immediately, and document the condition/situation on the time ticket. This is not the elevator company's responsibility.

Lockout/Tagout/Tryout Program

(Reference: Policy AXX-0018)

This Lockout/Tagout/Tryout (LOTOTO) policy is designed to ensure the safety of workers during maintenance and repair of machinery and equipment. The primary purposes of our LOTOTO policy are:

1 Preventing Accidental Start-Up: Ensuring that machinery and equipment are properly shut down and cannot be accidentally started up while maintenance or repair work is being performed.

2 Isolating Energy Sources: Safely isolating and controlling hazardous energy sources, such as electrical, mechanical, hydraulic, pneumatic, chemical, thermal, or other energy types, to prevent accidental release that could cause injury.

3 Ensuring Worker Safety: Providing clear procedures and guidelines to protect workers from the release of hazardous energy during servicing and maintenance activities.

4 Compliance with Regulations: Ensuring compliance with occupational safety and health regulations, such as those mandated by the Occupational Safety and Health Administration (OSHA) in the United States.

5 Standardizing Procedures: Establishing a standardized approach for applying locks and tags to machinery and equipment, ensuring consistency and clarity across the organization.

6 Training and Awareness: Educating employees about the importance of LOTOTO procedures and ensuring they understand how to properly implement these procedures to maintain a safe work environment.

Definitions

Affected Employee - An employee whose job requires him/her to operate or use a machine or equipment on which service or maintenance is being performed under lock-out/tag-out, or whose job requires him/her to work in an area in which such work is being performed.

Authorized Person - The employee who applies the lock-out/tag-out device, or if that person is not available, the authorized person will meet the following guidelines:

- An employee directed and trained by the Company.
- Shall document the removal of the lock-out/tag-out device.
- Follow the steps listed in this program under re-energizing equipment.

Energy Source - Any source of electrical, mechanical, hydraulic, pneumatic, chemical, thermal, or other energy.

Energized - Connected to an energy source or containing residual or stored energy.

Energy Isolation Device A mechanical device that physically prevents the transmission or release of energy, including but not limited to the following: a manually operated electrical circuit breaker; a disconnect switch; a manually operated switch by which the conductors of a circuit can be disconnected from all ungrounded supply conductors, and, in addition, no pole can be operated independently; a line valve; a block; and any similar device used to block or isolate energy. Push buttons, selector switches and other control circuit type devices are not energy isolating devices.

De-Energize - Disconnected from an energy source and NOT containing residual or stored energy.

Lock-Out - The placement of an approved lock on an energy isolating device, in accordance with an established procedure, ensuring that the energy isolating device and the equipment being controlled, cannot be operated until the lock is removed.

Lockout Device - A device that utilizes a positive means such as a lock, either key or combination type, to hold an energy isolating device in the safe position and prevent the energizing of a machine or equipment. Included are blank flanges and bolted slip blinds that can be secured and tagged.

Tag-Out - The placement of a labeled tag or labeled hasp on an energy-isolating device, in accordance with an established procedure, to indicate that the energy isolating device and the equipment being controlled, may not be operated until the tag or hasp is removed.

Tagout Device - A warning tag standardized in size, color, with wording warning of hazardous energy (Do Not Start; Do Not Open; Do Not Close; Do Not Energize; Do Not Operate, etc.). A tagout system must be utilized at all times, whenever a Lockout/Tagout is being performed. The tag must be constructed and printed so that exposure to weather conditions or wet and damp locations will not cause the tag to deteriorate or the message on the tag to become illegible. The tag must be substantial enough to prevent inadvertent or accidental removal. The Tagout device attachment means shall be of a non-reusable type, attachable by hand, self-locking, and non-releasable with a minimum unlocking strength of no less than 50 pounds and having the general design and basic characteristics of being at least equivalent to a one-piece, all environment-tolerant nylon cable tie.

Try-Out – The verification that a machine or system is at a zero-energy state, by trying to restart the equipment or machine, or the application of a meter (that has been verified to be working properly) and showing a zero-energy state.

Company LOTOTO Statement

Unless it is not feasible (i.e., inspecting, troubleshooting, or observing function), employees will not perform any work on equipment where there is a potential to come in contact with energized mechanical or electrical hazards, until all energy sources have been de-energized, grounded, or guarded to protect the employees. ALL EMPLOYEES will follow the proper procedures, as outlined in this policy, when working with energized electrical or mechanical equipment. If an employee finds it more hazardous to work under such conditions, the employee must contact a supervisor before proceeding. If for any reason a device, equipment, or a system cannot be locked out or tagged out, the supervisor must be consulted and a Job Hazard Analysis must be performed before proceeding.

Supervisor's Responsibilities

Supervisors are responsible for:

1. Instructing their workers about the procedures contained in this policy.
2. Periodically inspecting work practices to ensure compliance with these procedures.
3. Immediately initiating corrective action when an employee is not following the proper safety procedures under this policy.

A Supervisor may delegate the instruction of employees and compliance follow-up to a "mechanic" or to the "mechanic in charge". That Supervisor must document, in writing, this delegation to upper management prior to delegation.

Locking and Tagging Circuits

The Company will provide personal locks and tags to be used for locking and tagging circuits. If the energy source has a lockout point, always use locks over tags. Only when an energy source does not have a lockout point, should a tag be used as the Company does not believe that tags are as secure as locks. Place personal locks / tags on deactivated controls while working on energized or de-energized equipment or circuits. Equipment or circuits that are de-energized shall have personal locks / tags attached at all points where such equipment or circuits can be energized. If it is not possible to affix a lock, contact a supervisor for guidance as there may be other ways to lock out the energy source. In an instance that you cannot put a lock or tag on something that needs to be worked on, a Job Hazard Analysis must be conducted, and your supervisor consulted prior to proceeding with the work.

If any employee is working on equipment that may become accidentally energized, then each employee working on the equipment shall affix a personal lock on the locked-out device. If a piece of equipment is being secured overnight due to repairs, and no employees are working on the equipment, then employees may use a Company Lock until work on the equipment begins again.

Do not give out keys and combinations for personal locks to any employee, except the employee to whom the lock belongs. Personal locks will have a distinct color from Company locks.

Lockout/Tagout Devices

All Lockout/Tagout devices must be:

- Capable of withstanding the environment to which they are exposed for the maximum period that exposure is expected.
- Standardized within the facility in at least one of the following criteria: Color; shape; or size; and additionally, in the case of tagout devices, print and format shall be standardized.
- Substantial enough to prevent removal without the use of excessive force or unusual techniques, such as with the use of bolt cutters or other metal cutting tools.
- Clearly able to identify the employee applying the device.

De-energizing Procedure

ALL EMPLOYEES MUST COMPLY WITH THE FOLLOWING PROCEDURE BEFORE WORKING ON ENERGIZED MECHANICAL OR ELECTRICAL EQUIPMENT. ANY EMPLOYEE OBSERVED NOT COMPLYING WITH THIS PROCEDURE IS SUBJECT TO DISCIPLINE, INCLUDING POSSIBLE TERMINATION. IF AN EMPLOYEE FINDS IT MORE HAZARDOUS TO WORK UNDER SUCH CONDITIONS, THEY MUST CONTACT A SUPERVISOR BEFORE

PROCEEDING. It is also important to continually LOTOTO whenever trying out machinery after trying any resolution to a problem.

It is important to remember that many types of equipment are locked out differently or may have different configurations within the same machine room or hoistway. It is important to conduct a Job Hazard Analysis to ensure that all energy sources are identified so they can be properly isolated.

1. De-Energizing Equipment

De-energize all live parts prior to performing work, unless it is not possible or feasible to do so (no illumination, troubleshooting, etc.).

1. **Identify** machinery/equipment that needs to be locked out.
2. **Alert** any personnel, trades, customers, etc. that the equipment will be shut down.
3. **Shut off** the machinery/equipment. Make sure all moving parts have come to a complete stop or safe state. (Do not use control circuit devices, switches, or interlocks as a substitute for de-energizing equipment.)
4. **De-energize** all energy sources. Energy-isolating devices must be used on the main power circuit, not on a control power circuit.
5. Always remember to **release stored energy**, if stored energy could endanger employees (i.e., hydraulic equipment).
6. **Apply personal lock / tag.** Apply a personal lock to the energy-isolating device for each hazardous energy source and keep the keys to the lock in your possession. These locks ensure personal lockout protection. Since no other worker has a key for that lock, the lock can't be removed inadvertently. Tags shall contain a statement prohibiting unauthorized operation and removal of the tag and lock. The tag must also list the name of the person responsible for the tag and lock. Each worker must apply a personal lock (or a tag if the equipment does not allow for a personal lock).
7. **Try Out.** Verify the effectiveness of the de-energization and lockout process after each energy-isolating device is locked out or after a group of nearby devices is locked out. Before testing, ensure that all other workers are clear of the hazards and that no hazard will be created if the process isn't effective. The most common way to test the effectiveness is to attempt to start the equipment and confirm it does not start. Use a multi-meter or other test equipment on the electrical circuit to assure no unrelated back-feed or inadvertently induced voltage exists.

Re-Energizing Equipment

ALL EMPLOYEES MUST COMPLY WITH THE FOLLOWING PROCEDURE BEFORE RE-ENERGIZING MECHANICAL OR ELECTRICAL EQUIPMENT. ANY EMPLOYEE OBSERVED NOT COMPLYING WITH THIS PROCEDURE IS SUBJECT TO DISCIPLINE, INCLUDING POSSIBLE TERMINATION. IF AN EMPLOYEE FINDS IT MORE HAZARDOUS TO WORK

UNDER SUCH CONDITIONS, THEY MUST CONTACT A SUPERVISOR BEFORE PROCEEDING.

Before re-energizing, it is important to note that a lot of conditions may have changed to the equipment, especially when on an active jobsite and the equipment has been off for an extended period of time. It is important to physically inspect all areas where the equipment operation will affect to ensure safety and avoid damage.

1. Re-energizing Equipment

1. Check the machine or equipment and the immediate area around the machine to ensure that nonessential items have been removed. Ensure all guarding and covers are replaced.
2. Confirm that all employees or other trades have been safely positioned or removed from the area. When working in teams, elevator personnel should echo preparedness for the equipment coming back online.
3. Verify that the controls are in neutral, set to off, or in a ready state, if applicable.
4. Remove the lockout devices and energy isolating devices and re-energize the machine or equipment.
5. Notify affected employees, customers, or trades that the servicing or maintenance is completed, and the machine or equipment is ready for use.

Lockout/Tagout Procedures for Escalators and Moving Walks

The applicable procedures listed above apply to all employees who work on escalators and moving walks. Anytime work is to be performed within the interior plane of the steps / pallets, it shall be locked out / tagged out to prevent the unit from starting unexpectedly.

- In addition to lockout/tagout, whenever working within the truss where 10% or more of the escalator steps are removed, a mechanical blocking device shall be activated to prevent the escalator from moving. If the unit is not equipped with a mechanical blocking device, the drive chain and/or step axles must be secured to the truss braces to prevent movement in either direction.
- Whenever the steps are removed and the unit is going to be left out-of-service, the steps/pallets should be moved to cover the openings (whenever possible).

MRL Car Movement Locking Devices

Some machine-room-less (MRL) elevators are equipped with special car movement locking devices that provide an independent method of securing the car. The cartop can then be safely used for performing maintenance or inspection of overhead equipment.

- Elevator suspension must be in place.

- The device is typically located on the crosshead.
- It shall have a sign stating “WARNING! Engage before maintaining or inspecting brake, emergency brake, or controller”.

Some MRLs with controllers in the hoistway have multiple lockout locations. Prior to performing lockout/tagout, determine the best location to de-energize the equipment that will be worked on. Refer to the manufacturer’s documentation for further information.

Supervisory / Emergency Removal of Lockout/Tagout

If it is necessary to operate a piece of equipment which is locked out, every effort must be made to locate the employee whose lock is on the equipment. If the employee cannot be located, and after positive assurance is made that no one is working on the locked-out equipment, and that the equipment is safe, the supervisor/manager may personally remove the lock. The supervisor/manager must remember that there is a danger of the employee returning, thinking the machine is still locked out. The supervisor/manager must either ensure the equipment is once again locked out before the employee resumes work or make contact with the employee and make them aware of the situation and issue the employee a new lock.

Whenever there is need for a lock to be cut by a supervisor, that supervisor is required to complete the Supervisory Lock Removal form (Appendix H) and send to the Safety Manager.

Authorized Employees: If a machine is locked out and it becomes necessary to leave, recheck upon returning to make sure the machine is still locked out. While supervisors will make every effort to avoid removing locks, there may be situations when it must be done. The recheck is for your protection.

Group Lockout Procedure

For jobs where more than one individual is required to lockout or tagout equipment, each worker shall place his/her own personal lockout or tagout device on the energy isolating device(s). When an energy isolating device cannot accept multiple locks or tags, a multiple lockout or tagout device (hasp) may be used. Alternatively, a single lock may be used to lockout the machine or equipment with the key being placed in a lockout box which allows the use of multiple locks to secure the box or cabinet. As each person no longer needs to maintain his/her lockout protection, that person will remove their lock from the box.

Shift Change / Personnel Lock Transfer

If a job is not complete at the end of a work shift, there must be continuity of the lock-out. In most cases it will not be safe or possible to re-test the effectiveness of the lockout. Workers coming on shift must place their personal locks on all the lock-out points before the workers going off shift remove their locks. Alternatively, the Supervisor may lock the lock-out points

before workers going off shift remove their locks to ensure continuity of the lock-out until workers coming on shift can apply their personal locks. One other alternative is for the workers going off shift to remove their personal locks and the lead Mechanic or Mechanic in Charge applies the Company lock. This must be communicated to those coming on shift that personal locks must be applied before the company lock comes off and that only the oncoming lead or Mechanic in Charge is to remove the company lock once all personal locks are applied.

Inspecting, Troubleshooting or Observation

When it is necessary to enter an elevator pit for inspection, troubleshooting or observation and the movement of the elevator is required, you shall tag the controller OUT OF SERVICE by placing a sign on the controller stating that the "ELEVATOR IS UNDER CONTROL OF SERVICE PERSON - DO NOT OPERATE". Use the pit stop switch to eliminate any existing hazards. Ensure the pit stop switch is tested before entering the pit. Any and all other pit entry needed with the car running must have a Job Hazard Analysis completed in accordance with pit access policies and procedures.

Procedures for Outside Personnel / Contractors

Outside personnel/contractors shall be advised that the company has and enforces the use of lockout/tagout/tryout procedures. They will be informed of the use of locks and tags and notified about the prohibition of attempts to restart or re-energize machines or equipment that are locked out / tagged out.

The company will obtain information from the outside personnel/contractor about their lockout/tagout/tryout procedures and inform affected employees of their procedures. The contract employer must advise the host employer of any unique hazards presented by the work, unanticipated hazards, and any measures taken to correct hazards reported to them.

The outside personnel/contractor will be required to sign a certification form (see Appendix F). If outside personnel/contractor has previously signed a certification that is on file, additional signed certification is not necessary.

Training

Lockout/Tagout/Tryout training shall be provided upon hire/assignment and annually thereafter. Training resources and forms are available from the Safety Manager or in the LOTOTO Program Binder. All instructor-led training must be documented and contain the employee's name, date of training, training materials used, and trainer's name. Training may be carried out via the Learning Management System as well.

Employees shall be selected and trained as either "Manager", "Authorized", "Affected", or "Other" employees.

1. **Managers:** Includes upper management and Superintendents/Supervisors. They shall be trained initially and annually in the elements and requirements of an effective LOTOTO program. Managers will complete the Control of Hazardous Energy (Lockout/Tagout/Tryout) for Management course.
2. **Authorized Employees:** Shall be trained initially and at least annually in the requirements of identifying hazardous energy sources and following proper LOTOTO procedures. Authorized employees will complete the Control of Hazardous Energy (Lockout/Tagout/Tryout) for Authorized Employees course.

Hands-On Demonstration: All authorized employees must also demonstrate their understanding of LOTOTO procedures by successfully performing LOTOTO procedures on hydraulic equipment under the supervision of an experienced authorized employee before the new or transferred employee is allowed to conduct LOTOTO procedures independently. Observe the employee perform hydraulic LOTOTO procedures and evaluate performance and enhance training as needed to correct deficiencies. The hands-on demonstration form is in Appendix A. In addition to initial and annual training, authorized employees should be retrained when:

- Observations and inspections reveal deficiencies.
- There is a change in job assignments.
- There is a change in machines, equipment or process that presents a new hazard.
- There is a change in LOTOTO procedures.

Authorized Employees must complete training (including Hands-On Demo) before they can start using LOTOTO procedures. The cut-off mark for this requirement is 90 days. For instance, if an employee were to leave the company on January 1st, but rejoined the company on February 28th, the Hands-On demo will not need to be repeated. However, if the employee were to rejoin the company on May 1st, the training and demo would need to be repeated.

Lockout/Tagout/Tryout Program Acknowledgement Form: All Authorized employees must sign the "LOTOTO Program Acknowledgement Form" upon successful completion of their initial training and demonstration, and prior to being assigned LOTOTO work. The form is in Appendix B.

Signed acknowledge forms should be completed within Dayforce.

1. **Affected Employees** are those who operate equipment and are affected by changes to that machinery during LOTOTO activities. Other Employees are those who work in an area where LOTOTO procedures may be utilized. Train (initial and annual) all Affected and Other employees in the following:
 - Understand the potential hazardous energy sources of equipment they are working on.
 - Why and when LOTOTO is needed to protect personnel performing certain tasks.

- How to identify locks and tags used in LOTOTO; never attempt to restart or reenergize equipment that is locked or tagged out.
- Safety personnel or their representatives will use the Lockout/Tagout Awareness Training document (Appendix C) to conduct this training. A copy of the Lockout/Tagout Awareness training is available in the LOTOTO binder and on Dayforce.

Annual Review

The LOTOTO program shall be reviewed annually to ensure program policies and procedures are being followed. Management shall ensure that an annual review of the program is being performed. The review must include an evaluation of:

- Energy Control Procedures
- Changes in equipment or processes
- Training activities
- Documentation practices, and
- Incident Investigations

Critical elements of the annual review include getting feedback from employees observing Energy Control Procedures. If needed, the program should be updated by revising LOTOTO procedures, improving the content of training sessions, and modifying recordkeeping practices.

The annual review must be certified using the “Annual Review” form. Remember that this document may be requested as part of a LOTOTO audit, either by an OSHA inspector or internal company auditor.

Documentation

Documentation is essential for tracking and verifying that our LOTOTO program meets Axxiom Elevator and regulatory requirements. Records and documents are also very useful during compliance audits, whether they are performed by management or OSHA inspectors. All items listed below must be recorded and kept in accordance with this plan. Records kept relating to this plan shall include:

Document

Length of Time to Keep

Where to Keep

LOTOTO Policy

Retain until documents are superseded

Dayforce and LOTOTO binder

Training Records

Retain for 3 years

Dayforce

Annual Review Forms

Retain for 3 years

Dayforce and LOTOTO binder

Observation Forms

Retain for 3 years

Dayforce

LOTOTO Acknowledgement Forms

Retain for 3 years

Dayforce

The LOTOTO binder shall be always stored locally in an area accessible to all employees.

Revision Tracking Table

Date

Revision Description

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Jumper Safety

(Reference: Policy AXX-0019)

This program is instituted by Axxiom Elevator to prevent accidents caused by the incorrect installation or removal of jumpers. Jumpers are temporary electrical connections used to bypass certain elevator safety circuits during installation, modernization, maintenance, or repair work.

Scope

This Jumper Policy covers how the Company will train on jumper use, documentation of jumper use, inspection of jumpers, verification of safety circuits, and rules and procedures for jumper use.

Since Axxiom Elevator routinely operates jobsites which may expose employees to jumper use, or circuits that have jumpers applied, compliance with the directives of this Jumper Policy is mandatory.

Responsibilities

It is the responsibility of the **Safety Manager** to administer the Jumper program. A coordinator may also be assigned to the program. It is their responsibility to ensure Department Managers and Superintendents are trained and ultimately that all authorized employees are trained.

Responsibility for various aspects of the Program may be delegated to management and supervisory personnel best suited to the task. However, the Safety Director must monitor and ensure compliance.

Each department **Supervisor/Manager/Superintendent** will be responsible for ensuring their teams and individual field personnel are trained and have the necessary resources concerning the proper use of jumpers for defeating safety circuits as well as employee understanding what effect using or removing a jumper will have.

The **Foreman, Mechanic in Charge (MIC), or Mechanic** onsite will be responsible for determining the need for jumper use or removal as opposed to full Lock Out Tag Out Try Out (LOTOTO) for diagnosing problems.

Employees are responsible for obeying safety rules, following proper procedures, using protective equipment as required and notifying their supervisor of any unsafe conditions they may observe. The active role of the employees and their responsibilities in the Program should be continuously and positively reinforced since their contribution to overall safety and the success of the Jumper Program is paramount.

Jumper Program Components

The Jumper Program consists of:

- Management Briefing – All Managers and Supervisors should be familiar with this guide
- Employee Training – Make sure new employees are trained in jumper use
- Control Measures – Implement various control measures as necessary (standard company issued jumper kits, jumper logs, numbering, etc.) Jumpers are to be numbered with employee number, checked on an annual basis for correct number, and check for wear and tear during the annual Safety Bag check.
- Annual Review – Using the Annual Review Form, confirm program elements are in place

Jumper Use Overview

Jumpers are temporary electrical connections used in the elevator industry to bypass certain elevator safety circuits during installation, modernization, maintenance, or repair work. They are typically used when the elevator needs to be taken out of service or when certain safety features need to be temporarily disabled for maintenance or repair purposes. For new installation jobs, jumpers may come pre-wired in the various elevator packages for installation use and diagnostics.

Jumper Hazards

Jumpers can pose serious safety hazards if they are not installed or removed correctly. They can create electrical shock or fire hazards, and they can also cause elevator systems to malfunction or fail, which can result in accidents and injuries to both elevator personnel and the general public.

- To ensure the safe use of jumpers in the elevator industry, it is important to follow all manufacturer instructions and safety procedures when installing, removing, and storing jumpers.
- Proper jumper training is critical to ensure that elevator personnel understand the risks associated with jumpers and how to use them safely.
- Use appropriate personal protective equipment (PPE) such as insulated gloves and safety glasses which can help prevent electrical shock and other hazards.
- Ensure that all lockout/tagout/tryout procedures are in place before installing or removing jumpers. This will help prevent energization of circuits and keep personnel safe.
- Inspect jumpers before use.
- Properly store jumpers: Jumpers should be stored in a designated location that is easily accessible and clearly labeled to prevent confusion and mistakes during installation.
- Insulate jumper connectors: Use insulation sleeves or covers on jumper connectors to prevent accidental contact with energized electrical components.
- Reconnect safety circuits: Always verify that all safety circuits are properly reconnected and functioning correctly after installing or removing jumpers.

By following appropriate safety procedures and best practices, elevator personnel can help prevent accidents and injuries caused by incorrect jumper use.

General Jumper Use Rules

1. When jumpers are used, they shall have the following characteristics:
2. Jumpers shall be extra-long, tied in knots, with brightly colored wires and/or clips.
3. Jumpers shall be numbered in sequence.
4. Each employee shall have his/her name marked in indelible ink on a label permanently attached to each jumper.

5. Make sure you understand what effect using or removing a jumper will have on the entire elevator or escalator system prior to use.
6. Jumper wires shall never be placed or configured to resemble permanent wiring.
7. The number of jumpers carried shall be limited so that all jumpers can be accounted for at all times (this is why numbering in sequence will help). Jumpers shall be removed and accounted for when returning equipment to service.
8. Equipment shall never be returned to service with jumpers left on. Equipment found in this condition shall be reported to your Superintendent/Manager immediately.
9. Jumpers shall not be used as a diagnostic tool. Always use a meter to troubleshoot circuits.
10. Temporary bridging (i.e., tomahawk) devices shall not be used to short out hall door contacts.
11. Do not jump out a door and gate at the same time. NOTE: You may deviate from this requirement only when a second qualified person is on site and in direct communication.
USE EXTREME CAUTION WHEN THESE CIRCUITS ARE JUMPED OUT.
12. Ensure the elevator is on inspection before placing jumpers on door, gate, or safety circuits. Perform a visual inspection and verify that all hoistway doors are mechanically closed.
13. Ensure all jumpers are removed before placing equipment back in service.

Procedures for Jumper Use

1. Remove elevator(s) from public use and ensure no passengers are in the elevator.
2. Place jumper kit on the machine room door handle, controller door, or other conspicuous place.
3. Verbally communicate to all other elevator personnel on the jobsite when jumper(s) are to be used. This communication provides needed information on circuits bypassed and equipment affected.
4. When the work is complete, all jumpers shall be removed, counted and returned to the jumper kit.
5. Never leave jumpers on equipment or in the machine room.

Safety circuits, pit stop switches should never be left jumped out. In any instance where an elevator is found to have circuits jumped out, or there is a temporary need to leave a site with circuits jumped out, supervisor approval is required, and the unit must be LOTOTO and noted on time ticket.

Modernization and New Construction

On modernization and new construction jobs, many jumpers are used to operate elevators.

- Jumpers shall be brightly colored, easily identifiable and shall be long and conspicuous.
- A Jumper Log (Appendix A) shall be established to ensure that affected personnel are aware of the safety circuits which are not functioning. It is the responsibility of the

mechanic / mechanic in charge to ensure that the Jumper Log is kept with the controller and shall not be removed until all jumpers are removed.

- Door, gate, or other safety circuits shall never be jumped out unless the car is on inspection in the controller.
- Jumpers shall be removed as soon as they are no longer needed, and maintenance and troubleshooting procedures outlined above shall be followed.
- Lead Mechanic or MIC on New Construction and Modernization projects shall return all jumpers to the office and note on time ticket that jumpers were delivered.
- 2 sets of mechanics on 1 job/diff time...how do we acct for all jumpers

In addition to the above, stickers shall be placed in the controller cabinet, in close proximity to the inspection/automatic switches.

Before moving the car with open doors, verify the car is on inspection.

Door Bypass Procedure

1. Inspection operation with open door circuits (ASME 17.1, Requirement 2.26.1.5) shall only be utilized when it is necessary to move an elevator when the car gate and/or door lock circuit is electrically open. Before utilizing this, steps shall be taken to identify the problem circuit (car gate or door lock) and bypass only that circuit. Once a decision has been made to utilize door bypass circuitry, a Job Hazard Analysis must be performed, and barricades installed. Then the elevator shall be placed on inspection operation via the car top or the in-car inspection switches. Once in inspection operation, the pertinent switch(es) can be switched to “bypass” position. Then, whenever possible, the elevator shall always be operated from the top of the car.
2. Extreme caution shall be taken prior to and while moving the elevator to ensure the safety of the public and elevator personnel, i.e., unprotected openings, body parts clear of moving equipment, equipment unobstructed, etc.
3. Once the purpose for using the door bypass circuitry has been met, the switch(es) are to be switched to the “open” position. A thorough check of the car gate/door lock circuitry shall then be performed to ensure proper operation. Once this is confirmed, the elevator shall be placed back in automatic operation and returned to service.

Inspecting Jumpers

Inspecting jumpers each time prior to use is an important maintenance task that helps ensure safe and reliable operation. Here are the general steps to inspect jumpers:

1. Visually inspect jumpers: Visually inspect the jumpers for any signs of damage or wear. Look for signs of corrosion, cracking, or fraying, and check that the connectors are properly attached and secure.

2. **Check insulation:** Check the insulation around the jumpers for any signs of damage or wear. Look for cracks or holes in the insulation and verify that the insulation is not frayed or exposed.
3. **Inspect connectors:** Inspect the connectors on the jumpers for any signs of damage or wear. Look for corrosion damage to the connectors and ensure that they are properly attached and secure.

If after inspection, any issues are found with any of the jumpers, do not use the jumpers. Contact your supervisor to be issued a new kit.

Testing of Jumpers

Testing jumpers involves verifying that the circuits are properly connected and functioning properly. Here are the general steps to test jumpers:

1. **Review safety procedures:** Before testing jumpers, authorized employees should review all appropriate safety procedures and guidelines to ensure the testing is performed safely.
2. **De-energize circuits:** Before testing jumpers, ensure that all relevant circuits are de-energized to prevent electrical shock or fire hazards.
3. **Inspect jumpers:** Inspect the jumpers for any signs of damage or wear that could cause circuit failures or other issues.
4. **Perform continuity test:** Use a continuity tester or multimeter to test the jumpers for continuity. This will verify that the circuits are properly connected and that there are no open circuits or breaks in the jumper wire.
5. **Test resistance:** Use a multimeter to test the resistance of the jumper circuit. This will verify that the circuit is functioning correctly and that there are no shorts or other issues that could cause problems.
6. **Verify circuits:** Verify that all circuits are properly connected and functioning correctly before and after jumper testing.
7. **Document testing:** Document the results of testing in the jumper log, including the date and time of testing, the location of the jumper, the elevator system and components involved, and the results of the test.

Jumper Log

A jumper log (Appendix A) is a record that documents the installation, removal, and testing of jumpers. For multi-car jobs, a Jumper Log shall be attached to each controller. Here is an example of the jumper log process:

1. **Record installation:** When jumpers are installed, the installation should be recorded in the jumper log. The log should include the date and time of installation, the location of

the jumper, the elevator system and components involved, and the name of the technician who performed the installation.

2. **Record removal:** When jumpers are removed, the removal should be recorded in the jumper log. The log should include the date and time of removal, the location of the jumper, the elevator system and components involved, and the name of the technician who performed the removal.
3. **Test jumpers:** Jumpers should be tested regularly to ensure that they are functioning correctly. The testing should be recorded in the jumper log, including the date and time of testing, the location of the jumper, the elevator system and components involved, and the results of the test.
4. **Review logs:** The jumper log should be reviewed regularly to ensure that all installations, removals, and testing are properly recorded.
5. **Verify safety:** Personnel should verify that all safety circuits and components are properly connected and functioning before and after jumper installation or removal.
6. **Maintain records:** The jumper log should be kept up-to-date and maintained in a secure location as part of the job file.

Recordkeeping

Training records are required to be kept for three years. Any other prevalent records regarding the Jumper Program should also be kept for three years.

Training

The Department Manager is responsible for ensuring that all employees within their department have completed their Jumper training, although the actual training itself may be carried out by others.

After an employee has completed the training program, the instructor will determine whether the employee can safely perform the job. All employees who receive training must be documented with either a paper roster or tracked within the Safety Management System. Each training roster must include the name(s) of the employee(s) trained, the date(s) of the training, and the instructor's name who administered the training.

Training records must be maintained for three years.

Revision Tracking Table

Date

Revision Description

Page # of Revision

Powered Industrial Trucks

(Reference: Policy AXX-0020)

The purpose of this document is to define minimum safety standards governing Powered Industrial Truck (PIT) operations for all Axxiom Elevator facilities and jobsites.

All Axxiom Elevator facilities will use this document and OSHA standard 29 CFR 1910.178 to comply with regulations and create a safe work environment for employees operating and working around powered industrial trucks.

Scope

This document applies to all Axxiom Elevator employees, contractors, visitors, and facilities and supersedes any previously published Company or facility documents. This policy also applies to third party transportation carriers delivering or transporting products to or from Axxiom Elevator facilities or jobsites.

Policy Statement

It is the policy of the Company to permit only trained and authorized personnel to operate Powered Industrial Trucks. This policy is applicable to daily operators, contractors, visitors, as well as those employees who may only occasionally use a powered industrial truck. Under no circumstance is anyone permitted to operate an industrial lift truck without completion of training and certification.

Roles, Responsibility and Authority

1. Axxiom Elevator Safety Responsibilities
 - Administer and coordinate the Powered Industrial Truck Safety Policy.
 - Monitor the program for compliance to Axxiom Elevator and regulatory requirements.

Regional Manager/ Branch Manger Responsibilities

- Ensures the Powered Industrial Truck Safety Policy is followed.
- Designates an individual to be the site Powered Industrial Truck Safety Policy Administrator.

Site Powered Industrial Truck Safety Policy Administrator Responsibilities

- Maintain and provide periodic training program for all employees.
- Monitoring employee re-training.
- Maintain employee training records.
- Ensure periodic and risky behavior refresher training is conducted.

Supervisors or Managers Responsibilities

- Know the hazards in their shops and areas.
- Assure all industrial truck operators are properly trained and certified.
- Assure all operators understand the safe operation of an industrial truck.
- Assure all lift trucks are inspected at the beginning of each shift.
- Conduct random reviews of Inspection paperwork to ascertain compliance.
- Assure Lift Truck operators follow and adhere to Company safety rules.
- Conduct routine behavioral observations to ensure safe operation of PIT.

PIT Trainer

- Ability to understand lift truck manuals, manufacturer's specifications, drawings, instrumentation, and parts lists.
- Demonstrated knowledge of any State-specific regulation/legislation applicable to the operation of powered industrial trucks that may exceed the minimum standards set forth in the policy and program.
- Demonstrated knowledge of workplace-specific hazards and how to train students to recognize and correct those hazards.
- Demonstrated knowledge of the purpose and function of all components, devices and accessories commonly used on lift trucks (such as man cages), and how to carry out an inspection to determine that all are functioning properly.
- A working knowledge of electrical, hydraulic, and mechanical systems to have sufficient background information to explain the elements of a powered industrial truck.
- A working knowledge of the various energy sources associated with P.I.T.s (such as industrial batteries, natural gas, propane, gasoline, diesel) and how to handle them including how to fuel, change, charge, store, handle and maintain them.
- Responsible for implementing annual maintenance on PIT equipment with qualified outside vendor(s) and maintaining all records for local PIT equipment.

Operator's Responsibilities

- Use the PIT in accordance with manufacturer's instructions.
- Follow the Company rules when operating an industrial lift truck.
- Understanding and adhering to the safe operation of a powered industrial lift truck.
- Operate the industrial lift truck in a safe manner to prevent accidents from occurring.

- Assure that the industrial lift truck he/she is about to use has been inspected during the shift for safe operation.
- Immediately place a forklift out of service if the forklift fails the pre-operational checklist.

Procedure

1. Operator Training Program

1. PIT Trainer Program

The PIT Trainer must be certified by completing an approved train the trainer program authorized by Axxiom Elevator Safety Management. A licensed operator or long-term operator is preferred. Trainers will go through all required training through the current Learning Management System (LMS) program Training Courses. Axxiom Elevator Safety Managers will manage the PIT Trainer program. Trainers will be observed periodically for effectiveness as required. Axxiom Elevator Safety Managers will have overall approval and maintain an active roster of certified PIT Trainers.

1. New Operator Training

Note:

All applicants and current operators shall be physically capable able under medical criteria to perform all duties of a competent operator. Questions, please bring to the attention of your Safety Manager.

Prior to operating any PIT outside a training environment, all new employees (including those who have previous PIT experience) will complete the requirements below. Any employee entering a new role or classification or planning to use PIT for any reason who has not previously been trained must also complete the requirements below in the following order:

1. Complete required operator classroom training for any equipment type which they will use in the execution of their responsibilities.
2. Review site layout and site-specific safety rules or regulations with a supervisor.
3. Pass a written test administered by PIT Trainer with a score of at least 80%.
4. Perform On the Job Training (OJT) until Supervisor is satisfied with the operator's performance.
5. Pass an operator skills test (hands on evaluation) administered by PIT Trainer.

Documentation of the operator's completion of this process will be maintained in the individual's personnel file for the duration of the individual's employment (refer to related documents section for current documents).

New hires (and all other employees who are newly trained on PIT) will be closely monitored during their first thirty days operating new equipment. During that time, operators will periodically meet with a PIT Trainer to review their performance and address any safety concerns.

1. Operator Re-Certification

Operator certifications are good for a three-year period. Re-certification of an operator will consist of the following steps in order as numbered:

1. Pass an operator skills test (hands on evaluation) administered by PIT Trainer.
2. Pass an inspection assessment (hands-on) on specific equipment with PIT Trainer.

1. Re-training

It is the Company's policy to provide re-training for all drivers to ensure that they review the required safe operating practices and are made aware of any changes to the Company's Policies and Procedures.

The Company also requires either complete re-training or specific area re-training when:

- The operator has been observed to operate the industrial lift truck in an unsafe manner.
- The operator has been involved in a preventable accident or a near miss.
- The operator has received an evaluation that reveals that the operator is not operating the industrial lift truck safely.
- The operator is assigned to operate a different type of industrial lift truck.
- Significant changes in the workplace occur which may affect the safe operation of the truck.

All re-training of PIT operators shall be documented and retained in the PIT operators' file.

Awareness Training

All employees who will be working around forklift operations must complete awareness training. Managers/supervisors shall use the "Working Safely Around Forklifts" Awareness Training (Appendix I) guidance document for this training activity; awareness training must be documented for affected employees.

Operation of PIT Equipment

1. Maintenance, Storage of Equipment, and Safety Prior to Beginning Operation

- All PIT equipment must have annual maintenance performed by a qualified 3rd party vendor where all aspects of the equipment are inspected for wear. This inspection must be documented with all repair recommendations. This documentation must be kept in a file in the local branch by the PIT Trainer or Warehouse Manager.
- Only drivers authorized by the employer and trained in the safe operation of PIT shall be permitted to operate specified PIT equipment. Operator is responsible for knowing all safety devices, functions, and operations of the equipment.

- Prior to operating any PIT, the operator shall complete a pre-shift inspection. Immediately place a PIT out of service if the pre-shift inspection identifies items that would prevent the safe operation of the vehicle, or if a problem arises during operation during shift. Pre-trip inspections will be maintained for one year at each PIT location.
- When unattended, PIT's will be turned off, forks lowered to the ground and the parking brake applied (engaged). Unattended is when an operator is 25 ft or more away from PIT or it is out of view of operator. Park in authorized areas. Do not block exits.
- The use of cellphones, ear buds, headphones, or non-work-related equipment is prohibited while operating a Powered Industrial Truck.
- Smoking is always prohibited in all Axxiom Elevator facilities; no smoking is allowed while operating any PIT or anywhere near PIT charging stations or where compressed gas tanks will be stored or exchanged.
- Changing PIT batteries may only be done by trained personnel. Complete a risk assessment to determine the need for PPE during a battery change. If needed, the following requirements govern safe exchange of PIT batteries:
- If using an approved battery changing equipment, all functions, operations, and safety features shall be in working order before use.
- PPE (face shield, rubber gloves) must be worn while changing batteries. (If the equipment is equipped with a preventative shield between the battery and operators' station, a face shield is not required.)
- Eye wash stations are required in area of battery filling or charging station(s)
- The facility must provide a spill kit.
- A conveyor or overhead hoist shall be used to change batteries.
- Reinstalled batteries must be properly positioned and secured in the truck.
- Tools and other metallic objects must be kept away from the tops of batteries.
- Exchanging the tank for a propane or gas-powered PIT should be done outside if possible. Operators must first turn off the PIT, then shut off the valve and disconnect the tank prior to removing the tank. Replacement tank should be installed in reverse order (position tank, connect tank, open valve, resume operation).
- Propane tanks should be stored outdoors in a dry, well-ventilated area. The tanks should be protected from direct sunlight, not in an area of excessive heat (120°F or higher) and away from ignition sources and other combustibles. Tanks are not to be stored upside down. Propane and flammable liquid tanks shall be separated by at least 20 feet.

1. General Rules for Operation of Equipment

- The PIT Speed limit will be set to a maximum of 7 mph or less on all equipment.
- Forklifts and other PIT equipment are to be used only as intended, PIT equipment is not to be used for hoisting and rigging purposes.
- Ensure the proper length forks are equipped as needed. Do not attempt to pick up unbalanced loads with forks that are too short.
- Horse play is strictly forbidden.
- No riders are allowed on the forks of any PIT at any time.
- Seatbelts must be equipped with all sit-down lifts and shall be worn according to the manufacturer's specifications while in operation. Tampering with a seatbelt in any way will result in disciplinary action.

- Horns shall be used for awareness. Horns shall be used at all intersections, in and out of trailers, high traffic areas and in the presence of pedestrians.
- The operator must slow down for wet and slippery floors.
- Only loads within the rated capacity of the truck can be handled. Lift capacity will be marked on all PIT's. The operator is responsible for ensuring that the load does not exceed rated weight limits.
- Operators must look in the direction of travel and monitor left and right for potential hazards (other trucks, pedestrians, and obstructions). Pedestrians have the right of way.
- When traveling up or down ramps and inclines, the operator must travel with the load facing uphill. Parking on an incline should be avoided; if unavoidable, PIT should be positioned with forks directly uphill, and wheels shall be chocked.
- Employees shall not place any part of their bodies outside the running lines of any industrial truck, loaded or empty.
- Other trucks moving in the same direction shall not be passed at intersections, blind spots, or dangerous locations. Parked equipment may be passed, if safe to complete.
- Operators must maintain a safe distance from other vehicles, at least two seconds or three lengths of the PIT.
- Trucks equipped with manufacturer approved attachments must be operated based on the modified lift capacity based on the attachment addition. The nameplate on the PIT must be updated with modified lift capacity.
- An overhead guard must be used as protection against falling objects. An overhead guard is intended to offer protection from the impact of small packages, boxes, bagged material, etc., but not to withstand the impact of a falling capacity load.
- All incidents, near misses and collisions shall be reported to the direct supervisor or manager on duty immediately.

1. Pedestrian Safety in the Vicinity of PIT Operation

- Pedestrians must stay approximately 2 ft from a lift while in operation. This will be known as the "zone of safety".
- While Pedestrians have the right of way, they shall ensure that the path they are taking is free and clear of all obstacles, including moving PIT equipment before proceeding.
- Employees shall not cross under, stand, or work beneath the elevated portion of any industrial truck, loaded or empty.
- Trucks shall not be driven up to anyone standing in front of a bench or other fixed object of such size that the person could be caught between the truck and object.
- Parked PIT shall not block fire extinguishers, doors/exits, or other areas which should be kept clear to allow flow of pedestrian traffic.

1. Picking Up and Carrying a Load

- Prior to travelling with a load, the operator must lift the load 1 to 2 inches to ensure the load is stable, wheels stay in firm contact with ground and steering is not impacted. Never raise or lower while in motion, a slight tilt back should be done to help with stability.
- Keep load/forks as low as possible when moving (forks no more than 6 inches above the floor).

- A load backrest must be used whenever necessary to minimize the possibility of the load or part of the load falling rearward.
- Operation will occur at a safe speed for the pallet being moved. This may be slower than the PIT top operating speed and must be determined by the operator.
- Only stable and or safely arranged loads can be moved. Cautions must be taken for off-centered loads that cannot be centered.
- The “stability triangle” is the imaginary triangle between the front two wheels of the PIT and the center of the rear wheels. When the center of gravity of the PIT or the truck/load combination falls within the stability triangle, the vehicle is stable and will not tip over. However, if the load and PIT fall outside of the stability triangle, the vehicle is unstable and may tip over. Operators must remain cognizant of the weight rating of the PIT and the weight of the load they are carrying.

1. Loading and Unloading a Trailer

- Operators must inspect the floor, walls, and ceiling of the trailer for safe operation prior to performing their assigned task.
- When loading/unloading trailers, dock plates must be used. Operators will ensure dock plates are in good condition prior to use.
- Mobile dock boards or mobile bridge plates must be properly secured before they are driven over.
- Dock boards or bridge plates must be driven over carefully and slowly at their rated capacity never exceeded. Bent, cracked or any other damage to a bridge pallet makes it inoperable.
- When operating on a liftgate, the PIT shall be pulled out of the trailer onto the liftgate with the steer tire first. The lift gate guard must be up and used whenever loading or unloading empty or loaded PIT. (Forklifts shall not operate on liftgates)
- Bridging, when necessary, will be allowed. A walkie pallet jack may cross an extended liftgate to a dock plate, but a forklift shall never cross an extended lift gate from a dock.

1. Racking, Stacking and Storage

- Pallets must be wrapped prior to racking or stacking. Extreme care must be used when tilting the load forward or backward, particularly when high tiering. An elevated load may not be tilted forward except when the load is in a deposit position over a rack or stack. When stacking or tiering, use only enough backwards tilt to stabilize the load.
- Empty pallets should not be stacked higher than 8 feet high.
- At no time shall material be stacked higher than 2 pallets.

1. PIT Operation on Jobsites

- Pallets must be wrapped prior to placing in racking or rafters. Take care not to stack pallets too high.
- Axxiom employees must comply with all Axxiom safety rules as well as customer/contractor safety rules.
- Axxiom employees must be certified through the Axxiom certification process outlined above, as well as any customer-specific certification or training mandates.
- Axxiom employees shall ensure they do not block fire lanes, handicap spots, safety egress routes or any other pedestrian traffic areas.

- Axxiom employees should avoid potholes, curbs and any other surfaces which may make PIT unstable or otherwise unsafe to operate. Use a guide if necessary.

Inspection, Repairs, and Modifications of PIT Equipment

- Operators are responsible for completing a pre-shift inspection prior to operating PIT.
- Documentation of pre-shift inspections must be maintained for one year after the inspection.
- Operators should not attempt to conduct maintenance on PIT equipment. Any operational or functional issues or safety concerns should be immediately identified to an authorized third-party vendor via the employee's supervisor.
- Modifications and additions which affect capacity and safe operation shall not be performed without prior written approval from the manufacturer and the Safety Department.
- All PIT equipment must have annual maintenance performed by a qualified 3rd party vendor where all aspects of the equipment are inspected for wear. This inspection must be documented with all repair recommendations. This documentation must be kept in a file in the local branch by the PIT Trainer or Warehouse Manager.
- Facilities must maintain a maintenance record for each piece of PIT equipment which includes documentation of maintenance, repairs, and inspections.

Aerial Lifts (including Scissor Lifts)

1. Overview

Aerial lifts are commonly used in construction, maintenance, inspection, and repair services to lift Axxiom employees to an elevated work position. Proper operation and use of aerial lifts can make completion of tasks at elevation safer and more efficient. However, unsafe use, operation and aerial lift work practices can result in serious injury. This program has been developed due to the hazards associated with improper use and Axxiom's concern for the safety of individuals in and around this type of equipment. In addition, this program outlines general, operating, maintenance, inspection and training requirements governing safe aerial lift use at Axxiom facilities and jobsites.

1. Aerial Lift Policy

All Axxiom employees must successfully complete a training program and receive certification prior to the operation of any aerial lift. Contractors operating aerial lifts on Axxiom projects are expected to meet or exceed the requirements described in this Plan and comply with all applicable statutes and regulations governing the use of powered industrial trucks as listed above.

1. Aerial Lift Scope

This program applies to the operation of all aerial lifts operated by Axxiom employees. Please see Appendix Q – Examples of Aerial Lifts for specific examples.

1. Maintenance

- All PIT equipment must have annual maintenance performed by a qualified 3rd party vendor where all aspects of the equipment are inspected for wear. This inspection must be documented with all repair recommendations. This documentation must be kept in a file in the local branch by the PIT Trainer or Warehouse Manager.
- Facilities must maintain a maintenance record for each piece of PIT equipment which includes documentation of maintenance, repairs, and inspections.

1. Pre-Use Inspection

- Prior to the operation of any aerial lift the Pre-Use Inspection Checklist found in Appendix Q must be completed. This applies at the beginning of every work period, and whenever a new equipment operator takes control of the aerial lift.
- Any safety defects (such as hydraulic fluid leaks; defective brakes, steering, lights, or horn; and/or missing fire extinguisher, lights, seat belt, or back-up alarm) must be reported for immediate repair. The lift must also be locked/tagged and taken out of service.

1. General Safe Work Practices

- Operators shall not wear any loose clothing or any accessory that can catch in moving parts.
- Before the machine is started, the operator must walk completely around the machine to ensure everyone, and everything is clear of the machine.
- Aerial lifts shall never be used as hoisting and rigging platforms to lift other materials.
- Articulating boom and extendable boom platforms, primarily designed as personnel carriers, shall have both platform (upper) and lower controls. Upper controls shall be in or beside the platform within easy reach of the operator. Lower controls shall provide for overriding the upper controls. Controls shall be plainly marked as to their function. Lower-level controls shall not be operated unless permission has been obtained from the employee in the lift, except in case of emergency.
- Modifications and additions that may affect the capacity or safe operation of an aerial/scissor lift are strictly prohibited without the manufacturer's written approval. Capacity, operation, and maintenance instruction markings will be changed as necessary if the manufacturer approves a modification.
- The insulated portion (if applicable) of an aerial / scissor lift shall not be altered in any manner that might reduce its insulating value.
- Any signs, plates, or decals which are missing or illegible must be replaced.
- If the aerial / scissor lift becomes disabled, an "out of service" tag or equivalent shall be attached to the controls inside the platform in a conspicuous location.
- Aerial/scissor lift devices with noted, reported deficiencies shall not be operated until repairs are made and equipment is authorized for use.
- Operators must report all accidents, regardless of fault and severity, to their supervisor.

1. Safe Work Practices Before Operation

- Consideration shall be given to the amount of wind. Follow the manufacturer's instruction regarding operation in windy conditions. As a rule, aerial lifts shall not be operated in winds exceeding 25mph although this can vary depending on the model of equipment.
- At 20mph wind speeds or anticipated gusts, lifts will be lowered to a maximum height of 20 feet.
- At 25mph wind speeds or anticipated gusts, lifts will be grounded.
- If at any time, personnel/staff feels unsafe in lifts, they may make decision to ground the lift and cease their task...no questions asked. •
- Guardrails must be installed, and access gates or openings must be closed before raising the platform.
- Boom and platform load limits specified by the manufacturer shall not be exceeded.
- Before moving an aerial lift for travel, the boom(s) shall be inspected to see that it is properly cradled, and outriggers are in stowed position (if equipped).
- Consideration shall be given to the protection of bystanders via barricading, having another employee keep bystanders at a safe distance or by other means.
- Aerial lifts shall not be operated from trucks, scaffolds, or similar equipment.

1. Safe Procedures During Operation

- Attention shall be given to the direction of travel, clearances above, below and on all sides.
- Employees shall not sit or climb on the guardrails of the aerial lift.
- Planks, ladders or other devices shall not be used on the work platform.
- An aerial lift shall not be moved when the boom is elevated in a working position with employees in the basket.
- Aerial lift shall not be placed against another object to steady the elevated platform.
- Aerial lift shall not be used as a crane or other lifting device.
- Aerial lift devices shall not be operated on grades, side slopes or ramps that exceed the manufacturer's recommendations.
- The brakes shall be set and outriggers, when used, shall be positioned on pads or a solid surface.
- The speed of aerial lift devices shall be limited according to the conditions of the ground surface, congestion, visibility, slope, location of personnel and other factors that may cause hazards to other nearby personnel.
- Stunt driving and horseplay shall not be permitted.
- Booms and elevated platform devices shall not be positioned in an attempt to jack the wheels off the ground.
- The area surrounding the elevated platform shall be cleared of personnel and equipment prior to lowering the elevated platform.
- All equipment must be secured on the inside of the aerial lift
- Operators are to call for assistance if the platform or any part of the machine becomes entangled.

1. Safe Work Practices After Operation

- Safe shutdown shall be achieved by utilizing a suitable parking area, placing the platform in the stowed position, placing controls in neutral, idling engine for gradual cooling, turning off electrical power, and taking the necessary steps to prevent unauthorized use.

- Aerial lifts shall be shut off prior to fueling. Fueling must be completed in well-ventilated areas free of flames, sparks or other hazards which may cause fires or explosions.
 1. **Changing or Charging Batteries**
- Battery charging installations must be located in areas designated for that purpose
- Facilities must provide for: flushing and neutralizing spilled electrolyte, fire protection, protection of charging apparatus from damage by trucks, adequate ventilation for dispersal of fumes from gassing batteries.
- Precautions must be taken to prevent open flames, sparks, or electric arcs in battery charging areas.
- Employees charging and changing batteries shall be authorized to do the work, trained in the proper handling, and required to wear protective clothing, including face shields, long sleeves, rubber boots, aprons, and gloves.

Related Documents

- Appendix A – PIT Operator Log (List of Authorized PIT Operators)
- Appendix B – PIT Inspection Forms (and examples)
- Appendix C – Aerial Lift Inspection Forms (and examples)
- Appendix D – Trainer Guide
- Appendix E – PIT Quiz (English and Spanish)
- Appendix F – Skills Evaluation Form
- Appendix G – PIT Operator Certification Cards
- Appendix H – “Working Safely Around Forklifts” Awareness Training
- Appendix I – Safe Operation of PITs
- Appendix J – Aerial Lift Quiz
- Appendix K – Aerial Lift Training
- Appendix L – Forklift Operator Training (English)
- Appendix M – Forklift Operator Training (Spanish)

Revision Tracking Table

Date

Revision Description

Page # of Revision

Escalator & Moving Walks Safety

(Reference: Policy AXX-0021)

The purpose of this document is to define minimum safety standards governing Escalator and Moving Walk service, repair, modernization and new installation operations for all Axxiom Elevator facilities and jobsites. These standards will define certain job hazards known on/around this type of equipment, as well as methods to counteract these hazards in order to prevent injuries and fatalities.

Scope

This document applies to all Axxiom Elevator employees, contractors, visitors, and facilities and supersedes any previously published Company or facility documents.

Policy Statement

Escalators and moving walks can pose as additional hazards typically because troubleshooting, disassembly, and reassembly of these units is not performed as often as doing the same on elevators for most technicians. This policy will serve to outline and counteract many of these known hazards, however there may be hazards that are unique to each piece of equipment and the location of that equipment. To identify all hazards, a Job Hazard Analysis (JHA) is required before starting any escalator or moving walk repair, modernization, or new installation job.

NOTE: Using an escalator as a stairway could cause injuries because there are several design differences between escalator steps and stairs. For example, the height of the step riser on an escalator is greater than typical stairs. Escalator steps are deeper (front to back) than normal stairs. These differences could result in a person catching their foot on the edge of the steps. In addition, the height of the step riser is not consistent throughout the length of the unit. Therefore, inform customers that it is unwise to allow personnel access to an escalator that is shut off. Unless overruled by a customer, in writing, barricade the escalator.

Before you Begin

- You shall familiarize yourself with the safety procedures for the particular piece of equipment pertaining to the job including safety features of the unit, proper Lockout/Tag Out/Tryout procedures (including how to safely access the controller and disconnects).
- Before work begins, the mechanic shall instruct all employees inexperienced with escalator or moving walk service on the location of the emergency STOP button(s) and safety switches.
- Technicians arriving at a location shall alert the owner or maintenance team and place company-approved signs and proper barricades at both ends of escalators and walks to restrict all non-authorized persons from entering the work area or stairways.
- Always establish and maintain control of the escalator/moving walk prior to entering and working in the machine room or pit areas.
- Always test emergency stop switches and operate the escalator using inspection controls, when they exist, or key controls.

Escalator / Moving Walk Barricades

- Barricades shall be positioned to completely surround the escalator/moving walk from public access.
- Barricades shall be positioned to surround, from public access, any floor opening created when equipment access plate(s) are removed.
- Barricades shall be a minimum of 42 inches high.
- Barricades shall be securely attached to the balustrades, handrails, and/or floor.
- All sections shall be connected.
- A system shall be in place to keep the barricade rigid.
- Before leaving the building, if the unit is to be returned to service, remove all 'Out of Service' signs and barricades. Check out with the building manager.

When Removing any Components of the Escalator / Moving Walk

- Always mechanically secure the escalator from movement by two independent means, prior to working in the truss.
- When step, step treads, pallets, comb plates, comb plate teeth, floor plates or trap doors have been removed, and power is not required, the equipment shall be locked out, tagged out and secured from movement with barricades at both ends.
- Always use barriers, warnings and redundant controls by eliminating power and securing mechanical energy whenever leaving the unit unattended with steps/pallets removed.
- Communicate with all employees before moving an escalator or moving walk. Repeat commands or direction signals. Make sure their actions show they understand your message. **CAUTION:** Due to the length of some escalators and moving walks, normal voice and visual communication may not be adequate. A clear means of communication must be established.
- Never start, or attempt to start, a unit while anyone is on, in, or about to step on the unit.
- Never ride the escalator or moving walk when the steps (pallets) are removed.
- Always control live electricity and rotating equipment when working within close proximity of the hazard. Barriers or other measures must be used to prevent inadvertent contact of the lower legs with returning steps/pallets.

Remaining in the pit with moving parts

Staying in the pit or machine room of an escalator/moving walk with unguarded moving parts is only allowed under the following conditions:

- The person in the pit or machine room must operate the unit in inspection control by using two hands positioned outside the pit or machine room.

- Jog circuits can only be used when guarding of moving and/or electrical parts are in place.
- The inspection box must be operated with two hands.
- The person must be protected from moving parts.

Removing Steps or Pallets

Fatal and serious incidents have resulted when employees stepped into openings due to missing steps or pallets. **NEVER RIDE THE UNIT** when any steps or pallets are removed. Riding is allowed if the unit is on an Inspection circuit, with only **ONE OPENING** and barriers are in place. The standard work must include clear modes of communication to ensure all personnel are aware of when and how the unit is to be removed. When walking on a partially disassembled unit, do not walk on the step axles as some axles rotate. Install a walking board and always hold the handrail. Always conduct a written Job Hazard Analysis to assess these hazards.

1. Procedure for Removal of Escalator Steps

The following is an example of two methods for the safe removal of steps from an escalator using the LOTOTO procedure. (Deviation from this description must be accompanied by a Job Hazard Analysis, approved by your supervisor).

Method 1:

1. Install and secure barricades at the top and bottom landings to restrict access by unauthorized persons.
2. Stop the escalator (ensure that no one is riding on it).
3. Open machine room access.
4. Remove the lower landing plate to gain access to the pit.
5. From a safe position outside the escalator, operate the escalator to position a step for insertion of the step lifter tool.
6. Turn off the pit switch and mainline disconnect, LOTOTO the mainline disconnect.
7. Install step lifters and disengage sleeves.
8. Remove step if crab casting is cut to allow same.
9. Reconnect mainline disconnect and pit switch.
10. (Alternate) Move step to removal position.
11. Turn off pit switch and mainline disconnect. LOTOTO the mainline disconnect.
12. Remove step and place in a safe storage area.

Method 2: Removal of the steps with more than one person

1. Install and secure barricades at the top and bottom landings to restrict access by unauthorized persons.
2. Stop the escalator (ensure that no one is riding on it).
3. Open machine room access.

4. Remove the lower landing plate to gain access to the pit.
5. From a safe position outside the escalator, operate the escalator to position a step for insertion of the step lifter tool.
6. Turn off the pit switch and mainline disconnect, position the second person at the main line to make sure it remains in the off position. If the JHA establishes that redundant circuits (step, E-stop, carriage circuits) are tested and verified and there are no electrical hazards, then LOTOTO may not be necessary if no electrical hazard exists.
7. The primary person installs the step lifters and disengages sleeves.
8. Remove step if crab casting is cut to allow same.
9. The primary person who is removing the step clearly signals that they are in a safe position for the restoration of power. The second person will now reconnect the mainline and the primary will restore the pit switch.
10. (Alternate) Move step to removal position.
11. Turn off pit switch and mainline disconnect, position the second person at the main line to make sure it remains in the off position. Use redundant tested circuits as an added precaution.
12. Remove step and place in a safe storage area.
13. If the escalator has an inspection or jog circuit feature where we have control of the motion and this feature is engaged, it will be acceptable to test and verify the stop switch and remove or replace a step. (If the escalator is not quipped with an inspection feature, LOTOTO or alternate means to control the equipment must be followed)

Lockout / Tagout / Tryout (LOTOTO)

- Whenever work must be performed within the interior plane of the steps/pallets (truss) or machinery space that does not require movement of the unit, lockout and tagout the mainline power disconnect in the 'OFF' position. See Lockout Tagout Tryout (LOTOTO) procedures under policy # AXX-0018.
- Before entering the unit or pit, personally verify that the unit will not start by testing and verifying all relied upon circuits and switches.
- If the configuration of the escalator or moving walk is such that there is a danger of entanglement from an adjacent unit, all units which pose a hazard will be locked out and tagged out and barricaded at both ends.
- If not equipped with an automatic lubricator, the escalator must be under control by means of an inspection or job circuit. If means of control is not available, the unit must be LOTOTO to lubricate.
- Once any steps have been removed and work is being performed within the truss, the unit must be electrically LOTOTO, and mechanically secured by two independent means. Typically, this can be achieved by adding an appropriate padded sling(s) around the step axle and a structural member of the truss in both directions in addition to the machine brake which is applied when the power is shut off. Test and verify movement.
- Be aware that the 110-volt power supply may still be live at the junction box, even after the mainline power has been disconnected. Use a circuit tester to test all circuits before working on them.

Removeable Controllers

Escalator controllers must be covered on all sides, including the bottom, and have feet that allow the controller to be placed upright when out of the machine room.

Ensure the power is off when removing and replacing the controller. Verify that the power has been removed by using a voltage sensor prior to removing the controller from the pit. Due to limited space in the machine room, it may not be practical to test and verify voltage by opening the controller. If conditions allow, the following procedure should be used:

1. Test the AC Voltage Sensor on a 110VAC outlet to ensure it is working properly.
2. Prior to moving the controller, use the AC Voltage Sensor to test the electrical cable entering the escalator main line disconnect to confirm it is live ("hot").
3. Standing (or kneeling) to the side, turn off the main line disconnect.
4. Apply your personal lock(s).
5. Using the AC Sensor, test the same electrical cable that is entering the main line disconnect to confirm it is de-energized.
6. Test the AC Voltage Sensor to a 110VAC outlet to confirm it is still working.
7. This will confirm the controller is de-energized and safe to remove from the machine room.
8. Depending on the model/manufacturer, a small hoist and stand may be needed to lift it out of the truss.
9. Care should be taken to make sure the controller remains in an upright position throughout the process.

Make sure the controller is always in its designated operational position (e.g. upright) when moved from its normal location (e.g., if positioned improperly, circuits may function due to gravity). Older controllers that are not equipped with covers on the rear and the bottom must be retrofitted. Contact your supervisor if this is the case.

Live Electrical Work

Whenever service must be performed on the unit, lockout and tag the mainline power disconnect in the 'OFF' position. Test and verify for power at the controller. Exceptions to this may include inspection, surveys, troubleshooting, etc. If power is required to be on, a Job Hazard Analysis is to be performed before progressing into the job.

Where power is required such as inspections, assess the hazards before you start. If the inspection cannot be performed without the risk of entrapment, then actions must be taken to eliminate the hazards, such as barriers or guards. If the hazards cannot be eliminated, then LOTOTO may be the only solution. Remember: Always mechanically secure the escalator from movement by two independent means prior to working in the truss.

While escalators or walks are being serviced, the holding or maintaining circuit must be opened so that the person in attendance has complete control of the equipment. This is normally done by a switch being turned to an 'INSPECT' position, which changes the start switch to a 'dead-man control' switch. To move the unit in the 'INSPECT' mode, the key switch will have to be held in the up or down position. If a unit is equipped with a hand-held pendant control, this should be used to service the unit. **CAUTION:** Always ensure that you are in a safe position outside of the escalator/moving walk before operating the unit. Beware of instant start units.

Temporary circuit jumpers or clips may be the only means available for conducting some service work on escalators or moving walks. Always use safe jumper procedures. Refer to Policy # AXX-0019 for proper procedures. Jumpers should only be used as a last resort.

Placing Equipment Back in Service

Before equipment is placed back in service, ensure that:

- All jumpers are to be removed and accounted for.
- All locks, tags, and slings have been removed.
- All safety circuits relevant to the work performed must be tested to ensure they are operating.
- Remove the barricades and all 'Out of Service' signage.
- Before leaving the building, observe the correct operation of the unit and checkout with the building owner or representative.

Construction and Modernization Safety for Escalators and Moving Walks

- During construction or modernization, the entire landing should be protected with proper barricades.
- When balustrade is not in place, lifelines must be suspended down the center or down each side for lanyard connection.
- Do not apply power until truss assembly, and all electrical parts mounted on the truss and all other cabinets and equipment are properly grounded.
- Properly support and secure equipment during the movement, storage, hoisting and installation of equipment.
- Inspect hoisting and rigging equipment for wear or damage before use. Use safe work procedures when hoisting escalator equipment. The rigging could slip, or break and loads could move uncontrollably. Stand clear of openings when hoisting or moving equipment. All personnel must be kept out of the danger zones in the event of an incident. Review and use the JHA process.
- Wear personal protective equipment (approved safety footwear, fall protection, eye, head and hand protection) while rigging, hoisting, moving, and installing equipment.

- Gloves and protective glasses must be worn when tightening clamps on glass supports or doing any work involving balustrade glass. The glass could shatter or break with explosive force.
- Use working and walking boards when traveling on the truss.

If Abnormal Situations Arise:

Notify your supervisor or another worker for assistance in assessing the hazard and forming appropriate work procedure. Perform a Job Hazard Analysis.

Due to the restricted space, plan your lifts, use lifting equipment, assist when necessary, and proceed with caution.

Many back injuries have occurred while working on escalators.

Revision Tracking Table

Date

Revision Description

Page # of Revision

Hoistway Safety

(Reference: Policy AXX-0022)

This Hoistway Safety policy is designed to ensure the safety of workers during maintenance and repair of machinery and equipment. The primary purposes of the Hoistway Safety policy are to advise on procedures for car top and pit access and egress, how to work safely in the hoistway, and the following below:

1 Providing General Precautions: These precautions are typical when carrying out work on car tops, pits and within hoistways.

2 Establishing Procedures: While all hoistways are not the same, this policy will serve to establish procedures for control of equipment when accessing and egressing the hoistway.

3 Ensuring Worker Safety: Providing clear procedures and guidelines to protect workers from the release of hazardous energy, or exposure to bodily harm during servicing, repair, modernization and construction activities.

4 Compliance with Regulations: Ensuring compliance with occupational safety and health regulations, such as those mandated by the Occupational Safety and Health Administration (OSHA) in the United States.

5 Training and Awareness: Educating employees about the importance of Access and Egress procedures and ensuring they understand how to properly implement these procedures to maintain a safe work environment.

Fundamental Rules

1. ALWAYS lockout, tagout, tryout (LOTOTO) when power is not required.
2. ALWAYS establish and maintain control of the elevator prior to accessing/egressing the car top.
3. ALWAYS use fall protection when a fall hazard exists.
4. NEVER ride the car top with the elevator in normal operation.
5. NEVER work above or below others when working in the hoistway.

Car Top Work General Precautions

Before attempting to gain access to the hoistway, it should be decided if the work being done will require electrical power to be on or if special positioning of the elevator will be necessary. If not, follow the approved lockout procedure.

Some general hazards and conditions to be considered when working on car tops are:

- Make sure the car top is clean. Make sure that there are no grease or oil spills and no loose objects. Make sure nothing is left when you leave the car top.
- Secure tools or anything else needed for the job on the car top. Make sure that they will be out of your standing area, and that they do not project over the edge of the car top.
- Be sure you have a secure surface to stand on that will support your weight. Don't stand on the car top emergency exit or fan unit and use special care when the car tops are curved or domed.
- Ensure even footing on the car top prior to moving the car with two-hand controls.
- Never hold onto 2:1 ropes. All car top 2:1 sheaves must be guarded.
- When you are on the car top, make sure that any protective equipment you are wearing does not become entangled or catch on any protrusions. Don't wear a tool belt and never carry tools in your back pockets. Remove cell phones, radios, or key rings from your belt before working from the car top and store in a safe, out of the way location (e.g., under the crosshead, etc.).
- Always keep all parts of your body within the limits of the car top of the elevator being inspected. Be especially careful if there is an adjacent elevator running alongside.

- Never hang an electric extension cord or light from the car or counterweight ropes. Severe damage may result if the hoist ropes are exposed to electric current flow caused by a short or ground.
- Always be aware of any overhead obstructions where overhead clearance is limited.
- Movement from car top to car top is prohibited (e.g., stepping from one car top to another).
- Ensure you have enough light on the car top for safe working (10-foot candles is the code minimum measured at the point of any elevator part or equipment where maintenance or inspection is to be performed from the car top A17.1-2022 2.14.7.1.4).
- Finally, always be alert and be aware of your surroundings and the conditions that exist where you are working.

Access Key Switch, where available, shall be the primary method to gain access to a hoistway or pit.

Car Top Access and Egress Procedures

The following two example procedures are the most common. These procedures assure your safety by addressing all fundamental safety requirements for the features of the subject elevators in the examples. If either of these examples cannot be followed, a Job Hazard Analysis must be completed. Car top access and egress shall always employ a standardized process that addresses:

- Complete control of hoistway doors during access and egress procedures.
- Test and verification of three circuits prior to accessing the car top (e.g., Hall door contacts, Top of Car Emergency Stop Switch, Top of Car Inspection Switch (TOCI)).
- Test and verification if the TOCI operating buttons following safe access. Run the car in the down direction first, then test operation in the up direction.
- Maintaining complete control of the car and hall doors when egressing from the car top.
- Protection of the public.

For the following example the following will be needed:

- Barricades and Out of Service signage
 - Access key and approved hoistway door unlocking device
 - Mechanical device for holding doors open (door wedge tool)
1. **Example #1: For Cars without Access Key Switched & without In-Car Inspection:**

Access:

(If needed, erect barricades and signage before starting work.)

1. Capture the car at the top landing and verify no passengers are present. Then place two in-car calls for the next floor and the lowest landing.

2. Unlock the hoistway door and open no more than 6" to verify the safe location of the car and that the open-door contact has stopped the car in flight.
3. Insert mechanical blocking device (door wedge) in the hoistway door sill.
4. Maintain your body in a safe and balanced position.
5. Engage the car-top stop switch (if an auxiliary stop switch is provided and going to be used, it must be tested independently and in sequence).
6. Close the hoistway door and wait ten seconds. Reopen the doors no more than 6" to verify that the car has not moved, then open the door fully.
7. Insert the door wedge tool in sill under the hoistway door.
8. Turn top of car (TOC) Inspection switch on, disengage the stop switch, remove the door wedge tool and close the doors and wait ten seconds to verify the Top of Car Inspection switch is working.
9. Reopen the hoistway door no more than 6" to verify the car has not moved.
10. Insert the door wedge tool in sill under the hoistway door.
11. Engage the car-top stop switch.
12. Turn on the car top light.
13. Establish a central position avoiding weak areas (i.e., emergency exits, fans, lights, etc.) and carefully step onto the car.
14. Remove door block (wedge) tool.
15. Maintaining a safe and balanced position, close the hoistway door.
16. Test the function of the down, up and common buttons. To prevent snagging hazards and entrapments in the overhead, always test operation in the down direction first.
17. Always engage and verify the stop button when the car is not moving.

Egress:

1. If possible, return to the same floor where you accessed the car (because the hoistway door you entered is already verified).
2. If exiting at a different floor, test that the hoistway door contact will stop the car by opening it while the car is on top-of car inspection. You could also stop the car, block the lock open and try to run the car on inspection.
3. Place the Top of Car Inspection (TOCI) Emergency Stop button in the Stop position.
4. At the exit floor, open the hoistway door, exit the car top and assume a safe and balanced position on the landing. Ensure your lanyard is disconnected before stepping on the landing.
5. Mechanically block the door in the open position.
6. Verify that all tools and materials have been removed from the car top and turn the car top light off.
7. Restore the Inspection switch to normal operation; place the stop switch to run. Remove the mechanical blocking device (door wedge) and close the hoistway doors.
8. Always verify that the car is operating properly before leaving the building.

1. Example #2: For Cars with Access Key Switch(es) and In-Car Inspection:

Access:

1. Capture the car and take the car to the access floor.
2. Turn the In-Car Inspection switch to Inspection.
3. Verify that this action holds the car door open by observing for 10 seconds.
4. Block the hoistway door mechanically.
5. Position yourself in a safe and balanced position on the landing and insert the key into the Access switch.
6. Run the car in the DOWN direction until the vane is clear of the hoistway door interlock rollers and remove the keys and close the car door. Continue running the car down to a safe access position. To ensure the car does not move unexpectedly, remove the key from the access switch.
7. Engage the car-top stop switch (if an auxiliary stop switch is provided and going to be used, it must be tested independently and in sequence).
8. TEST and VERIFY the Stop switch by activating the key in the 'up' and 'down' positions. Remove key from Access key switch. Do not attempt to verify the stop switch while on the down access limit.
9. Turn Top of Car Inspection (TOCI) switch to inspection and reset the stop switch to the run position. Test and verify the Top of Car Inspection (TOCI) switch by attempting to move the car with the Access key switch. Remove the key from the access key switch and re-engage the Stop switch.
10. Turn on the car top light, remove the door wedge tool, and carefully step onto the car top.
11. Establish a central position avoiding weak areas (i.e., emergency exit, fans, lights, etc.).
12. Verify that the Inspection/Normal switch is still turned to Inspection.
13. Maintaining a safe and balanced position, return the Stop switch to the 'run' position.
14. Test the function of the down, up and common buttons (always test in the down direction first).
15. Always engage and verify the stop button when the car is not moving.

Egress:

1. Return to the Access Key floor.
2. If it is absolutely necessary to exit from a different floor, test that the door contact will stop the car by opening it while running the car on inspection. (Alternatively, stop the car, block the lock open and try to run the car on inspection.
3. Place the Top of Car Inspection (TOCI) Emergency Stop button in the 'stop' position.
4. At the exit floor, open the hoistway door, exit the car top and assume a safe and balanced position on the landing.
5. Mechanically block the door in the open position (using door wedge).
6. Turn the car light off and make sure that all tools and materials have been removed and the car top is clean.
7. Restore the Inspection switch to 'normal' and the Stop switch to 'run'.
8. Remove mechanical door blocking devices and maintain control of hall doors.
9. Using the access key, return the car to the landing (using extreme caution when re-engaging the car door vane with the hall door rollers).
10. Return the In-Car Inspection switch to 'normal'.

11. Verify that the car is operating properly before leaving the building.

Working on the Car Top

Before attempting to gain access to the hoistway, it should be decided if the work being done will require electrical power to be on or if special positioning of the elevator will be necessary. If not, follow the approved lockout procedure.

1. Machine Room-Less (MRL) Car Movement Locking Devices

- Some MRLs with controllers in the hoistway have multiple lockout locations. Prior to performing LOTOTO, determine the best location to de-energize the equipment you will be working on. Refer to manufacturer's documentation for more information.
- Some MRL elevators are equipped with special car movement blocking devices that provide an additional method of securing the car. The blocking device can be engaged while the car top is level with the top landing as long as movement of the car is not needed. The blocking device shall be used for work performed at this level.
- **NOTE:** MRL Car Blocking device is designed to resist the force of the unbalanced load and is not capable of supporting the weight of the car without the aid of suspension means (i.e., belts and chain fall, etc.).

If power is required, use the following procedures:

1. Cars Without Top-of-Car Inspection

If the elevator is not equipped with TOCI, position the car for safe access to the work to be performed and LOTOTO. Riding the car top on normal operation is a violation of the Fundamental Rules. If for some reason a technician has to ride the car top, a written Job Hazard Analysis (JHA) must be completed, reviewed, and approved by the Supervisor *and* Safety Manager and posted on the car top. At a minimum, the JHA must include:

- A safe method to remove the car from public use and access the car top.
- 'Out of Service' signs posted to advise the public.
- The employee assigned to the car top shall be familiar with and understand the operation of the elevator.
- A clear system of communication shall be established between the two parties.
- A system shall be implemented so that the car will not respond to hall calls.
- A system shall be implemented to reduce the speed of the elevator to 150 FPM or less.
- A manually operated safety switch designed to stop the motion of the elevator in either direction shall be in reach of the person on the car top (e.g., SOS switch).
- A safe refuge space shall exist for the person who is on the car top.
- A safe means to exit the car top and return the elevator to service.

Working in the Hoistway

When work is being performed within 20” of adjacent moving equipment, precautions shall be taken, such as removing and LOTOTO the adjacent elevator from service, installing hoistway screening, or screening/barricading the car top work area.

Bottom terminal door work shall be performed from inside the car.

1. General Precautions

- To gain access to the hoistway via landing doors, situate the car at a safe distance. Use an appropriate unlocking device, access key switch, or other approved device to unlock the landing door.
- Before entering the hoistway, a safe means of access and egress to and from the hoistway shall be clearly established.
- Always secure the landing doors with a blocking device (door wedge). Always use proper barricades to protect the public.
- Simultaneous work shall not be conducted in the hoistway by Axxiom employees, or others, where individuals are stacked in the hoistway (i.e., above and below each other).

During inspections, surveys, or conducting any work, employees are not to walk on or touch hoistway spreader beams without having complete control of the adjacent elevator. Fall protection shall be used if there is any potential of being in a fall hazard.

CAUTION: No more than two people will be permitted to work in each individual hoistway at the same time.

Any exception will require a JHA prior to the work being performed and approval by the supervisor or Branch Manager.

MATERIAL HANDLING:

Various elevator components such as cab walls and ceilings may require more than two people to carry them into hoistways or onto car platforms. These brief material handling jobs require a JHA approved by your supervisor.

1. Transportation of Personnel or Non-Elevator Equipment

Transportation of non-Axxiom personnel on car platforms or inside car cabs during construction or modernization is not permitted. Requests for these activities must be made through the office and must have a JHA completed with the contractor (or customer) and must have a signed Indemnity Agreement in place prior to work.

Pit Safety

Fundamental Rules:

ALWAYS lockout, tagout and tryout equipment when power is not required.

ALWAYS establish and maintain control of the elevator prior to accessing/egressing the pit.

ACCESS KEY SWITCH, where available, shall be the primary method to gain access.

ALWAYS control rotating equipment when working within close proximity (5 ft.) of a hazard.

1. General Precautions

Many serious injuries occur each year while entering and exiting pits. Every employee must be aware of the hazards before entering a pit. Some of the more common hazards are:

- Inadequate refuge space
- Inadequate lighting
- Improper access
- Fall and tripping hazards
- Unsafe or lack of pit ladders
- Moisture/water/oil
- Moving equipment

Take appropriate steps to eliminate or minimize these hazards and any others that are identified.

Caution: When entering the pit, at least 20" must be maintained from the ladder rungs to the adjacent moving equipment. If less than 20" is present, protection must be installed to prevent accidental contact, such as screening, ridged barrier, or another fixed barrier.

Before entering a pit, familiarize yourself with the position of the car and counterweights for the car being worked on as well as any other cars/counterweights in the vicinity.

If notified by the building owner or representative that the pit and/or hoistway has been classified as a permit required confined space (this notification could be verbal or the pit/hoistway may be labeled), immediately notify your supervisor for further instructions. In either case, DO NOT enter the pit/hoistway until consulting your supervisor and receiving authorization.

The lighting level in the pit shall be adequate for the work being performed. If the lighting level is inadequate, supplement with a portable light. If the light is corded, a GFCI must be used.

Where the lighting switch is co-located in the pit with the emergency stop switch, these switches shall be clearly identified and separately positioned so that they may not be confused.

All lighting shall be protected against mechanical damage.

When entering a pit from the lowest landing, a ladder shall be used in all cases where the pit floor is greater than 3 ft. below the landing sill. Always use a safe means of access. If necessary, use a portable ladder. (Portable ladders must extend 3 ft. above the landing and 'A'

frame ladders must be used in their open configuration.) Never jump into a pit. While the elevator is locked out per our standard work procedures, sit on the landing and use both hands to guide your body into the pit.

Do not enter into a wet pit when there is any source of electrical power present. The water shall be removed, and the pit dried out prior to the inspection or performance of any work.

Remember to protect yourself and the public. If entry is through a hoistway door, block the door from being fully closed – this will ensure the door lock contact remains open. To prevent the public from opening the door, block the door so that it cannot be opened inadvertently.

1. Pipe Stands

Pipe stands shall be installed under the following conditions:

- Hydraulic cylinder leak
- Working on the hydraulic system
- Working on all residential elevator systems or LULAs
- Working in the pit of a single bottom cylinder
- When pit refuge space is 20" or less

NOTE: Whenever unexplained oil loss occurs on a hydraulic elevator, do not enter the pit. Contact your supervisor for further instructions.

1. Deep Pits

Fall protection means shall be used to access/egress the pit in the following cases:

- Access to the ladder exposes a person to a fall hazard 6 ft. or greater AND the ladder is farther than 29.5 in. from the interior edge of the door sill.
- Access exposes a person to a fall hazard and the ladder (or handholds) extends to less than 42 in/ above the access landing.

Fall protection can be mitigated by improving the height with additional handholds or relocating ladders to meet requirements. If installing handholds or relocating the ladder is not feasible, fall protection systems should be installed.

Working in the Pit

When work is being performed with 5 ft. of adjacent moving equipment in the pit in a common hoistway (i.e., multiple units in the same hoistway of shaft), steps shall be taken to ensure protection from adjacent moving equipment.

This protection can be accomplished by:

- LOTOTO adjacent running equipment

- Installation of adequate screening between elevators in a common hoistway/pit
- Guarding of rotating and moving equipment

Many serious injuries have occurred while entering and exiting elevator pits. All employees must be aware of the hazards before they enter a pit. Some of the most common hazards that need to be managed are inadequate lighting, improper access keys, unsafe pit ladders, height of toe guards, wet pits, integrity of door lock chain and safety chains, and lack of refuge space. Take steps to minimize these and other hazards. When accessing and egressing on pit ladders, use both hands and always maintain 3-points of contact.

Always establish and maintain control of the elevator prior to and during inspections, lubrication, minor cleaning tasks, retrieval of keys, etc. Entering the pit through the lowest landing door will only be permitted when two independent means of securing the elevator are available and confirmed. For example, the use of an emergency stop switch and opening of the door lock.

For tasks where power is not required LOTOTO the main line switch.

When in the pit, particular attention should be given to the position of the car and counterweight of both the car you are working on and any adjacent cars to it if there is additional equipment operated by other cars (e.g., counterweight, governor sheaves, etc.).

Inspection Operation with a Person in the Pit

For work that involves two people and where the pit is equipped with a pit inspection station, refer to Pit Inspection Station procedure for accessing and egressing the pit. This work requires a Job Hazard Analysis to be completed before work begins.

The following precautions shall be observed when it is necessary to operate an elevator with a person in the pit.

When a pit inspection station is not installed, ALWAYS IDENTIFY A REFUGE SPACE BEFORE ENTERING THE PIT.

Be aware of possible hazards in the pit (e.g., counterweights, sump holes, brackets, sheaves, standing water, traveling cables, comp chains, etc.). Take steps to ensure that no part of the body inadvertently projects into portions of any adjacent elevator hoistway area.

1. The elevator shall only be operated on inspection.
2. Have a qualified person familiar with the inspection operation of the elevator stationed on the car top.
3. A clear means of communication shall be developed and used between the two parties. This communication system should include these minimum precautions: The operator and those in attendance utilize the <Statement, Repeat, Repeat> method before moving the car.

4. The employee on the car top shall gain control of the car by following the car top access procedure.
5. Prior to pit entry, the person who is about to enter the pit shall test the operation of the pit entry door contact and the pit stop switch(es).
6. To prevent accidental starting of the car, each time the car has been moved, the emergency stop switches on the car top and in the pit shall be in the stop position until the process is repeated.
7. When the task is complete, the person who is in the pit shall communicate their intentions and exit the hoistway by following the pit exit procedure.

NOTE: To prevent a miscommunication, the car shall only be returned to Normal Operation when the other technician from the pit is observed outside of the hoistway by the technician on the car top. This simple extra step can prevent a fatal miscommunication.

Cleaning and Repair of Elevator Pits

1. General

Hospitals and other medical organizations are required to have documented plans and procedures for handling and disposing of medical or biological waste materials on their premises.

In any cleaning operation where the employee is required to clean up sharp items (there is a potential for cuts to occur), the procedures outlined below and in our Bloodborne Pathogens Program must be followed to minimize the chances of the employee receiving a cut and becoming infected. Special customer training may be required. Notify your supervisor of any occurrence.

If there are foreign objects or matter that has adhered to the pit walls, that may be moved or disturbed through the course of normal work in the pit, the owner must be contacted to clean down the walls before our work can begin. For assistance with this, contact your supervisor if necessary.

1. Procedure

1. If the elevator pit or equipment (or similar equipment) needing cleaning is seen to contain blood, biological materials, syringes, needles, etc., contact the administration of the hospital or other organization; inform them of the problem and request that they remove all medical waste from the pit before commencing or continuing any work.

Entry of other than qualified elevator people must be supervised, and safety procedures must be strictly adhered to (e.g., LOTOTO).

1. The bloodborne pathogen control procedure must meet the following:
2. Have adequate lighting.
3. Use a brush and dustpan.

4. Wear protective, heavy gloves.
5. Wear protective shoes or boots.
6. Wear a dust mask when appropriate.
7. Use proper disposal methods.
8. Any skin surface which comes in contact with blood or other substances should immediately be washed thoroughly with soap and water. The same process should be taken with any type of skin penetration by possibly infectious substances, except in addition to washing the employee should contact a supervisor immediately.

Walk-In Pits

Every walk-in pit is different. Therefore, it is difficult to make one set of procedures that apply to all situations. Rules and procedures must be established for each building site by using job hazard analyses with assistance from supervision. Written JHAs or job hazard stickers must be posted in the pit at an obvious visible location.

For brief inspections with the equipment operating, to retrieve dropped items (e.g., keys, money, jewelry, small tools, etc.) or to walk from one pit to another to make observations, the JHA will determine if it is safe to enter. This usually depends mostly on the height of the pit (7 ft. or more of overhead clearance) and the guarding or location of related components such as tapes, governors, counterweights, traveling cables, etc.

For repair work, the unit should be locked out and tagged out. For brief work activities such as adjustments or adding oil to buffers, a JHA may determine that two tested and verified circuits provide sufficient safe control.

When working on elevated buffer stands (more than 6 ft. off the pit floor), fall protection (proper guardrails or PPE) is required.

Always wear a hard hat in multiple unit walk-in pits.

Confined Space in Elevator Pits

OSHA's general industry standard, 29 CFR 1910.146, defines a 'permit required confined space' as a space that has one or more of the following characteristics:

- Contains or has the potential to contain a hazardous atmosphere.
- Contains a material that has the potential for engulfing the entrants.
- Has an internal configuration such that an entrant could be trapped, or asphyxiated by inwardly converging walls, or by a floor which slopes downward and tapers to a smaller cross-section; or
- Contains any other recognized serious safety or health hazard.

Through the IUEC, NEIEP, the larger elevator companies and others, assessments have been made by Environmental Health and Safety professionals to determine if Permit Required Confined Spaces exist and where necessary, steps have been taken to implement programs where needed. All employees who enter these spaces will be trained to do so. Never enter a Permit Required Confined Space unless you have been trained and an entry permit has been arranged.

On customer sites, elevator pits under normal circumstances contain the same atmosphere as the buildings they serve. As a result, under normal circumstances, elevator pits do not present hazards from toxic, oxygen deficient or flammable atmospheres. Any hazard that may be associated with mechanical equipment are controlled or eliminated by elevator company employees in compliance with applicable OSHA safety regulations, elevator code requirements, standard industry work practices and/or company policies and procedures (e.g., Elevator Industry Field Employees' Safety Handbook and/or company policies and procedures and 29 CFR 1910.147). Thus, elevator pits are generally not classified as Permit Required Confined Spaces.

Customers occasionally request that Axxiom Elevator comply with the customer's confined space entry procedures when performing maintenance work in elevator pits or occasionally pits will be designated (signage may indicate) Permit Required Confined Spaces.

If a customer makes this request or if signage indicates this condition, contact your supervisor to resolve the situation with the customer. Do not enter the space without proper training and authorization.

Safe Refuge Space in Pits

When working in pits that do not provide safe refuge space of 20 in. (height) or more, a blocking device capable of preventing incidental motion must be deployed. A safety sign visible from all access places must be displayed to communicate the risk and the support mechanism must be in place prior to entering the pit. A blocking device is required even if the unit is locked out and tagged out.

Pit Access/Egress Procedures

1. Hoistway Door Unlocking Procedure

ACCESS:

1. Set up proper barricades.
2. Place two car calls (for the floor above the access landing and the top landing).
3. Step out of the elevator and wait for the doors to close.
4. Open the hoistway door and observe that this action has stopped the car.
5. Mechanically block the hoistway door.

6. Maintaining a safe position, reach in and place the pit switch into the stop position.
7. Remove the block, close the hoistway door and wait 10 seconds.
8. Re-open the door, confirm the car has not moved and turn on the pit light switch. **NOTE:** If there is an additional pit E-stop that will be used as part of the work activity, it must also be tested independently from the primary E-stop.
9. Mechanically block the hoistway door and enter the pit.
10. Close hoistway door to within a distance that prevents closing of the door contact and prevents falls into the pit and re-block it.

EGRESS:

1. Open the hoistway door and mechanically block it in the fully open position.
2. Switch off the pit light if located in the pit.
3. Carefully exit the pit and remove all tools and equipment.
4. Maintaining a safe position, reach in and turn off the light and place the pit switch into the run position.
5. Remove the mechanical block and close the hoistway door.
6. Make sure the elevator operates properly before leaving the building.

1. Access Key Switch Procedure

ACCESS:

1. Set up proper barricades.
2. Place car on in-car inspection.
3. Mechanically block the hoistway door.
4. Move the elevator in the up direction.
5. Remove key from switch.
6. Maintaining a safe position, reach in and place the pit switch into the stop position.
7. Re-insert the key and try to move the elevator in both directions to verify the pit E-stop is functional. **NOTE:** If there is an additional pit E-stop that will be used as part of the work activity it must be also tested independently from the primary E-stop.
8. Access the pit.
9. Close the hoistway door to within a distance that prevents closing of the door contact and prevents falls into the pit and re-block it.

EGRESS:

1. Open the hoistway door and mechanically block it in the fully open position.
2. Switch off the pit light if located in the pit.
3. Carefully exit the pit and remove all tools and equipment.
4. Maintaining a safe position, reach in and turn off the light and place the pit switch into the run position.
5. Move the elevator in the down direction level with the bottom landing.
6. Place the car on normal operation.
7. Remove the mechanical block and close the hoistway door.

8. Make sure the elevator operates properly before leaving the building.

1. **Pit Access with Pit Inspection Station**

THE PIT INSPECTION AND PIT INSPECTION EMERGENCY STOP MUST BE TESTED AND VERIFIED WHEN USE OF THOSE CIRCUITS IS REQUIRED.

Follow normal access procedures for tasks that do not require use of the pit inspection box.

Door Unlocking Device Procedure

1. Notify the customer, take control of equipment and install barricades.
2. Establish demand inside the elevator by placing 2 calls in the up direction and step out of the car.
3. Catch the elevator in flight to verify the door lock is working.
4. Insert door wedge tool and engage the landing pit Emergency Stop switch.
5. Close the door and wait 10 seconds. Open door, install door wedge tool. Verify car did not move.
6. Leave wedge tool in and E-stop engaged and climb down to access the pit inspection box.
7. Close the hoistway door to within a distance that prevents closing of the door contact and prevents falls into the pit and re-block it.
8. Place pit Inspection Station to inspection mode and egress pit.
9. Disengage the pit E-stop switch, close the door and wait 10 seconds.
10. Open the door and verify the car did not move. Install door wedge tool.
11. Engage landing E-stop, access pit, and remove door wedge tool and remove the door wedge tool and close the door.
12. Engage Pit Inspection Station E-stop.
13. Disengage landing E-stop.
14. While in safe refuge and on pit inspection, attempt to run the car in the up direction to verify the E-stop is working properly.

Keyed Access

1. Notify the customer, take control of equipment and install barricades.
2. Place the car on in-car inspection and install door wedge tool.
3. Move car in the up direction with your access key (then remove key).
4. Engage top landing Pit Emergency Stop switch.
5. Verify E-stop by attempting to move the car with your access key in both directions (hold for 10 seconds).
6. Verify car did not move and remove access key.
7. Leave wedge tool and landing E-stop engaged.
8. Climb down to access pit inspection box.
9. Close the hoistway door to within a distance that prevents closing of the door contact and prevents falls into the pit and re-block it.

10. Place Pit Inspection Box to inspection mode and egress pit, with door wedge tool in place.
11. Disengage landing pit E-stop switch and attempt to move car with access key in both directions.
12. Verify car did not move, remove key, engage landing E-stop and access pit.
13. Remove the door wedge tool and close the door and engage the Pit Inspection E-Stop.
14. Disengage the landing E-stop.
15. While in safe refuge and on pit inspection, attempt to run the car in the up direction to verify the E-stop is working properly.

Training

Hoistway access training shall be provided upon hire/assignment and annually thereafter. Training resources and forms are available from the Safety Manager or in the Car Top & Pit Safety Program Binder. All instructor-led training must be documented and contain the employee's name, date of training, training materials used, and trainer's name. Training may be carried out via the Learning Management System as well.

Hands-On Demonstration: All new employees must also demonstrate their understanding of Car Top and Pit Access procedures by successfully performing said procedures on equipment under the supervision of an experienced authorized employee before the new or transferred employee is allowed to conduct access/egress procedures independently. The hands-on demonstration form is in Appendix B. In addition to initial and annual training, employees should be retrained when:

- Observations and inspections reveal deficiencies.
- There is a change in job assignments.
- There is a change in machines, equipment or process that presents a new hazard.
- There has been a change in Car Top/Pit Access procedures.

Annual Review

The Car Top & Pit Safety program shall be reviewed annually to ensure program policies and procedures are being followed. Management shall ensure that an annual review of the program is being performed. The review must include an evaluation of:

- Energy Control Procedures
- Changes in equipment or processes
- Training activities
- Documentation practices, and
- Incident Investigations

The annual review will be certified using the "Car Top & Pit Safety Annual Review" form.

Revision Tracking Table

Date

Revision Description

Page # of Revision

Hoisting and Rigging

(Reference: Policy AXX-0023)

The objective of the Hoisting and Rigging Program is to provide safety procedures for Axxiom Elevator employees who work with or near cranes, hoists, and other mechanical lifting devices on any jobsite or shop areas. This program ensures safe practices, compliance with regulations, and assigns responsibility to relevant personnel.

Objectives

- Assign responsibility to relevant personnel for managing inspection of hoisting and rigging equipment in each branch.
- Provide employees with basic knowledge of principles relating to safe hoisting and rigging practices in the elevator industry.
- Identify common unsafe practices and probable consequences.
- Review of common hand signals utilized during hoisting operations.
- Selecting the right equipment for the job.
- Review of different types of rigging and hoisting equipment.

Responsibilities

It is the responsibility of the **Safety Manager** to administer the Hoisting and Rigging program. A coordinator may also be assigned to the program. It is their responsibility to ensure Department Managers and Superintendents are trained and ultimately that all field employees are trained. Responsibility for various aspects of the Program may be delegated to management and supervisory personnel best suited to the task. However, the Safety Manager must monitor and ensure compliance.

Management:

1. Establish Policies and Procedures:
 - Develop and implement comprehensive hoisting and rigging policies and procedures that comply with regulatory requirements and industry standards.

1. Provide Resources:
 - Allocate necessary resources, including funding, equipment, and personnel, to support safe hoisting and rigging operations.
1. Training and Development:
 - Ensure that employees receive appropriate training and development on hoisting and rigging safety practices, including initial training and ongoing refresher courses.
1. Safety Oversight:
 - Provide oversight and support for safety initiatives related to hoisting and rigging activities, including conducting audits and inspections.
1. Risk Management:
 - Collaborate with supervisors and safety professionals to identify and mitigate risks associated with hoisting and rigging operations.
1. Continuous Improvement:
 - Encourage a culture of continuous improvement by implementing feedback mechanisms and evaluating the effectiveness of hoisting and rigging procedures.

Supervisors:

1. Implementation of Policies:
 - Ensure that hoisting and rigging policies and procedures are implemented effectively within their area of responsibility.
1. Training and Supervision:
 - Provide supervision and guidance to employees involved in hoisting and rigging activities, ensuring they have the necessary skills and knowledge to perform their tasks safely.
1. Safety Monitoring:
 - Monitor and enforce compliance with safety protocols during hoisting and rigging operations, intervening if unsafe practices are observed.
1. Equipment Inspection:
 - Oversee pre-use inspections of hoisting and rigging equipment to verify that it is in good working condition and meets safety standards.
1. Incident Investigation:
 - Investigate any hoisting or rigging-related incidents or near misses, identify root causes, and implement corrective actions to prevent recurrence.
1. Communication:
 - Facilitate communication between management, employees, and safety personnel regarding hoisting and rigging safety issues, including reporting hazards and suggesting improvements.

Employees:

1. Compliance with Policies:
 - Follow all hoisting and rigging policies, procedures, and safety guidelines established by the organization.
1. Training and Competency:

- Participate in hoisting and rigging safety training programs to acquire and maintain the necessary knowledge and skills for safe operations.
- 1. Equipment Use:
 - Use hoisting and rigging equipment only if trained and authorized to do so, adhering to load capacity limits and safe operating procedures.
- 1. Pre-Use Inspections:
 - Conduct pre-use inspections of hoisting and rigging equipment before each use, reporting any defects or issues to supervisors.
- 1. Communication:
 - Communicate effectively with supervisors and colleagues during hoisting and rigging operations, using standardized hand signals, and following established protocols.
- 1. Safety Awareness:
 - Maintain a proactive approach to safety by identifying and reporting hazards, participating in safety meetings, and suggesting improvements to enhance hoisting and rigging safety.

Pre-Use Inspections of Hoisting & Rigging Equipment

A pre-use inspection for hoisting and rigging equipment is a critical step to ensure that the equipment is safe and in proper working condition before it is used to lift or move loads. The inspection process involves a thorough examination of the equipment to identify any defects, damage, or wear that could compromise its safety during operation. The specific items inspected may vary depending on the type of equipment, but generally, the inspection should cover the following aspects:

1. Visual Inspection:

Conduct a visual inspection of the equipment in use, looking for signs of damage, corrosion, or wear. Check for cracks, bends, or deformation in structural components.

1. Functional Inspection:

Test the functionality of all moving parts, controls, and safety devices. Ensure that mechanisms such as brakes, locking pins, and limit switches are working properly.

1. Load Capacity Verification:

Confirm that the equipment is rated and suitable for the intended load capacity. Check load rating labels and markings to ensure they are legible and accurate.

1. Wire Rope or Chain Inspection (if applicable):

Inspect wire ropes or chains for signs of kinks, broken strands, or excessive wear. Check for proper lubrication and ensure end connections are secure.

1. Hooks and Attachments Inspection:

Examine hooks, shackles, and other attachments for deformation, cracks, or wear. Ensure that safety latches and closures are functioning correctly.

1. Controls and Operating Mechanisms:

Test all control mechanisms, including pendant controls, remote controls, or levers, to ensure they operate smoothly and respond correctly.

1. Safety Devices Inspection:

Check safety devices such as overload protection systems, limit switches, and emergency stop buttons for proper functioning.

1. Structural Integrity:

Inspect the structural integrity of the equipment, including the frame, boom, or lifting arms. Look for signs of stress, fatigue, or misalignment.

1. Documentation and Certification:

Verify that the equipment has up-to-date inspection and maintenance records. Ensure that certifications or inspection tags are current and valid.

1. General Condition Check:

Assess the overall condition of the equipment, including cleanliness, lubrication, and general maintenance. Address any issues identified during the inspection.

Note: If the equipment is located or stored at any of our mining customers, special care must be taken to ensure slings are not being degraded due to chemicals in the atmosphere.

1. Operational Test (if applicable):

If possible, conduct a brief operational test under no-load conditions to verify smooth operation and responsiveness.

1. Annual Certification

Certain hoisting equipment is required to have a periodic (not to exceed 12 months) inspection. These include chain hoists and lever hoists. Currently, Axxiom Elevator has partnered with Fastenal Inc. to repair and certify any of our manual, electric, and lever chain hoists as well as cable pullers. Fastenal will pick-up and then deliver the hoists back to each branch. The Safety Manager organizes and manages the equipment timeframes and will direct each branch to submit specified hoists each month for repair and certification. If the branch cannot comply for any reason, they must let the Safety Manager know immediately.

All other equipment will be inspected annually by the Safety Manager or their designee, with the inspection being documented.

Documentation for all inspections and certifications is to be kept in the Hoisting and Rigging binder at each branch with the Safety Manager copied on all documentation.

Hoisting Hand Signals

Signals are used for hoisting to ensure safe communication between the signaler and the crane (hoist) operator, especially in noisy construction site environments. These are the reasons for standard signals:

Safety: Miscommunication between the signaler and operator can be dangerous for people and property.

Efficiency: Hand signals are a simple, low-tech, and universal way to communicate.

Coordination: Hand signals help keep the project running smoothly and on schedule.

Visibility: Hand signals allow the signaler to see the load area from a vantage point that's not visible to the operator.

Noise: Hand signals are effective in noisy environments where verbal communication can be difficult.

Standardization: Standardized hand signals help ensure safety across the construction industry.

Below is the Standard Method hand signals typically used in the U.S.; however, the spotter, crane operator, and lift director must agree on non-standard signals before the lift operation begins.

It is the signaler's responsibility to always remain in visual contact with the crane operator. The crane operator may refuse to lift any load at any time based on potential safety issues. If using radios to communicate, ensure there are fresh batteries in your radio as well as a freshly charged battery on your person in reserve. If, for any reason, communication is interrupted with the crane operator, immediately gain visual contact with the crane operator and pause the operation until proper communication can be regained.

Safe Hoisting Operations

For any lifting operation where multiple rigging set-ups are required, a Rigging and Hoisting Plan and Job Hazard Analysis must be created and implemented to ensure employee safety and discussion on how the equipment will move as well as the possible hazards. See below for safe hoisting tips to ensure safe hoisting operations:

- Read the manufacturer operating and maintenance instructions.
- Be familiar with operating controls, procedures, and warnings on the unit.
- Make sure that the unit is securely attached to a suitable support before applying load. Maintain firm footing when operating unit.
- Make sure that load slings, or other approved single attachments, are properly sized and seated in the hook saddle.
- Make sure the hook latches, if used, are closed, and not supporting any part of the load. Make sure that the load is free to move and will clear all obstructions.
- Take up slack carefully and check the load balance.
- Move the load a few inches and check load-holding action before continuing.
- Make sure all people stay clear from the suspended or supported load. Always stay clear of the load, never stand under the load, know the weight of the load, capacity of your rigging equipment, the structure to which you are rigging and the overall condition of these items. Stand uphill or to the side of a load that is on a ramp.
- When our employees are conducting hoisting operations, access to hoisting areas, especially hoistways and wellways, shall be restricted to elevator personnel involved in the lift. Warning signs shall be posted.
- Elevator personnel shall avoid areas where other trades are hoisting.
- The hoist shall be installed only in locations that permit the operator to stand clear of the load at all times.
- The overhead structure shall be padded where any choke or assembly is passed over steel. This padding is essential to keep the rigging in good condition.
- Only properly manufactured wire rope or nylon slings and chokers shall be used for rigging. Never use "homemade" or altered sling or choker.
- Avoid swinging the load or load hook.
- Protect load chain from weld spatter or other potential hazards.
- Avoid "fly-back", by keeping a firm grip on the lever (ratchet lever hoist), until the operating stroke is completed, and the lever is at rest.
- If, during the lift, something does not sound or seem right, all employees have Stop Work Authority to stop work, and to inspect and correct as needed. No retaliation will come, regardless of whether a hazard was found or not.
- Promptly report any malfunction, unusual performance, or damage to the unit.
- Inspect unit regularly, replace damaged or worn parts, and keep appropriate maintenance records.
- The use of two or more chain hoists for hoisting a single elevator equipment load is not permitted, unless any one of them will handle the entire load by itself. This is superseded by an elevator/escalator specific lifting plan.
- Whenever a load cannot be lifted by one person pulling on the chain, investigate for overload or a defect in the chain fall.
- A clevis-type shackle, with locking device or through bolt with jam nut, or nut with cotter pin, shall be used to hoist rails, brackets, etc.
- Use only manufacturer recommended wire rope of the proper size for powered hoists. Always examine such wire rope for defects. Wire ropes found to be defective shall not be used.

- Use hook latches wherever possible. Hooks which do not have safety latches shall be moused as a temporary measure.
- Apply lubricant to load chain as recommended by the hoist manufacturer.
- Always wear gloves when handling wire rope.
- When using a capstan hoist, do not stand in the coil of rope.
- Rated load divided by the number of parts of the rope shall not exceed 20% of the nominal breaking strength of the rope (i.e., safety factor of 5)
- Wire rope used for supporting the top block hoist rig shall have, as a minimum, one more turn than the number of sheaves used in the traveling hoist rope and shall be padded where it goes around a support beam. Remember: Wire rope strength deteriorates 25% when it is wrapped around supports and is secured with fist grip type clips.
- Hoist machines shall be substantially secured so that they will not shift under the load.
- When hoisting rails into a hoistway, and the guardrail system must be removed to allow access, if there is more than a 6 ft fall exposure, the employee feeding rails into the hoistway shall be protected from falling into the hoistway by a personal fall arrest system attached to a lifeline. Remember to keep the guardrail system up on the unused portions of the hoistway and keep others out of your hoisting area.
- Clear communications are required. All verbal commands shall be repeated by the receiving party and reconfirmed by the directing party.

Note: If there is a difference in opinion between Axxiom Elevator Hoisting and Rigging policies versus that of a General, Prime, or Subcontractor, note that in the hoistway, Axxiom policy shall prevail unless the other party has reasoning for any deviation. If this is the case, contact the Safety Manager for further clarification.

Hoisting Operations Pitfalls

Many times, workers cut corners either to save time, or because they believe the operation will be quick and the item just needs to get into place quickly. All Axxiom employees are required to complete a Job Hazard Analysis (JHA) before hoisting operations, regardless of how small or “quick” the operation may be. The purpose of completing the JHA is to slow down, consider all possible effects, plan travel paths, and to ensure all team members fully understand what will happen, where they need to be, and what they need to do. The following statements should be taken into consideration when conducting hoisting operations:

- **DO NOT** lift or pull more than rated load.
- **DO NOT** use hoist load-limiting device to measure the load.
- **DO NOT** use a damaged hoist or hoist that is not working correctly.
- **DO NOT** use a hoist with twisted, kinked, damaged, or worn chain.
- **DO NOT** lift a load unless the chain is properly seated in chain wheel(s) or sprocket(s).
- **DO NOT** use load chain as a sling or wrap load chain around the load.
- **DO NOT** lift a load if any binding prevents equal loading on all supporting chains.
- **DO NOT** apply the load to the tip of the hook.
- **DO NOT** operate unless the load is centered under hoist.

- **DO NOT** operate the unit when it is restricted from adjusting itself to form a straight line with the direction of loading.
- **DO NOT** operate hoist with anything other than manual power.
- **DO NOT** permit more than one operator to pull on a single hand chain or lever at one time.
- **DO NOT** drag slings, chains, etc., along the floor or across equipment.
- **DO NOT** operate unit with any lever extension (cheater bar).
- **DO NOT** allow your attention to be diverted from operating the hoist.
- **DO NOT** operate hoist beyond limits of load chain travel.
- **DO NOT** use hoists to lift, support, or transport people.
- **DO NOT** lift loads over people.
- **DO NOT** use the hoist chain as a choker.
- **DO NOT** leave a suspended load unattended unless specific precautions have been taken.
- **DO NOT** attempt to "free chain" unit with any load applied.
- **DO NOT** allow sharp contact between two hoists or between hoist and obstructions.
- **DO NOT** allow the chain or hook to be used as a ground for welding.
- **DO NOT** allow the chain or hook to be touched by a live welding electrode.
- **DO NOT** remove or obstruct the warnings on the hoist.
- **DO NOT** adjust or repair a hoist unless qualified to perform hoist maintenance.
- **DO NOT** attempt to lengthen the load chain or repair damaged load chain.
- **DO NOT** drill additional holes in the capstan hoist's mounting brackets, this could weaken the hoist.
- **DO NOT** allow non-Company personnel to use Axxiom Elevator hoisting and rigging equipment.

For specific operating, maintenance, and installation information, refer to Hoist Manufacturer Maintenance Manual.

It's important to perform the pre-use inspection in accordance with manufacturer recommendations and applicable safety standards. If any defects or concerns are identified during the inspection, the equipment should be taken out of service immediately and reported to the supervisor for further evaluation and corrective action. Regular and diligent pre-use inspections are essential to prevent accidents, ensure worker safety, and maintain the integrity of hoisting and rigging equipment.

Chain Slings

Due to the difficulties in inspecting and certifying alloy steel chain slings, Axxiom Elevator does NOT use chain slings for lifting.

Wire Rope

Wire rope is composed of individual wires that have been twisted to form strands. Strands are then twisted to form a wire rope. When wire rope has a fiber core, it is usually more flexible but is less resistant to environmental damage. Conversely, a core that is made of a wire rope strand tends to have greater strength and is more resistant to heat damage.

Wire rope may be further defined by the "lay." The lay of a wire rope describes the direction the wires and strands are twisted during the construction of the rope. Most wire rope is right lay, regular lay, which means that the strands pass from left-to-right across the rope, and the wires in the rope are laid opposite in direction to the lay of the strands. This type of rope has the widest range of applications.

Lang lay ropes are more flexible and have a greater wearing surface per wire than regular lay ropes. In addition, since the outside wires in lang lay rope lie at an angle to the rope axis, internal stress due to bending over sheaves and drums is reduced causing Lang lay ropes to be more resistant to bending fatigue.

A left lay rope is one in which the strands form a left-hand helix, similar to the threads of a left-hand screw thread. Left lay rope has its greatest usage in oil fields, on rod and tubing lines, blast hole rigs, and spudders, where rotation of right lay would loosen couplings. The rotation of a left lay rope tightens a standard coupling.

1. Wire Rope Sling Selection

When selecting a wire rope sling for a particular application, there are four characteristics to consider: strength, ability to bend without distortion, ability to withstand abrasive wear, and ability to withstand abuse.

Strength The strength of a wire rope is a function of its size, grade, and construction. It is sufficient to accommodate the applied maximum load. The maximum load limit is determined by means of an appropriate multiplier. This multiplier is the number by which the ultimate strength of a wire rope is divided to determine the working load limit. Thus, a wire rope sling, with a strength of 10,000 lbs. (4,545 kilograms) and a total working load of 2,000 lbs. (909 kilograms), has a design factor (multiplier) of five (5). New wire rope slings have a design factor of five (5). As a sling suffers from the rigors of continued service both the design factor and the sling's ultimate strength are proportionately reduced. If a sling is loaded beyond its ultimate strength, it will fail. Older slings shall be more rigorously inspected to ensure that rope conditions adversely affecting the strength of the sling are considered in determining if a wire rope sling should be allowed to continue in service.

Fatigue (Bending without Failure) - A wire rope shall have the ability to withstand repeated bending, without the wires failing from fatigue. Fatigue failure of the wires in a wire rope is the result of the development of small cracks from repeated applications of bending loads. It occurs when ropes make small radius bends. To prevent fatigue failure of wire rope slings, use blocking or padding to increase the radius of bend.

Abrasive Wear - The ability of a wire rope to withstand abrasion is determined by the size, number of wires, and construction of the rope. Smaller wires bend more readily, and therefore, offer greater flexibility, but are less able to withstand abrasive wear. Conversely, the larger wires of less flexible ropes are better able to withstand abrasion than the smaller wires of more flexible ropes.

Abuse Misuse or abuse of wire rope will cause a wire rope sling to become unsafe long before any other factor. Abusing a wire rope sling can cause serious structural damage to the wire rope, such as kinking or bird caging, which reduces the strength of the wire rope. (In bird caging, the wire rope strands are forcibly untwisted and become spread outward.) To prolong the life of the sling strictly adhere to the Manufacturer recommendations for safe and proper use of wire rope slings.

1. Wire Rope Life

Many operating conditions affect wire rope life, including bending, stresses, loading conditions, speed of load application (jerking), abrasion, corrosion, sling design, materials handled, environmental conditions, and history of previous usage.

In addition to the above operating conditions, the weight, size, and shape of the loads also affect the service life of a wire rope sling. Flexibility is also a factor. Select flexible ropes when smaller radius bending is required.

Use less flexible ropes when the rope will move through or over abrasive materials.

1. Wire Rope Sling Inspection

Visually inspect wire rope slings before use each day. The user should check the twists or lay of the sling. If six randomly distributed wires in one lay are broken, or three wires in one strand of a rope lay are damaged, the sling shall not be used. It is insufficient, however, to only check the condition of the wire rope. End fittings and other components should also be inspected for any damage that could make the sling unsafe.

Each user should keep a close watch on slings he or she is using. If any incident involving the movement of materials occurs, the user should immediately shut down the equipment and report the incident to a supervisor. The cause of the accident should be determined and corrected before resuming operations.

1. Allowable Number of Broken Strands for Wire Ropes used in Hoisting Equipment

1. Six randomly distributed broken wires in one lay.
2. Three broken wires in one strand on one lay.
3. If one-third of the outside wires show wear or disintegrate at any place on the rope.

1. Causes of Wire Rope Failure

4. Using wire rope of insufficient strength for the job.
5. Improper rigging.

6. Kinking.
7. Improper lubrication, storage and care, allowing rusting, corrosion, or internal abrasion.
8. Exposure to extreme heat.
9. Crushing on a winch drum.
10. Using drums or sheaves of insufficient diameter or incorrect tread.
11. Permitting ropes to abrade over sharp corners or other fixed objects.
12. Contact with electrical current.

1. Signs of Wire Rope Deterioration

13. Reduction of rope diameter below nominal diameter due to loss of core support, internal/external corrosion, or wear of outside wire.
14. Broken outside wires and degree of distribution or concentration of broken wires.
15. Worn outside wires.
16. Corroded, rusty or broken wires at end connections.
17. Severe kinking, crushing, cutting, and/or unstranding.

1. Wire Rope Fastenings

- When it is necessary to make a short bend, as in attaching wire rope or when it is to be looped, thimbles shall always be used.
- Fist grips are preferred fasteners because the wire rope does not become damaged.
- In forming an eye, the loose or “dead” end is clamped against the main part of the rope, with wire rope clips spaced a distance equal to six times the diameter of the rope.
- Wire-rope clip fastenings seldom develop more than 80% of rope strength, at best.
- The point of greatest fatigue and/or wear in a rope usually develops at or near the end where it becomes attached to the bucket on the block. Clips shall be inspected at least daily and tightened if they show signs of loosening. All connections shall be periodically disassembled, and the wire rope inspected for damage on a regular basis.

Chain Hoists

A chain hoist (sometimes known as a chain fall) is a mechanical device that lifts heavy loads by wrapping a chain or rope around a drum or lift wheel. Chain hoists can be operated manually, electrically, or pneumatically.

- Generally, chain hoists are more durable, more adaptable, and stronger than block and tackle.
- All chain hoists shall be visually inspected for defects prior to first use, and daily when in use.
- The lower hook on all chain hoists is designed to be the weakest part and will spread when overloaded. When this occurs, the entire hoist shall be inspected.
- Damaged hoists shall be removed from service.
- Repairs shall only be made by an approved third party.
- Always use proper hand signals when hoisting.
- Discard hooks that spread beyond allowable spread.

Slings and Hitches

- Chain slings shall not be used when hoisting material.
- Nylon slings are to be used for hoisting equipment within their rated load. Label with rating shall be attached to the sling.
- Nylon slings shall be inspected for cuts or tears before using and damaged slings discarded.
- The type of sling or hitch to be used shall be determined from the shape of the load and by the flexibility and condition of the rope. In lifting multiple objects, such as load of lumber or steel sheathing, the sling must bind on the load sufficiently to prevent slipping of the individual pieces. In handling single pieces, such as timbers, posts or piles, a timber hitch with two half-hitches (or similar hitch) shall be used.
- Only approved slings of proper size shall be used for slinging loads.
- If using wire rope as strands for hooking onto tackle blocks, there shall be the same number of parts of rope in the strap as there are moving parts in the tackle. For instance, if triple-block tackle is used, there shall be six parts of rope for the strap.
- Endless wire slings shall be made using a minimum of six fist grip rope clips as shown below. If the rope is greater than 5/8 in. additional fist grip rope clips are required.

Synthetic Web Slings

This section applies to slings fabricated by sewing of woven synthetic webbing of nylon or polyester type yarns, for the purpose of hoisting, lifting, and general material handling.

1. Webbing

Webbing should be of fabric woven of high tenacity synthetic yarns, offering suitable characteristics for use in the fabrication of web slings. Webbing shall have the following characteristics:

1. Sufficient certified tensile strength to meet the sling manufacturer's requirements.
2. Uniform thickness and width.
3. Full woven width, including selvage edges.
4. Webbing ends shall be sealed by heat, or other suitable means, to prevent raveling.

1. Thread

The thread used in the manufacture of synthetic web slings shall be of the same generic type of yarn as the sling webbing.

1. Stitching

1. Stitching shall be the only method used to fabricate synthetic web slings within the scope of this standard.
2. The stitching pattern and length of stitching shall be in accordance with the manufacturer's standard practice.

1. Fittings

3. The material selected shall be compatible with the mechanical and environmental requirements imposed on the fitting. Material selected should be carbon steel, alloy steel, aluminum, or other suitable material.
4. Fitting shall have sufficient strength to sustain twice the rated load of the sling without permanent deformation and a minimum breaking strength equal to five times the rated load of the sling.
5. All surfaces shall be cleanly finished and sharp edges removed so as not to cause damage to the webbing.
6. Slings incorporating aluminum fittings shall not be used where fumes, vapors, sprays, mists, or liquids of caustic, or acids are present.
7. The eye opening in the fitting shall be the proper shape and size to ensure that the fitting will seat properly in the hook or other attachment.

1. Marking (Sling Identification)

Each sling shall be permanently marked to show:

1. Name of trademark or manufacturer.
2. Manufacturer's code or stock number.
3. Rated loads for types of hitches used.
4. Type of synthetic web material.

1. Safety Factor

The safety factor for synthetic web slings shall be a minimum of 5.

1. Rated Load

1. A sling shall not be used at a load greater than that shown on its tags. Slings shall be used in accordance with the manufacturer's recommended rated loads.
2. Each manufacturer shall make available on request test data to justify these recommended rated loads.

1. Effects of Environment

3. Chemically active environments, such as acids and caustics, can affect the strength of slings. The manufacturer should be consulted before slings are used in chemically active environments.
4. Nylon and polyester slings shall not be used at temperatures in excess of 194°(F) 90°(C).

1. Inspection

5. Before using any new or repaired sling, it shall be inspected to ensure that the correct sling is being used as well as to determine that the sling meets the requirements of the standard.
6. Frequent inspections should be made by the person handling the sling each day the sling is used.
7. Periodic inspections should be conducted by a Competent Person. Frequency of inspection should be based on:
 1. Frequency of sling use

2. Severity of service conditions
3. Experience gained on the service life of slings used in similar applications.
4. Periodic inspections should be conducted at least annually.

1. Removal Criteria

A sling should be removed from service if damage such as the following is visible and shall only be returned to service when approved by a Qualified Person.

1. Acid or caustic burns.
2. Melting or charring of any part of the sling.
3. Holes, tears, cuts, or snags.
4. Broken or worn stitching in load bearing splices.
5. Excessive abrasive wear.
6. Knots in any part of the sling.
7. Excessive pitting or corrosion, or cracked, distorted, or broken fittings.
8. Other visible damage that causes doubt as to the strength of the sling.

1. Operating Practices

9. The weight of the load shall be within the rated load of the sling.
10. Slings shall not be shortened or lengthened by knotting or other methods not approved by the sling manufacturer.
11. Slings that appear to be damaged shall not be used unless inspected and accepted as usable by a Qualified Person.
12. Sling shall be hitched in a manner providing control of the load.
13. Sharp corners in contact with the sling should be padded with material of sufficient strength to minimize damage to the sling.
14. Personnel should stand clear of the suspended load.
15. Personnel shall not ride the sling.
16. Shock loading should be avoided.
17. Slings should not be pulled from under a load when the load is resting on the sling.
18. Slings should be stored in a cool, dry, and dark place to prevent environmental damage.
19. Twisting and kinking the legs shall be avoided.
20. The load applied to the hook should be centered in the base (bowl) of hook to prevent point loading on the hook.
21. While lifting, with or without a load, personnel shall be alert for possible snagging.
22. In a basket hitch, the load should be balanced to prevent slippage.
23. The sling's legs should contain or support the load from the sides above the center of gravity when using a basket hitch.
24. Slings should be long enough so that the rated load is adequate when the angle of the legs is taken into consideration.
25. Slings should not be dragged on the floor or over an abrasive surface.
26. In a choker hitch, slings shall be long enough so the choker fitting chokes on the webbing and never on the other fitting.
27. When extensive exposure to sunlight or ultraviolet light is experienced by nylon or polyester web slings, the sling manufacturer should be consulted for recommended inspection procedure because of loss of strength.

Beam Clamps

1. A beam clamp used for rigging shall be engineered to properly support the expected load.
2. Confirm with your supervisor that the hoisting beam is rated to carry the maximum load to be hoisted. Do not load the lower flange to more than 50% of the beam's capacity.
3. Do not use a choker through the eye of the beam clamp while hoisting.
4. Beam clamps shall be properly sized for the beam to which it is attached.
5. Use only case-hardened bolts with lock nuts or nuts with lock washers for the beam clamp assembly.

Manila Rope

1. Frozen manila rope shall not be used in load-carrying service.
2. Manila rope shall be protected from abrasion by padding where it is fastened or drawn over square corners, or sharp or rough surfaces.
3. Even the finest-quality rope deteriorates very rapidly when not given the best of care. Kinking, overheating, moisture, and acid all cause deterioration, which is not readily noticeable upon casual inspection.
4. Manila rope shall be stored in a clean, dry location. Keep off the pit floor and coil into a protective device such as a drum.
5. Manila rope is not a substitute for wire rope. It shall only be used for lashings, tackles, tag lines, straps on light leads and temporary guy lines and light hoisting.
6. Good characteristics to look for during manila rope inspection are:
 1. Hard but pliant
 2. Silvery or pearly luster
 3. Inner fibers bright and clean
 4. Individual yarn strong
 5. Uncut and unabraded outer and inner fibers
 6. Good stretch and spring

Poor characteristics of manila rope include:

1. Brown spots – weak, soft
2. Black or dark spots – weak
3. Abrasion of fibers
4. Loss of stretch
5. Cuts or burns
6. Dirt between inner fibers
7. Freezing of rope

Synthetic Rope

Synthetic fibers are made from nylon, polypropylene, or polyester. Synthetic fiber ropes consist of individual threads and fibers that run the full length of the rope (natural fibers are not continuous – in fact they are short and overlapped).

1. Do not choose synthetic rope when burning and welding. Synthetic rope is also more likely to be affected by chemicals and it tends to be slippery.
2. Do not use clamps for splicing synthetic rope unless it is specifically designed for that purpose.
3. Good practices when using synthetic rope:
 1. Keep the rope dry and clean and away from chemicals.
 2. Never overload a rope.
 3. Never use frozen rope.
 4. Don't drag a rope on the ground. This will damage the outside surface of the rope.
 5. Never allow the rope to bend over sharp edges.
 6. Don't permit the rope to drag against itself.
 7. Observe proper picking angles.
 8. Pad all corners when lifting materials.
 9. When coupling ropes, use thimbles.
 10. Inspect the rope often by twisting to expose the inside yarns.
 11. Synthetic rope shall be removed from service if it shows signs of:
 1. Abnormal wear.
 2. Powder between strands.
 3. Broken or cut fibers.
 4. Variations in the size or roundness of strands.
 5. Discoloration or rotting.
 6. Distortion of hardware.
 7. When rope is damaged and taken out of service, it shall be completely destroyed to prevent others from using it.

Knots

Knots are weak. If a knot or hitch of any kind is tied in a rope, its failure under stress is sure to occur at that place. Each fiber in the straight part of the rope takes proper share of the load. In all knots, the rope is cramped or has a short bend, which throws an overload on those fibers that are on the outside of the bend, and one fiber after another breaks until the rope is torn apart. The shorter the bend in the standing rope, the weaker the knot.

Visual Rigging Inspection (Appendix A)

For most topics discussed above, refer to Appendix A for a visual representation on what to inspect rigging for and proper hoisting techniques.

Training

Under no circumstance may an employee work with hoisting and rigging until he/she has successfully completed the training through the IUEC Apprenticeship Rigging and Hoisting course as well as shown management that rigging and hoisting knowledge is sufficient. This includes all new employees, regardless of claimed previous experience.

The Department Manager is responsible for ensuring that all employees within their department have completed their Hoisting and Rigging training, although the actual training itself may be carried out by others.

After an employee has completed the training program, the instructor or supervisor will determine whether the employee can safely perform the job. All employees who receive training must be documented with either a paper roster or tracked within the Safety Management System. Each training roster must include the name(s) of the employee(s) trained, the date(s) of the training, and the instructor's name who administered the training.

Training records must be maintained for three years.

Plan Evaluation

It is inherent that problems may occasionally arise in the Hoisting and Rigging Program. Although we may not be able to eliminate all problems, we try to eliminate as many problems as possible to improve employee protection and encourage employee safe practices. The Hoisting and Rigging Program will be evaluated annually by the Safety Manager and revised as necessary. This will include any received input from all levels of the organization.

Appendices

A – Visual Rigging and Hoisting Guide

B – Hoisting and Rigging Inventory & Inspection Log

Revision Tracking Table

Date

Revision Description

Page # of Revision

Stored Energy

(Reference: Policy AXX-0024)

Axxiom Elevator is committed to the safe operation and maintenance of equipment that utilizes stored energy of all types. This policy outlines the procedures and practices necessary to prevent accidents, injuries, and property damage associated with the release of stored energy.

Policy Objectives

1. **Safety Priority:** The safety of our employees, contractors, and visitors/customers is our top priority. We are dedicated to providing a work environment free from hazards related to stored mechanical or electrical energy.
2. **Compliance:** We will comply with all applicable local, state, and federal regulations governing the use of mechanical and electrical stored energy.
3. **Training and Education:** We will provide comprehensive training and education programs to ensure that all personnel are knowledgeable about the safe handling, operation, and maintenance of equipment with stored mechanical/electrical energy.
4. **Lockout/Tagout/Tryout Procedures:** We will implement and enforce robust lockout/tagout/tryout procedures to isolate and secure energy sources before any maintenance or servicing activities.
5. **Preventative Maintenance:** We will establish regular inspection and maintenance schedules to identify and address wear, damage, or deterioration of components that store mechanical energy on our premises.
6. **Continuous Improvement:** We are committed to regular audits, reviews, and updates of our safety program to incorporate best practices and adapt to changing circumstances.

ROLES AND RESPONSIBILITIES

Management: It is the responsibility of management to provide the necessary resources, training, and oversight to ensure compliance with this policy.

Supervisors: Supervisors are responsible for enforcing safety protocols, conducting regular inspections, and addressing any non-compliance promptly.

Employees and Contractors: All personnel are expected to follow established safety procedures and report any unsafe conditions or incidents promptly.

Mechanical Stored Energy

Mechanical stored energy refers to energy that is stored within a mechanical system and can be released or harnessed for use later.

There are several common types of mechanical stored energy:

1. **Kinetic Energy:**
 - **Rotational Kinetic Energy:** Stored in rotating objects like flywheels. When a flywheel spins, it stores kinetic energy. This energy can be used to smooth out variations in power input/output.
 - **Linear Kinetic Energy:** Associated with the motion of an object in a straight line. For example, a moving vehicle possesses kinetic energy due to its motion.
2. **Potential Energy:**
 - **Gravitational Potential Energy:** This is energy stored due to an object's position in a gravitational field. The higher an object is elevated, the more potential energy it possesses. For example, a raised weight or an elevated water reservoir.
 - **Elastic Potential Energy:** This is energy stored in elastic materials like springs or rubber bands when they are stretched or compressed. When released, this energy can be used to perform work.
3. **Compressed Air or Gas:**
 - **Compressed Air Energy Storage (CAES):** In this method, air is compressed and stored in underground reservoirs. When needed, the compressed air is released and expanded through a turbine to generate electricity.
4. **Torsional Energy:**
 - **Torsion Springs:** These springs store energy by twisting. They are commonly used in various mechanical systems like doors, clutches, and more.
5. **Hydraulic Accumulators:**
 - **Hydraulic Energy Storage:** Hydraulic accumulators store energy by compressing a gas (usually nitrogen) within a chamber filled with hydraulic fluid. When the gas is compressed, it stores potential energy. When needed, the compressed gas pushes the fluid back into the system, releasing the stored energy.
6. **Flywheels:**
 - **Mechanical Flywheels:** These are heavy rotating discs used to store rotational energy. They are often used in machines where a constant power output is required, like in engines or generators.
7. **Pneumatic Springs:**
 - **Pneumatic Cylinders:** These use compressed air to store and release energy. When the air is compressed, it stores energy. When released, it can perform tasks like lifting or moving objects.
8. **Capacitors (Electro-Mechanical)**
 - **Electro-Mechanical Capacitors:** These are devices that can store and release electrical energy, but they have mechanical elements as well. For example, a supercapacitor can store energy mechanically by separating charges.

Note: The stacking of material is a major hazard in our industry. As such:

- Pallets of material are never to be stacked more than two pallets high.
- All material on pallets that goes anywhere above the floor into racking must be banded or wrapped.

Electrical Stored Energy

Electrical stored energy refers to energy that is stored in an electrical system and can be released or harnessed for use at a later time.

Here are some common types of electrical stored energy:

1. **Batteries:**

- **Chemical Energy Storage:** Batteries convert chemical energy into electrical energy. They come in various types (e.g., lead-acid, lithium-ion) and sizes for different applications. Batteries have a few hazards to be aware of:

1. **Battery Acid** – Sulfuric acid in a battery is corrosive and can severely burn the skin or eyes and eat through clothing. Personal Protective equipment such as goggles, a face shield, rubber gloves, or rubber apron can be worn to protect yourself. It should also be known where the closest first aid kit and eyewash station are located.
2. **Flammable Gases** – Batteries emit flammable hydrogen gas during charge and discharge cycles. Hydrogen is not toxic, but at high concentrations is a highly explosive gas. Adequate ventilation should be in place to disperse fumes given off during charging. Open flames, sparks, welding and electric arcs should be prohibited in areas where batteries are charging. Fire protection equipment, such as fire extinguishers should be stored in an easily accessible location nearby.
3. **Electrical Shock** – Batteries are stored energy devices meaning no overload protection is available if the battery is connected improperly or short-circuited. Always keep a voltmeter handy to verify correct polarity and expected voltage levels when connecting strings of batteries. Also exercise caution when working with metallic tools or conductors to prevent short circuits and sparks. Remove rings, watches and loose jewelry when working with batteries. Never lay tools or other metal parts on top of a battery.
4. **Battery Weight** – Cells used in large industrial standby-power applications can weigh anywhere from 20 to 100+ pounds apiece. Use safe lifting practices while exercising caution of electric shock as discussed above. If need be, work in teams, always lifting with your legs, and use mechanical aids as necessary. Do not attempt to stop a battery if it slides out of the equipment.

5. **Capacitors:**

- **Electrostatic Energy Storage:** Capacitors store electrical energy in an electric field between two conductive plates separated by an insulating material (dielectric). They can release this energy quickly. Always assume capacitors are charged unless you have confirmed that they are discharged. The dangers capacitors may pose are:

1. **Shock** – With covers open or circuitry unguarded, contact with the conductors of a capacitor could result in an instantaneous discharge of the capacitor. Many circuits have a discharge unit to bleed energy from a capacitor, but there is no visual way to verify this has occurred. If boards must be removed or work done around exposed capacitors, discharge them in accordance with the manufacturer's directions.
2. **Short Circuit (thermal and arc flash)** – If an object gets near the capacitor conductors, capacitors have been known to discharge through the air looking for ground. This is

known as arc flash and arc blast. This can be deadly. Shock PPE (safety glasses and electrical gloves rated for the highest potential of voltage (either input or output)) with leather protectors) should be used.

3. **Physical (internal ruptures, fires)** – Rupture of a capacitor can create a fire hazard from the ignition of dielectric fluid. Dielectric fluids can release toxic gases when decomposed by fire or the heat of an electric arc. Again, PPE can aid in protecting from this hazard. Leather covers, especially to prevent thermal hazards.
 - **Discharge Method:** The standard way to discharge a capacitor is to connect a resistor across its terminals, effectively creating a path for the stored energy to dissipate as heat through the resistor; this is often called a “bleeder resistor” and is considered the safest method, as it allows for controlled discharge without sudden surges. Voltage must be considered when discharging any capacitor. If there is any question on how to discharge properly, please call your supervisor. It is also important to note discharge methods in the Job Hazard Analysis.

The following are lesser-known sources of electrical stored energy in the elevator industry, but still worth noting:

1. **Superconducting Magnetic Energy Storage (SMES):**
 - **Magnetic Energy Storage:** SMES systems store energy in a superconducting coil, creating a strong magnetic field. When needed, the energy is released by discharging the magnetic field.
2. **Flywheels (Electro-Mechanical):**
 - **Rotational Energy Storage:** While primarily mechanical, flywheels can also store electrical energy through a generator-motor system. The spinning flywheel stores energy kinetically and can be converted back to electricity when needed.
3. **Hydrogen Fuel Cells:**
 - **Chemical Energy Storage and Conversion:** Hydrogen fuel cells convert hydrogen and oxygen into electricity, releasing electrical energy. Hydrogen can be stored for later use.
4. **Supercapacitors:**
 - **Electrochemical Double-Layer Capacitors:** These are energy storage devices that bridge the gap between traditional capacitors and batteries. They store energy electrostatically and can release it rapidly.
5. **Compressed Air Energy Storage (CAES) (Electro-Mechanical):**
 - **Electro-Mechanical Energy Storage:** CAES systems use electricity to compress and store air in underground reservoirs. When needed, the compressed air is expanded through a turbine to generate electricity.
6. **Thermal Energy Storage (Electro-Thermal):**
 - **Thermal Energy Conversion:** Electrical energy can be converted into thermal energy and stored in materials like phase-change materials or hot water. This thermal energy can later be used for heating or power generation.
7. **Electrochemical Storage (e.g., Flow Batteries):**
 - **Redox Reactions:** These batteries use reversible chemical reactions to store and release electrical energy. Flow batteries, for example, store energy in chemical solutions.
8. **Gravitational Potential Energy (Electro-Gravitational):**

- **Hydroelectric Pumped Storage:** This involves using electrical energy to pump water uphill to a reservoir. When energy is needed, the water is released, flowing downhill through turbines to generate electricity.
- 9. **Nuclear Energy Storage (Electro-Nuclear):**
- **Nuclear Batteries:** These are devices that use radioactive isotopes to generate a small but long-lasting electrical current. They are often used in situations where long-term power is needed, such as in spacecraft.

Safety While Working Around Stored Energy

Working around mechanical and electrical stored energy requires careful precautions to prevent accidents or injuries. There can be more than one type of stored energy within an elevator machine room/workspace. Utilize a Job Hazard Analysis to make sure all potential stored energy hazards are at a zero-state prior to beginning work. Here are important safety steps to take for both management and employees:

1. **Training and Education:** Ensure that personnel are properly trained in elevator operation, maintenance, and the handling of mechanical and electrical stored energy.
2. **LOTOTO Procedures:** Implement and follow lockout/tagout/tryout procedures before performing any maintenance, modernization, repair, or servicing activities on elevators. This includes isolating and securing energy sources to prevent accidental activation.
3. **Wear Appropriate PPE:** Use the necessary Personal Protective Equipment (PPE) such as hard hats, safety goggles, appropriate gloves, and hard-toed boots to protect against potential hazards.
4. **Proper Ventilation:** Ensure that elevator shafts and machine rooms are well-ventilated, especially in cases where compressed air or gases are involved.
5. **Use Safety Barriers and Guards:** Install and maintain physical barriers and guards to prevent unauthorized access to elevator machinery and shafts.
6. **Inspect Elevator Components Regularly:** Conduct routine inspections of elevator components, including cables, pulleys, brakes, and counterweights, to identify and address any signs of wear or damage.
7. **Emergency Stop Devices:** Familiarize yourself with the location and operation of emergency stop devices. Use them in case of an emergency or if a potentially dangerous situation arises.
8. **Avoid Overloading:** Ensure that elevators are not overloaded beyond their rated capacity to prevent strain on mechanical components.
9. **Follow Manufacturer Guidelines:** Adhere to the manufacturer's guidelines and recommendations for maintenance, operation, and safety precautions related to the specific elevator model.
10. **Use Caution in Elevator Pit:** If working in the elevator pit, be aware of potential pinch points, moving parts, and hydraulic systems. Use caution and ensure the elevator is properly secured. DO NOT prop cars/equipment with timbers, utilize appropriate metal stands.

11. **Ask Questions:** If you are unsure of but may suspect a potential stored energy release, STOP WORK, clear the area, and seek assistance before proceeding. Be 100% sure of the results of your current activity before proceeding.

By following these safety steps, you can help ensure a secure working environment when dealing with mechanical and electrical stored energy. Always prioritize safety and consult with qualified experts or supervisors if you have any questions or concerns.

Personal Protective Equipment

When dealing with mechanical or electrical stored energy, it is important to use appropriate Personal Protective Equipment (PPE) to minimize the risk of accidents and injuries. The specific PPE required can vary depending on the type of mechanical or electrical stored energy and the associated equipment. Here are some common types of PPE that may be necessary:

- **Eye Protection:** Safety goggles or glasses can help protect against flying debris, particles, or chemical splashes.
- **Hand Protection:** Heavy-duty gloves can provide protection against sharp edges, hot surfaces, and contact with moving parts.
- **Foot Protection:** Steel-toed boots or safety shoes can help prevent injuries from falling objects or crush hazards.
- **Head Protection:** Hard hats are essential to protect against falling objects, especially in areas where potential energy is stored.
- **Respiratory Protection:** Depending on the environment, respiratory protection like masks or respirators may be necessary to protect against dust, fumes, or other airborne contaminants.
- **Ear Protection:** Earplugs or earmuffs can help protect against excessive noise levels, which can be associated with some mechanical systems.
- **Body Protection:** Coveralls or other protective clothing may be needed to shield against exposure to hazardous materials or to prevent entanglement in moving parts.
- **High-Visibility Clothing:** In environments where visibility is crucial, such as around heavy machinery or in low-light conditions, high-visibility clothing helps make workers more visible to others.
- **Lockout/Tagout Devices:** Lockout devices are not exactly PPE, but they are crucial for the safe handling of stored energy. They are used to physically lock equipment in a safe position to prevent inadvertent activation during maintenance or servicing.
- **Fall Protection:** In elevated work environments, fall protection equipment such as harnesses, lanyards, and anchor points may be necessary to prevent falls.
- **Electrically Insulated Gloves and Tools:** For work involving electrical stored energy, especially in capacitors or electrical systems, insulated gloves and tools may be required to protect against electric shock.
- **Chemical-Resistant Gloves and Clothing:** In situations where exposure to chemicals is a concern, appropriate chemical-resistant gloves and clothing should be used.

It is crucial to conduct a thorough risk assessment to determine the specific PPE needed for a given task or work environment involving mechanical or electrical stored energy. Additionally, training on the proper selection, use, and maintenance of PPE is essential to ensure its effectiveness. Always follow relevant safety regulations and guidelines provided by the equipment manufacturer and local regulatory authorities.

Plan Evaluation

This policy will be reviewed periodically to ensure its effectiveness and relevance. Updates will be made as needed to align with changes in regulations, technology, or work processes.

Recordkeeping

Training records must be maintained for a minimum of three years. The most recent record of training must always be kept on file, even if older than three years.

Revision Tracking Table

Date

Revision Description

Page # of Revision

Heat Illness Prevention Program

(Reference: Policy AXX-0025)

This program is instituted by Axxiom Elevator to protect employees subject to heat stress. The information in this plan will facilitate implementation of measures necessary to prevent heat related illnesses.

The primary objective of the Heat Illness Prevention Policy is the education of employees about the hazards of heat stress. Incidents involving heat illness can be prevented. Axxiom Elevator encourages all suggestions to this plan to be voiced because we are committed to the success of our written plan. We strive for a comprehensive, integrated prevention system that establishes clear understanding, safe behavior, and involvement in the plan from every position at Axxiom Elevator.

Scope

The Heat Illness Prevention Policy covers how the company will train on heat stress related illness awareness, prevention of heat illness, acclimatization, as well as some engineering and administrative controls for managing heat stress.

Axxiom Elevator intends to comply with all federal, state, and local laws and ordinances with respect to Heat Illness Preventions initiatives.

Since Axxiom Elevator has jobsites which may expose employees to heat stress, compliance with the directives of the Heat Illness Prevention Policy is mandatory. This policy has been designed to help Axxiom Elevator achieve and maintain compliance with these requirements.

Requirements

The basic requirements of the Heat Illness Prevention Policy are:

- Develop and make a written Heat Illness Prevention Program (“Program”). The written plan must list the signs of Heat Illness, prevention techniques (both engineering and administrative), indicate who is responsible for the various aspects of the Program, and indicate where written materials will be made available to employees. This binder, when completed, will serve as your written Program. This needs to be made available to all employees.
- Obtain and maintain hygrometers for employee teams commonly working outdoors or in non-air-conditioned jobsites.
- Train each employee who may be exposed to heat stress when working.

Responsibilities

Physical factors that contribute to heat related illness should be taken into consideration before assigning a task. The most common physical factors that can contribute to heat related illness are type of work, level and duration of physical activity, and clothing color, weight, and breathability. The following departments affected by this plan include:

1. New Installation
2. Modernization
3. Shop / Truck Drivers
4. Repair

It is the responsibility of the **Safety Manager** to administer the Heat Illness Prevention program. A coordinator may also be assigned to the program. It is their responsibility to ensure Department Managers and Superintendents are trained and ultimately that all employees are trained. Responsibility for various aspects of the Program may be delegated to management and supervisory personnel best suited to the task. However, the Safety Manager must monitor and assure compliance.

Each department Supervisor/Manager/Superintendent will be responsible to ensure that their teams and individual field personnel are trained and have the necessary resources concerning the prevention and recognition of heat illness. This includes water and ice, sport drinks, fans, etc. The Supervisor/Manager/Superintendent will also work with the Building Owner or General Contractor to ensure site coverage.

The Foreman, Mechanic in Charge (MIC), or mechanic onsite will be responsible for determining the heat index value at their specific jobsite and ensuring this policy is followed at the different heat ranges encountered. The Foreman, Mechanic in Charge, or Mechanic will also delegate workload as directed by the Supervisor to appropriately allow acclimatization for new employees not acclimated to the heat.

Employees are responsible for obeying safety rules, following proper procedures, using protective equipment as required and notifying their supervisor of any unsafe conditions they may observe either in the facility or out in the field. The active role of the employees and their responsibilities in the Program should be continuously and positively reinforced since their contribution to overall safety and the success of the Heat Stress Prevention Program is paramount.

Program Components

The Heat Illness Prevention Program consists of:

- Management Briefing – All Managers and Supervisors should be familiar with this guide
- Acclimation – Make sure new employees are acclimated to the heat
- Control Measures – Implement various control measures as necessary
- Recognition – Make sure employees are familiar with the causes, symptoms, and first aid procedures
- Training – Include managers, supervisors/superintendents, and field/shop employees
- Reminders – Field Supervisors to remind and reinforce Heat Illness awareness during heat waves
- Annual Review – Using the Annual Review Form, confirm program elements are in place

Heat Illness Program Surveillance

Inspect job sites periodically to ensure employees are following safe work practices and not displaying symptoms of heat illness or heat stroke. The warmer the work environment the more frequently the worksites should be inspected. Because some jobsites are many hours away, a verifiable electronic means may be used to check in on employees. These can include cell phone and text messaging. In these conditions, a “buddy” system will be used by the teams on worksites to keep an eye on each other for symptoms.

A Heat Index in the range of 80-90° F the team should check in on each other at least every two hours. A Heat Index of 91° F or greater requires the team to check on one another every hour.

Refer to Appendix D for a Summary of Risk Levels and Associated Protective Measures for details of the actions needed.

Supervisors must ensure personal factors that contribute to heat related illness are taken into consideration before assigning work duties where there is a possibility of a heat related illness occurring. The most common factors are age, weight/fitness, drug/alcohol use, prior heat related illness, etc.

Control Measures

Implementation of control measures is the key to success. Start implementing the various control measures at a Heat Index of 80°F. A hygrometer should be located in the central area of the shop and either assign a hygrometer or have the Heat Index application installed for each New Installation and Modernization mechanic. If a hygrometer is opted for, it should be magnetic so they can be easily moved and hung on road boxes. See Appendix B for where to purchase. Next post the Heat Index Table (Appendix C) near the wall hygrometer and assign each New Installation and Modernization mechanic a laminated copy. Finally, assign a safety watch to monitor the temperature and implement the various heat stress controls, including:

Acclimation

Acclimation of employees to warm temperatures can help prevent heat-related problems. Employees must be introduced to the work environment in gradual increments. When you suspect a possible heat injury, allow employees time to cool off in cool or shaded areas and drink water. Refer to Appendix E for more details on Acclimation.

Jobsite Scheduling

Work with Building Owners and General Contractors to start work shifts earlier in the day and schedule strenuous job tasks during the coolest time of the day.

Water

The most important preventative measure on a hot day is cool water consumption. Encourage employees to drink water 15 minutes before beginning work and about one quart (4 cups, 2 bottles) per hour during their shift. *Employees shall have access to potable drinking water. Where water is not plumbed or supplied, provide water in sufficient quantity throughout the work shift.*

Employees should not wait to drink water until they feel thirsty. Consume water at regular intervals to avoid dehydration. Encourage employees to watch out for each other and be alert to extreme stages of heat-related problems.

Sports Drinks

Provide sports drinks such as Gatorade or Squencher (individual containers). If you provide “barrel” style coolers, check and clean them for sanitary conditions. You may fill these with powdered drinks.

Ice

Consider ice availability. Ice cubes are better than crushed ice. If possible, the Supervisor or MIC might fill the cooler at the beginning of the day with fresh ice.

Air Circulation

Provide air circulation with fans or spot coolers. It may also be possible to work with Building Owners or General Contractors to provide swamp coolers.

Clothing

Employees should wear lightweight, loose fitting, light-colored clothing. Employees must utilize appropriate PPE though and appropriate work attire throughout the shift (no shorts).

Heat Related Illness and First Aid

Heat stroke, the most serious form of heat-related illness, happens when the body becomes unable to regulate its core temperature. Sweating stops and the body can no longer rid itself of excess heat. Signs include confusion, loss of consciousness, and seizures. **Heat stroke is a medical emergency that may result in death! Call 911 immediately.**

Heat exhaustion is the body’s response to loss of water and salt from heavy sweating. Signs include nausea, dizziness, weakness, irritability, thirst, and heavy sweating.

Heat cramps are caused by the loss of body salts and fluid during sweating. Low salt levels in muscles cause painful cramps. Tired muscles – those used for performing work – are usually the ones most affected by cramps. Cramps may occur during or after work hours.

Heat rash, also known as prickly heat, is skin irritation caused by sweat that does not evaporate from the skin. Heat rash is the most common problem in hot work environments.

First Aid Procedures:

- Employees suffering from heat illness or believing that a preventative recovery period is required; provide access to an area with shade that is open to the air or with ventilation or cooling.
- Take the employee to a cool dry place and provide cool liquids to restore body temperature.
- Lie employee down to relieve physical or mental fatigue.

- Provide emergency medical treatment for heat stroke and severe cases of heat stress. For heat stroke, always remove the employee to a cool area, drench clothing with cool water and fan the body. Call 9-1-1!
- Use a thermometer to determine core temperature and ice packs to lower temperature while waiting on paramedics.
- The Company will provide CPR and First Aid Training to interested employees using an official formal entity like the American Red Cross or other such organization. Training will be maintained every two years.

Postings

Create awareness to remind employees to drink water, replenish electrolytes, eat healthy, avoid caffeine, etc. Place postings on the Safety Board, near the shop entrance, and anywhere else where the temperatures begin to warm up in the spring. At least the Heat Index Table should be posted near the hygrometer in the Shop.

Recordkeeping

Training records are required to be kept for three years. Any other prevalent records regarding the Heat Illness Prevention Program should also be kept for three years.

Training

Under no circumstance may an employee work under heat stress conditions until he/she has successfully completed the Axxiom Elevator training program under the Heat Illness Prevention Program. This includes all new employees, regardless of claimed previous experience.

Supervisors must receive training in the prevention of heat related illnesses prior to supervising employees working in the heat. Supervisors trained in heat illness prevention need to know what to do when an employee exhibits symptoms consistent with possible heat illness, including emergency response procedures.

The Department Manager is responsible for ensuring that all employees within their department have completed their Heat Illness Prevention training, although the actual training itself may be carried out by others.

The Axxiom training program includes Heat Illness Prevention and Heat Stress Reminders.

After an employee has completed the training program, the instructor will determine whether the employee can safely perform the job. All employees who receive training must be documented with either a paper roster or tracked within the Learning Management System. Each training roster must include the name(s) of the employee(s) trained, the date(s) of the training, and the instructor's name who administered the training.

Plan Evaluation

It is inherent that problems may occasionally arise in the Heat Illness Prevention Program. Although we may not be able to eliminate all problems, we try to eliminate as many problems as possible to improve employee protection and encourage employee safe practices. The Heat Illness Prevention Program will be evaluated annually by the Safety Manager and revised as necessary. This will include any received input from all levels of the organization.

Appendices

A – Heat Stress Program Activities

B – Hygrometer purchasing options

C – Heat Index Table Posting

D – Acclimation Process

E – Summary of Risk Levels and Associated Protective Measures

Revision Tracking Table

Date

Revision Description

Page # of Revision

Personal Protective Equipment

(Reference: Policy AXX-0026)

As is generally known, Personal Protective Equipment (PPE) is the last line of defense, or last barrier between a worker and a hazard. To that end, Axxiom Elevator shall ensure the PPE for eyes, face, head, extremities, protective clothing, respiratory devices, and protective shields and barriers, are provided, used, and maintained in a sanitary and reliable condition. All employees shall have their PPE on the jobsite at all times.

All PPE issued to employees shall be documented on Appendix A – *EE PPE Checklist* and sent to the Safety Manager.

Employee-Owned Equipment

Where employees wish to provide their own protective equipment, the Safety Manager is responsible for ensuring its adequacy, including proper maintenance, and sanitation. The employee will need written permission from the Safety Manager to utilize their own PPE. In all other cases, employees shall only use Company approved PPE.

Various PPE Components

PPE is made up for each part of the body. This section will delineate what is needed:

1. Proper Clothing

- Do not wear shorts, tank tops, or trousers with cuffs. Long sleeved shirts are recommended.
- Cotton over other fabrics is recommended when working around live electricity. However, understand that cotton is not true fire resistant clothing
- Wear class-specific reflective vest on jobsites where mandated or required. Follow current ANSI 107 standards for reflective vests.
- Hand tools and other objects shall not be placed in the pockets of trousers due to the possibility of snagging on moving objects or contacting energized equipment.
- Finger rings, loose jewelry and torn clothing are hazardous and shall not be worn.
- When working around energized equipment, conductive jewelry, metal-framed glasses, and large belt buckles shall be removed or protected to prevent contacting live components.
- Personal protective equipment shall not be traded or exchanged unless sanitized.
- Oil free clothing shall be worn when performing welding or cutting operations.
- Always wear appropriate clothing when welding, grinding, cutting or working around an open flame.

1. Eye and Face Protection

OSHA standards require employers to provide suitable eye and face protection based on anticipated hazards. Eye protection is required whenever on a jobsite, or performing work where there is a risk of eye injury, such as working with hazardous substances or airborne particles and flying debris.

Types of protection include:

1. Safety glasses with side protection (used for most situations).
2. Goggles (used when dust or chemical splash hazards exist)
3. Welder's Goggles – Used for gas cutting and welding operations (see filter table at the end of this section for proper shade and number)
4. Welder's Shield – Used for electric arc-welding work. Lens shade chosen as above and protected by a clear cover glass.
5. Face Shields – Shall be worn when entire face needs protection, (i.e., flying sparks, chemical splash, etc.)

All employees must possess eye protection on every job. All employees must wear eye protection 100% of the time on jobsites.

- Employees shall wear eye and face protection equipment when machines or operations present potential eye or face injury from physical, chemical, or radiant agents (i.e. grinding, drilling, chiseling, babbitting, welding, dusty and windy atmosphere, etc).
- Eye and face protection shall meet the requirements specified in ANSI Z87.1-1989.
- Eye and face protection shall be kept clean and in good repair. The use of this type of equipment with structural or optical defects is prohibited.
- Protection shall meet with the following requirements:
 - Be adequate for the hazard
 - Fit snugly
 - Not interfere with the wearer's movements
 - Be kept in good repair
- Non-metallic eye protection shall be worn while working on or near circuits or electrical apparatus.
- Employees are not permitted to wear tinted or shaded eyewear on the job (inside of buildings), except for proper welding and cutting shields. Do not use sunglasses on any jobsite.
- Contact lenses are not recommended at work.
- For prescription safety glasses, ensure the prescription is incorporated in its design, or the user wears eye protection that can be worn over the prescription lenses without disturbing the proper position of the prescription lenses or the protective lenses.

Filter lens shade numbers for protection against Radiant Energy

Welding Operation

Shade #

Shielded metal-arc welding 1/16, 3/32, 1/8, 5/32" diameter electrodes

10

Gas-shielded arc welding (nonferrous) 1/16, 3/32, 1/8, 5/32" diameter electrodes

11

Gas-shielded arc welding (ferrous) 1/16, 3/32, 1/8, 5/32" diameter electrodes

12

Shielded metal arc welding (ferrous) 3/16, 7/32, 1/4" diameter electrodes

12

5/16, 3/8" diameter electrodes

14

Atomic hydrogen welding

10-14

Carbon-arc welding

14

Soldering

2

Torch brazing

3 or 4

Light cutting, up to 1 inch

3 or 4

Medium cutting, 1 inch to 6 inches

4 or 5

Heavy cutting, over 6 inches

5 or 6

Gas welding (light), up to 1/8"

4 or 5

Gas welding (medium), 1/8" to 1/2"

5 or 6

Gas welding (heavy), over 1/2"

6 or 8

1. Occupational Head Protection

Employees working in areas where there is a possible head injury hazard (i.e. from falling or flying objects, or from electrical shock and burns) must wear a hard hat at all times.

All hard hats must meet ANSI Z89.1-1969 and Z89.2-1971.

Employees must wear hard hats during all phases of construction and modernization when the potential for head injury exists. Employees must also wear hard hats during service/repair when performing hoisting and rigging operations, and any time there is work being conducted overhead, regardless of conditions.

- Hard hats shall be worn with the suspension properly fitted.
- Before using your hard hat, inspect it for cracks or other damage.
- Do not store hard hats in direct sunlight.
- The application of decals, stickers, etc. is not recommended unless required by the jobsite General Contractor. Painting hard hats is not permitted.

1. Occupational Foot Protection

All Axxiom Elevator personnel must wear foot protection before going onto any jobsite, warehouse, or anywhere else where there is a risk of injury to your feet. The Company adheres to ATSM F2413-18 which calls for:

- Compression rating of at least C75
- Impact Resistance rating of at least 75 (I)
- Electrical Hazard protection (EH)
- Puncture Resistance protection (PR)
- Oil/Slip Resistant Soles

Metatarsal protection is not required.

The Company will reimburse eligible employees up to \$100 per year on qualifying footwear with receipt and proof of qualification.

1. Hearing Protection

Hearing protection will be provided and used whenever it is not feasible to reduce noise levels or duration of exposure below 85 dB on a time-weighted average.

- There is no cure for noise-induced hearing loss.
- Clean non-disposable earplugs after each use.
- Earmuffs need to make a perfect seal around the ear to be effective.
- For extremely noisy situations, wear earplugs and earmuffs.

As a rule of thumb if you must raise your voice to speak to others at a distance of 5 ft or less – hearing protection should be worn.

For more information on the Axxiom Elevator Hearing Conservation Program, see Policy # AXX-0033.

1. Personal Fall Arrest Equipment

The entire Axxiom Elevator Fall Protection policy can be found in Policy # AXX-0028.

- Personal fall arrest equipment shall be worn, and lanyards attached to lifelines at all times when working where there is a potential to fall more than 6 ft (unless the regulatory authority requires 4 ft).
- Personal fall arrest equipment shall be used only for employee safeguarding.
- Any lifeline, harness, anchor, or lanyard subjected to shock loading (disengages), as distinguished from static loading, shall be immediately removed from service and shall not be used again.
- Lifelines shall be secured above the point of operation to an anchorage or structural member capable of supporting a minimum dead weight of 5,000 lbs.
- Each employee is to be properly trained on proper use, maintenance and inspection of personal fall arrest systems.
- Lanyards shall be attached to lifeline or suitable anchorage such that the employee cannot free fall more than 6 ft, exceed calculated forces, nor contact any lower level.

1. Hand Protection

The correct glove for the job should be utilized, such as welding gloves for welding or chemical resistant gloves for working with hazardous chemicals. Employees are expected to bring gloves to all jobsites or other area where hand protection should be required. Proper hand protection shall be worn when:

- Handling material
- Using power tools
- Using impact tools
- Using cutting tools, except when terminating wires
- Handling solvents or chemicals
- Any other activity that presents a hazard
- When working on equipment covered under the Arc Flash policy (AXX-0017)
- And on any jobsite

Gloves and other types of hand protection can be obtained from the office when needed. Ask your supervisor when in need of hand protection.

1. Respiratory Protection

In general, elevator industry work environments do not pose a health risk from breathing contaminated air. However, if respiratory protection is required by jobsite conditions or building owner requirements, contact your supervisor for more information if needed.

To control those occupational diseases caused by breathing air contaminated with harmful dusts, fogs, fumes, mist, gases, smokes, sprays, or vapors, the primary objective shall be to eliminate the atmospheric contamination. This shall be accomplished as far as feasible by accepted engineering control measures (for example, enclosure or confinement of the operation, general and local ventilation, and substitution of less toxic materials). When effective

engineering controls are not feasible, appropriate respirators shall be used or the area shall be completely avoided. In all but a few rare instances, Company policy will direct employees to completely avoid these areas. For the rare instances where respirators will be required, refer to Policy # AXX-0027 Respiratory Protection for full details.

Dust Mask Usage:

For employees that wish to voluntarily wear a dust mask to protect themselves from dust or nuisance odors, the Company is required to train this group of employees on these masks. This training is laid out in Appendix D along with a form the employee must sign. If the employee refuses to sign, they will not be permitted to wear a dust mask. Once the training is completed and the form is signed, provide each employee with a copy.

Training

Employee training is an essential part of the PPE program. Training on all required PPE shall be performed on the employee's first day of employment (or before deployment to the field), and an annual refresher training. Training will also be carried out if an item of PPE changes, or functions in a different way due to a modification or issuance of a different model. Similarly, if employees are found to be using PPE incorrectly, retraining shall be carried out on the spot.

All training shall be documented with a roster including the trainee's name, date, instructor's name, and a brief description of the training performed.

In addition, when employees are issued PPE (Safety Bag), Appendix A - *EE PPE Checklist* shall be completed by the employee, with the immediate supervisor. If any critical items are missing or do not fit, the supervisor will order these items as soon as possible and the employee is not to enter the related hazard until the PPE is issued.

Recordkeeping

Training records must be maintained for a minimum of three years. The most recent record of training must always be kept on file, even if older than three years. Appendix A, and Appendix D must be kept on file with the Safety Manager, even after separation of employment.

Review and Update

This policy will be reviewed periodically to ensure its effectiveness and relevance. Updates will be made as needed to align with changes in regulations, technology, or work processes.

Revision Tracking Table

Date

Revision Description

Page # of Revision

Respiratory Protection

(Reference: Policy AXX-0027)

At Axxiom Elevator, we are dedicated to the safety and health of our employees. We have determined that employees may be exposed to atmospheric hazards in the work environment. Our primary objective is to prevent atmospheric contamination through the use of accepted engineering control methods. When engineering controls are not feasible, or while they are being instituted, appropriate respirators will be used in accordance with this plan.

This program has been designed to comply with OSHA's respiratory protection requirements set forth in 29 CFR 1910.134. A copy of these standards can be found in Appendix A of this program. Exposure to contaminants shall not exceed the threshold limit values adopted by the American Conference of Industrial Hygienists (ACGIH), or OSHA specified levels (whichever is more stringent). This plan will be maintained electronically and is available for review upon request.

This program applies to all employees who are required to wear respirators during normal work operations or during some non-routine or emergency operations, as well as employees who voluntarily use respiratory protection other than dust masks.

Responsibility

When it has been determined that respiratory protection is necessary to protect the safety and health of our employees, respirators will be provided to affected employees. Respirators which are applicable and suitable for the intended purpose will be provided by Axxiom Elevator at no cost to affected employees. In addition, any expenses associated with training, medical evaluations, and respiratory protection equipment will be borne by Axxiom Elevator.

1. Program Administrator – The Safety Manager

- Identify work areas, processes, or tasks that require workers to wear respirators
- Evaluate hazards & select respiratory protection options
- Monitor respirator use to ensure that respirators are used in accordance with their specifications
- Arrange for and/or conduct training
- Ensure proper storage and maintenance of respiratory protection equipment
- Conduct or arrange qualitative fit testing

- Administer the medical surveillance program
- Maintain records required by the program
- Evaluate and update the written program, as needed

1. **Supervisors**

- Ensure that employees under their supervision (including new hires) receive appropriate training, fit testing, and annual medical evaluation
- Ensure the availability of appropriate respirators and accessories
- Be aware of tasks requiring the use of respiratory protection
- Enforce the proper use of respiratory protection, when necessary.
- Ensure that respirators are properly cleaned, maintained, and stored according to this program
- Ensure that respirators fit well and do not cause discomfort
- Continually monitor work areas and operations to identify respiratory hazards
- Coordinate with the Program Administrator on how to address respiratory hazards or other concerns regarding this program

1. **Employees**

- Wear respirators when and where required and in the manner in which they are trained.
- Employees must also:
 - Care for and maintain respirators as instructed, guard them against damage, and store them in a clean, sanitary location
 - Inform the Supervisor if the respirator no longer fits well and request a new one that fits properly
 - Inform the Supervisor or the Program Administrator of any respiratory hazards that may not be adequately addressed in the workplace and of any other concerns regarding this program
 - Use the respiratory protection in accordance with the manufacturer's instructions and the training received

Axxiom Elevator employees are NOT to enter Immediately Dangerous to Life or Health (IDLH) areas for any reason. We do not possess the training nor the PPE to perform entries into these areas safely. If need be, reach out to a supervisor for clarification. Any entry to an IDLH atmosphere will be contracted out by the customer.

Exposure Assessment

An exposure assessment will be conducted whenever it is anticipated that our employees are exposed to hazardous atmospheres. Assessments will be performed whenever there is an OSHA standard that applies to a specific substance that our employees are exposed to, when work processes change, when employees report symptoms or complain of respirator effects, or when the workplace contains visible emissions such as fumes, dust, aerosols, vapors, gasses, or mists.

The exposure assessment will serve as a tool to categorize the nature and magnitude of employee exposures to respiratory hazards and will include:

- Identification of hazardous substances used on the jobsite by the customer or work process.
- A review of work processes to determine where potential exposures to hazardous substances may occur. This review will be conducted by surveying the jobsite, reviewing the process records, and talking with employees and Supervisors.
- Exposure monitoring to quantify potential hazardous exposures.

The exposure assessment will be conducted and reviewed by the Program Administrator to evaluate the potential hazard and arrange for outside assistance as necessary. The results of the exposure assessment will be communicated to all affected employees. If it is determined that respiratory protection is necessary, all other elements of the respiratory protection program will be in effect for those tasks/sites, and this respiratory program will be updated accordingly. Hazard evaluations will be maintained in Appendix B of this plan. At times, these assessments will be carried out by our customers requiring respiratory protection to complete our work on their jobsites. This is common for mines. The Program Administrator will request the results of customer-ordered assessments for their sites.

Controls

Engineering and administrative controls will be utilized to the extent feasible to prevent contamination of the jobsite atmosphere. When engineering controls are not feasible, or while engineering controls are being put in place, appropriate respirators must be used.

Engineering controls physically change the work environment to reduce employee exposure to air contaminants. Such engineering controls may include:

- Change of the work process
- Isolation or enclosure of the work process or of employees
- Local exhaust or general dilution ventilation
- Substitution of less hazardous substances for harmful materials

Administrative controls involve changes in the length of time or the time of day in which an employee can be exposed. Examples of administrative controls are:

- Employee rotation
- Rescheduling work in area to times when air contaminant levels are low

Respirator Selection

The Program Administrator shall identify and evaluate the respiratory hazard(s) in the workplace; this evaluation shall include a reasonable estimate of employee exposures to respiratory hazard(s) and an identification of the contaminant's chemical state and physical form. Where the employee exposure cannot be identified or reasonably estimated, the atmosphere will be considered Immediately Dangerous to Life & Health (IDLH).

Respirators, to be used on site, will be selected based on the hazards to which workers are exposed in accordance with the OSHA Respiratory Protection Standard. The proper type of respirator for the specific hazard involved will be selected in accordance with the manufacturer's instructions. A list of employees and appropriate respiratory protection will be maintained in Appendix C of this document. The program administrator will select and provide appropriate respirators to affected employees. The following guidelines will be followed when making this selection:

- Respirators will be selected based on the respiratory hazards that employees are exposed to along with workplace and user factors that affect respirator performance and reliability.
- All respirators will be certified by the National Institute for Occupational Safety and Health (NIOSH) and shall be used in compliance with the conditions of its certification.
- All filters, cartridges, and canisters must also be labeled with the appropriate NIOSH approval label. The label must not be removed or defaced while the respirator is in use.
- Respirators will be selected from a sufficient number of respirator models and sizes so that the respirator is acceptable to, and correctly fits, the user.

Selection of respirators will be made in accordance with the assigned protection factor (APF) of the respirator, as well as the workplace and exposure factors. The APF of a respirator reflects the level of protection that a properly functioning respirator would be expected to provide to a population of properly fitted and trained users. The APFs listed in Table 1 of 29 CFR [1910.134\(d\)\(3\)\(i\)\(A\) will](#) be used to select a respirator that meets or exceeds the required level of employee protection.

Types of Respirators Used

Air Purifying Respirators

- Particulate Respirator ("N-95" or "dust mask")
- Capture particles in the air, such as dusts, mists, and fumes
- Does not protect against gases or vapors
- Generally become more effective as particles accumulate on the filter and plug spaces between the fibers
- Filters should be replaced when user finds it difficult to breathe through them
- Combination Respirators
- Are normally used in atmospheres that contain hazards of both particulates and gases
- Have both particulate filters and gas/vapor filters
- May be heavier than other types of respirators
- Gas & Vapor Respirators
- Are normally used when there are only hazardous gases and vapors in the air
- Use chemical filters (called cartridges or canisters) to remove dangerous gases or vapors
- Does not protect against airborne particles

- Are made to protect against specific gases or vapors
- Provide protection only as long as the filter's absorbing capacity is not depleted
- The service life of the filter depends upon many factors and can be estimated in various ways

Respirator Use

1. Voluntary Respirator Use

Employees who wish to wear respirators on a voluntary basis must make a request to the Program Administrator or their direct supervisor. Voluntary use will be determined on a case-by-case basis, depending on specific workplace conditions and the results of medical evaluations.

All employees who choose to wear respirators voluntarily will be provided with a copy of Appendix D of the OSHA Respiratory Protection Standard, which details the requirements for voluntary use of respirators by employees. Employees who choose to wear a half face piece APR must comply with the procedures for medical evaluation, respirator use, cleaning, storage, and maintenance portions of this program.

Employees who voluntarily wear filtering face pieces (dust masks) are not subject to the medical evaluation, cleaning, storage, and maintenance provisions of this program. However, the following applies:

- Dust masks may only be worn by an employee on a voluntary basis for the purpose of comfort against nuisance-level particulates.
- All dust masks must be NIOSH certified.
- Dust masks must be stored in a bag or box or otherwise protected from dirt or contamination.
- All instructions provided by the manufacturer on use, maintenance, cleaning and care, and warnings regarding the dust mask's limitations must be read and adhered to
- Dust masks may not be worn into atmospheres containing contaminants for which the dust mask is not designed to protect against.
- Dust masks must be stored properly so that an employee does not mistakenly use someone else's dust mask.
- All dust masks should be replaced whenever they are damaged, soiled, or causing noticeable increased breathing resistance.

1. General Respirator Use Procedures

- Employees will use their respirators under conditions specified in this program, and in accordance with the training they receive on the use of each particular model. In addition, the respirator shall not be used in a manner for which it is not certified by NIOSH or by its manufacturer.

- Employees may work for reasonable periods of time in concentrations exceeding permissible levels only if they are protected by appropriate respiratory protective equipment.
- All employees shall conduct user seal checks each time they wear their respirators. Employees shall use either the positive or negative pressure check (depending on which test works best for them) as specified in the OSHA Respiratory Protection Standard.
- **Positive Pressure Test:** This test is performed by closing off the exhalation valve with your hand. Breathe air into the mask. The face fit is satisfactory if some pressure can be built up inside the mask without any air leaking out between the mask and the face of the wearer.
- **Negative Pressure Test:** This test is performed by closing of the inlet openings of the cartridge with the palm of your hand. Some masks may require that the filter holder be removed to seal off the intake valve. Inhale gently so that a vacuum occurs within the face piece. Hold your breath for ten (10) seconds. If the vacuum remains, and no inward leakage is detected, the respirator is fit properly.
- After notifying their Supervisor, employees shall be permitted to leave the work area to maintain their respirator for the following reasons:
 - to clean their respirator if it is impeding their ability to work;
 - to change filters or cartridges;
 - to replace parts; or
 - to inspect respirator if it stops functioning as intended.
- Employees are not permitted to wear tight-fitting respirators if they have any condition, such as facial scars, facial hair, or missing dentures that would prevent a proper seal.
- Employees are not permitted to wear headphones, jewelry, or other items that may interfere with the seal between the face and the face piece.
- Before and after each use of a respirator, an employee or immediate supervisor must make an inspection of tightness or connections and the condition of the face piece, headbands, valves, filter holders and filters. Questionable items must be addressed immediately by the supervisor and/or Program Administrator.

Medical Evaluations

Respirator use generally places a burden on the employee by restricting breathing, causing claustrophobia, or increasing neck strain from the weight of the respirator. Each of these conditions may adversely affect the health of some employees who wear respirators. Therefore, employees who are either required to wear respirators or who choose to wear a half face piece APR voluntarily must pass a medical exam before being permitted to wear a respirator on the job. Any employee who refuses the medical evaluation will not be allowed to work in an area requiring respirator use.

Medical evaluations will be obtained from a physician or other licensed health care professional (PLHCP). Evaluations will be conducted through the use of a medical questionnaire or an initial medical examination that obtains the same information as the medical questionnaire. All examinations and questionnaires are to remain confidential between the employee and

physician. Only the physician's written recommendations may be released to the company and retained in accordance with the Documentation & Recordkeeping section of this plan. A list of employees currently included in the medical surveillance program is provided in Appendix E of this program.

The following procedure will be followed to obtain medical evaluations:

- Affected employees will be provided with the OSHA medical evaluation questionnaire located in Appendix C of 29 CFR 1910.134 along with a stamped envelope addressed to the company physician. Employees will be permitted to complete the questionnaire confidentially during the employee's normal working hours or at a time and place convenient to the employee.
- If feasible, employees who are unable to read the questionnaire will be provided with assistance. When this is not possible, the employee will be sent directly to the PLHCP for a medical evaluation.
- If requested by the employee, they will be provided an opportunity to discuss the medical evaluation with the PLHCP.

Follow-up evaluations will be conducted for the following circumstances and will include any medical tests, consultations, or diagnostic procedures that the PLHCP deems necessary to make a final determination.

- The employee gave a positive response among questions 1 through 8 in Section 2, Part A of Appendix C
- The initial medical examination demonstrates the need for a follow-up medical examination.

The following information will be provided to the PLHCP before s/he makes a recommendation concerning the employee's ability to use a respirator:

- A copy of this written program
- A copy of the OSHA Respiratory Protection Standard
- The type and weight of the respirator to be used by the employee
- The duration and frequency of respirator use (including use for rescue and escape)
- The expected physical work effort
- Additional protective clothing and equipment to be worn
- Temperature and humidity extremes that may be encountered; and
- Any additional protective clothing required

A written recommendation regarding the employee's ability to use the respirator will be obtained from the PLHCP. This recommendation will be kept on file and will contain the following information:

- Any limitations on respirator use related to the medical condition of the employee, or relating to the workplace conditions in which the respirator will be used, including whether or not the employee is medically able to use the respirator;

- The need, if any, for follow-up medical evaluations; and
- A statement that the PLHCP has provided the employee with a copy of the PLHCP's written recommendation

Additional medical evaluations will be completed in the following circumstances:

- The employee reports signs and/or symptoms related to their ability to use the respirator, such as shortness of breath, dizziness, chest pains or wheezing.
- The evaluating physician or Supervisor informs the Program Administrator that the employee needs to be reevaluated.
- Information found during the implementation of this program, including observations made during the fit testing and program evaluation, indicates a need for reevaluation.
- A change occurs in workplace conditions that may result in an increased physiological burden on the employee.

Fit Testing

All respirators that rely on a mask-to-face seal need to be annually checked to determine whether the mask provides an acceptable fit to a wearer. Employees who are required to wear half-face piece APRs will be fit tested annually. Voluntary respirator-wearers may be fit tested at the company's discretion. Acceptable methods of fit testing include:

- **Qualitative:** A pass/fail fit test to assess the adequacy of respirator fit that relies on the individual's response to the test agent, (like smoke)
- OSHA has required standardized procedures
- Required upon issue and annually thereafter
- Irritant smoke, isoamyl acetate, saccharin solution, or bitrex solution can be used
- Usually required to recite the "Rainbow Passage"
- Has a subjective response
- **Quantitative:** An assessment of the adequacy of respirator fit by numerically measuring the amount of leakage into the respirator.
- OSHA has required standardized procedures
- Measures concentration inside mask
- Takes approximately 15 minutes per mask
- Requires expensive equipment and trained personnel
- Ensures correct fit for model

Employees will be fit tested with the make, model, and size of respirator that they wear. Several models and sizes of respirators will be available so that an optimal fit can be obtained for each person. Fit testing will be conducted in accordance with the OSHA Respiratory Protection standard at the following intervals:

- Prior to being allowed to wear any respirator with a tight-fitting face piece
- Annually

- When there are changes in the employee's physical condition that could affect respiratory fit (e.g., obvious change in body weight, facial scarring, etc.).

Respirator Care and Maintenance

Respirators are to be properly maintained at all times in order to ensure that they function properly and protect employees adequately. Maintenance involves a thorough visual inspection for cleanliness and defects before and after each use. Worn or deteriorated parts will be replaced prior to use. No components will be replaced, or repairs made beyond those recommended by the manufacturer.

1. Cleaning and Disinfecting

Each respirator user is initially provided with an individual respirator that is clean, sanitary, and in good working condition. Respirators must be cleaned and disinfected using OSHA's Respiratory Cleaning Procedures (Appendix B-2 to 29 CFR 1910.134) or by procedures recommended by the respirator manufacturer, as long as such procedures are of equivalent effectiveness.

The Program Administrator or supervisor will ensure an adequate supply of appropriate cleaning and disinfection materials are available. If supplies are low, the supervisor should be notified.

1. Storage

All respirators will be stored in such a manner as to protect them from damage, contamination, dust, sunlight, extreme temperatures, excessive moisture, and damaging chemicals. Respirators will be stored in accordance with the manufacturer's recommendations in order to prevent deformation of the facepiece and exhalation valve. Respirators will not be placed in places such as lockers or toolboxes unless they are in carrying cartons.

Each employee will clean and inspect their own air-purifying respirator in accordance with the provisions of this program and will store their respirator in a plastic bag in the designated area. Each employee will have his/her name on the bag and that bag will only be used to store that employee's respirator.

Respirator components will be stored in their original manufacturer's packaging in a safe location.

1. Respirator Malfunctions and Defects

Respirators that are defective or have defective parts shall be taken out of service immediately. If, during an inspection, an employee discovers a defect in a respirator, he/she is to bring the defect to the attention of his/her supervisor. Supervisors will give all defective respirators to the Program Administrator. The Program Administrator will decide whether to:

- Temporarily take the respirator out of service until it can be repaired;
- Perform a simple fix on the spot, such as replacing a head strap; or
- Dispose of the respirator due to an irreparable problem or defect.

Change Schedules

Respirator cartridges shall be replaced as determined by the Program Administrator, supervisors, and manufacturers' recommendations. The following guidelines shall be followed:

- Replace cartridges when the end-of-service-life indicator (ESLI) indicates that replacement is required (if respirator is equipped with an ESLI certified by NIOSH)
- Replace cartridges on a change schedule for canisters and cartridges based on objective information that will ensure that canisters and cartridges are changed before the end of their service life. That information shall be based on: (1) experimental tests, (2) manufacturer's limitations, or (3) use of a math model.

Training

The Program Administrator will provide training to respirator users and their supervisors on the contents of the Axxiom Elevator Respiratory Protection Program and their responsibilities under it, and on the OSHA Respiratory Protection Standard. All affected employees and their supervisors will be trained prior to using a respirator in the workplace. Supervisors will also be trained prior to supervising employees that must wear respirators. The training course will cover the following topics:

- the Axxiom Elevator Respiratory Protection Program;
- the OSHA Respiratory Protection Standard (29 CFR 1910.134);
- respiratory hazards encountered at Axxiom Elevator and their health effects;
- proper selection and use of respirators;
- limitations of respirators;
- respirator donning and user seal (fit) checks;
- fit testing;
- emergency use procedures;
- maintenance and storage; and
- medical signs and symptoms limiting the effective use of respirators.

Employees will be retrained annually or as needed (e.g., if they change departments or work processes and need to use a different respirator). Employees must demonstrate their understanding of the topics covered in the training through hands-on exercises and a written test (administered through a Learning Management System). Respirator training will be documented by the Program Administrator and the documentation will include the type, model, and size of respirator for which each employee has been trained and fit tested.

Program Evaluation

The Program Administrator will conduct periodic evaluations of the workplace to ensure that the provisions of this program are being implemented. The evaluations will include regular consultations with employees who use respirators and their supervisors, site inspections, air monitoring and a review of records. Items to be considered will include:

- comfort;
- ability to breathe without objectionable effort;
- adequate visibility under all conditions;
- provisions for wearing prescription glasses;
- ability to perform all tasks without undue interference; and
- confidence in the face piece fit.

Identified problems will be noted in an inspection log and addressed by the Program Administrator. These findings will be reported to Axxiom Elevator management, and the report will list plans to correct deficiencies in the respirator program and target dates for the implementation of those corrections.

Documentation and Recordkeeping

Copies of training and fit test records shall be maintained by the Program Administrator. These records will be updated as new employees are trained, as existing employees receive refresher training, and as new fit tests are conducted.

For employees covered under the Respiratory Protection Program, the Program Administrator shall maintain copies of the physician's written recommendation regarding each employee's ability to wear a respirator. The completed medical questionnaires and evaluating physician's documented findings will remain confidential in the employee's medical records at the location of the evaluating physician's practice. Records of medical evaluations will be retained and made available in accordance with 29 CFR 1910.1020.

Revision Tracking Table

Date

Revision Description

Page # of Revision

Lead Awareness

(Reference: Policy AXX-0030)

This program is instituted by Axxiom Elevator to protect employees subject to minor lead exposure. Lead exposure in any amount in the elevator industry is very rare. The information in this plan will serve to inform workers on identifying areas with possible lead, typical exposure possibilities within the elevator industry, health effects of lead exposure, and lead exposure prevention.

It is the policy of Axxiom Elevator that if lead (at any level) is confirmed or suspected in the areas we operate on any jobsite, that we will cease work until the lead has been properly encapsulated or removed and no lead (or very low levels below the Permissible Exposure Limit) is present in the air. Lead encapsulation and removal will be the responsibility of the customer/building owner. At no point shall Axxiom Elevator employees knowingly expose themselves to lead.

What is Lead?

Lead is a toxic heavy metal that occurs naturally in the earth's crust. It has been widely used in various products and industrial processes due to its durability, malleability, and resistance to corrosion. While beneficial in certain applications, lead is highly toxic to humans and animals when ingested or inhaled, even in small amounts.

Lead was commonly used in gasoline, piping, solder, printing, shielding and water-proofing products. Lead was also used in paints and coatings until as recently as the 1980's. Lead is still used today in smelting operations, lead batteries, and some auto manufacturing products.

Typical Occupational Exposure to Lead

Typically, during construction activities, lead exposure can happen during:

- Demolition or salvage of structures containing lead
- Removal or encapsulation of lead materials (scraping, heating, sanding, grinding, blasting)
- Alteration, repair or renovation of structures containing lead
- Transportation, disposal or cleanup of lead materials
- Maintenance operations associated with construction activities

Ways in Which Lead Enters the Body

- **Inhalation:** Breathing lead fumes or dust. This is the most common route of entry in the workplace.
- **Ingestion:** Swallowing lead dust via food, cigarettes, etc.

Health Effects

Lead exposure can have both acute (short-term) and long-term (chronic) health effects, depending on the level of exposure, the duration of exposure, and the individual's age and health status. Lead is particularly harmful to children, as their developing bodies are more susceptible to its toxic effects.

Below is a summary of effects lead exposure can have on the various systems of the body:

System

Acute Effects

Long-Term Effects

Neurological

Headaches, confusion, seizures

Cognitive decline, memory loss, behavioral issues, learning disabilities (children)

Gastrointestinal

Abdominal pain, vomiting

Chronic constipation, digestive issues

Cardiovascular

Elevated blood pressure

Chronic hypertension, increased risk of heart disease

Reproductive

Sperm abnormalities, miscarriage

Reduced fertility, birth defects

Renal (Kidneys)

Kidney damage

Chronic kidney disease, renal failure

Hematological (Blood)

Anemia

Chronic anemia, fatigue

Musculoskeletal

Muscle pain, weakness

Reduced bone density, osteoporosis

Lead exposure, whether acute or chronic, is dangerous and can result in severe health consequences across multiple systems in the body. The key to managing lead exposure is prevention, early detection, and minimizing contact with lead in homes, workplaces, and the environment. Children and pregnant women are particularly vulnerable, and even low-level exposure can have lifelong effects.

Elevator Industry Lead Exposure

Lead exposure is exceedingly rare in the elevator industry, but it is still possible. Below are some activities that can expose a worker to lead.

1. Babbitting

Babbitt contains lead. The primary route of exposure is through the inhalation of fumes, though ingestion can be significant if hands are not washed before eating or smoking. Repeated overexposure can result in elevated concentrations in the body, which is slowly eliminated when exposure ceases. Significant lead fumes are only released when the lead approaches its boiling point, over 1000°F which is well above the temperature used in babbitting operations, around 400°F. Adequate ventilation shall be provided along with the following prevention methods:

- Use wedge shackles whenever possible.
- Protective face shields and gloves shall be worn while pouring babbitt.
- Preheat the bearing housing or shackle to be poured to be sure that it is dry. The presence of moisture will form captured steam and the hot babbitt will explode.
- Avoid breathing fumes. Melt and pour babbitt in well-ventilated areas.
- Wash hands prior to eating or smoking after handling babbitt.
- Do not use cutting torches to melt babbitt. An electric melting pot shall be used, as it will not allow babbitt temperatures to exceed a safe level.
- Care shall be taken when using resins for socketing. Use only Company-approved heaters for curing. Do not allow the material to come into contact with exposed skin. Do not store in direct sunlight.

1. Welding

Welding produces fumes from the metals being joined and the welding rods being used. In the elevator industry, iron oxide fumes are the primary exposure encountered. The recommended

exposure level is 5 mg/m³ (even in industrial environments where welding operations are continuous this level is seldom exceeded). In cases where welding is to be performed on painted surfaces, the paint shall first be removed in case the paint contains lead, which can rapidly vaporize during the welding process. The approved method of removing paint (only in the areas to be welded) is to grind//sand/chip with a HEPA vacuum going continuously at the point of operation. Adequate ventilation must always be provided.

General Exposure Prevention

- Use exhaust ventilation to capture dust/fumes whenever possible
- HEPA vacuum dust covered work surfaces (DO NOT use compressed air) or wet methods may be used
- Do not eat, drink, or smoke in areas where lead is present
- Wash hands and face after work
- Wear protective clothing to avoid getting dust on your clothes and then bringing it home to spouse and children

If Lead is Suspected on Jobsite

If lead is suspected on a jobsite, contact your supervisor immediately before resuming work. The supervisor will contact the customer/building owner and can have the area you need to work in or on tested and mitigated if need be.

Recordkeeping

Training records are required to be kept for three years. Any other prevalent records regarding the Heat Illness Prevention Program should also be kept for three years.

Training

All field employees will be trained in lead awareness at hire and annually. This training shall be in lead awareness only.

Revision Tracking Table

Date

Revision Description

Page # of Revision

Fall Protection Program

(Reference: Policy AXX-0028)

Fall protection is vital in the elevator industry to ensure worker safety, prevent serious accidents, and comply with safety regulations. A fall from height can result in catastrophic injuries, including head trauma, spinal injuries, or even fatalities. Proper use of fall protection systems and protocols can significantly reduce the risk of injury or death in these high-risk environments.

The purpose of the Fall Protection policy is to establish guidelines and procedures that protect employees from fall-related injuries and fatalities in the workplace. This policy outlines the necessary measures to prevent falls, reduce risks, and ensure compliance with safety regulations. Specifically, the key objectives of the Axxiom fall protection policy include:

1. **Preventing Falls:** The primary goal is to prevent falls from heights or elevated surfaces by implementing effective safety measures, such as guardrails, safety nets, harnesses, and fall arrest systems.
2. **Promoting Workplace Safety:** By establishing clear protocols for identifying fall hazards and using fall protection equipment, the policy fosters a culture of safety and ensures that both employees and managers take proactive steps to minimize risks.
3. **Complying with Regulations:** The Axxiom fall protection policy helps the Company comply with occupational safety regulations such as OSHA standards, and other state and local regulations.
4. **Clarifying Responsibilities:** This policy defines the roles and responsibilities of employees, supervisors, and safety personnel in relation to fall protection. This ensures that everyone understands their duties, such as wearing PPE, conducting inspections, and providing training.
5. **Providing Training and Education:** This policy ensures that workers receive the necessary training on how to recognize fall hazards, properly use fall protection equipment, and respond to fall incidents. This promotes worker awareness and competence in safety practices.

Responsibilities

Program Administrator – The Safety Manager

- Conduct continual observational safety checks of work operations
- Enforce safety policies and procedures
- Designate a Competent Person(s) at each Branch
- Approve changes or revisions to this plan

Management

- Provide fall protection systems for employees in work areas where injury from a fall to a lower level is a recognized hazard
- Conduct continual observational safety checks of work operations
- Enforce safety policies and procedures

Supervisors

- Correct any unsafe acts or conditions immediately
- Carry out training for new employees (and recurring annual training)
- Approve Site-Specific Fall Protection Plans

Employees

- Follow all Fall Protection site plans and policy
- Bring any unsafe/hazardous conditions or acts to management's attention in order to prevent injury to either themselves or any other employees
- Utilize Stop Work Authority for any perceived unsafe conditions

General Requirements

Fundamental Rule: Always use fall protection when a fall hazard exists.

Fall protection is required when a worker is exposed to a fall hazard, which exists when work takes place more than 6 ft. (unless the regulatory authority requires 4 ft.) above a lower level and an opening of 12 in. by 12 in. or greater and work activity places the worker within 6 ft. of the opening.

There are three ways of controlling fall hazards. Elimination of the fall hazard should be the first consideration. The second consideration is a guard rail system, and the third is a personal fall arrest system.

Controlling Fall Hazards

Elimination of fall hazards should always be the first consideration and the preferred option by covering all holes or openings.

1. Guardrails

A guardrail system must be used as the primary protection against falls. For barricades to be considered an adequate fall prevention system OSHA compliant guardrail system for car tops, open hoistways or escalator wellways must have:

- A top rail that is 42 in. $\pm$ 3 in. above the walking/working level
- A mid-rail at 21 in. above the walking/working level
- A toe board 3.5 in. high

- No greater than 8 ft. between uprights
- Top rail must be capable of sustaining a force equal to 200 lbs.
- Mid-rail must be capable of sustaining a force equal to 150 lbs.
- Toe board must be capable of sustaining a force of equal to 50 lbs.
- When 200 lbs. is applied to the top rail, the rail shall not deflect lower than 39 in.

1. **Construction Guardrails**

- OSHA Compliant removeable guardrail systems with toe boards must be installed at elevator hoistways or escalator wellways (typically by the General Contractor) after either rough or finished floors are in place.
- Signs must be installed warning against removal. It is also recommended that a sign indicate "Caution: Workers in Hoistway" be posted.
- After hoistways are enclosed, and before permanent doors are installed, openings must be protected by removeable guardrail systems (including toe boards).
- If it is necessary to remove the guardrails, be sure to replace them before leaving the area. When a guardrail is removed to perform a job, a personal fall arrest system must be utilized when a fall hazard is present.
- Wire-rope guardrail systems are not recommended for guarding hoistways. Where used, post spacing shall not be greater than 8 ft. and they shall not deflect to a height less than 39 in. above the walking/working level when a force of 200 lbs. is applied. Warning flags shall be attached at least every 6 ft., toe boards shall be provided, and they must be easily removeable for access to the hoistway at the terminal landings. Wire rope barricades typically sag and do not allow adequate protection. Given no choice regarding the use of wire rope barricades, a written JHA must be conducted, and a process (which includes daily inspections) must be implemented.
- If guardrails are not properly maintained notify your supervisor and the General Contractor immediately.
- On new installation, modernization, or major repair jobs where the general public is present, solid barricades at least 8 ft. high shall be used to fully enclose the work areas, open hoistways and escalator wellways. They must be properly secured to avoid unauthorized access.
- The methods shown below are recommended as a means of providing maximum protection and flexibility during construction. Do not use during modernization or major repair jobs where the general public is present – use only solid barricades.
- No materials or equipment shall be stored within 6 ft. of working edges.

1. **Cartop Guardrails**

Many cartops are equipped with guardrail systems. Never climb over or stand on guardrails. Be aware of pinch hazards and the risk of being caught between a guardrail and hoistway equipment. Below is a typical OSHA compliant cartop guardrail system:

1. **Escalator/Moving Walk Barricades**

- Barricades shall be positioned to completely surround the escalator/moving walk from public access.
- Barricades shall be positioned to surround, from public access, floor opening(s) created when equipment access plate(s) are removed.

- Barricades shall be a minimum of 42 in. high.
- Barricades must be securely attached to the balustrades, handrails, and/or floor.
- All sections must be connected.
- A system must be in place to keep the barricade rigid (see below).

1. **Elevator Maintenance Barricades**

- Barricades shall be positioned to restrict public access to the hoistway when doors are open greater than 5 in.
- Barricade shall be a minimum of 42 in. high.
- Barricade shall cover the entire entrance area.
- All sections must be connected.
- A system must be in place to keep the barricade rigid. See example below.

1. **Personal Fall Arrest System**

Depending on the application, a fall-arrest system typically consists of a full body harness, connectors, a shock absorber, and lanyard attached by a lifeline to an adequate anchor point.

Note: Body belts are prohibited and not to be used in place of a full-body harness.

1. **Fall Arrest System Performance Requirements**

Personal fall-arrest systems (PFAS) shall perform as follows:

- Limit maximum arresting force on the employee's body during a fall to 1,800 lbs. when used with a body harness.
- The system should be rigged so that the user cannot contact any lower level or free fall more than 6 ft.
- The system should bring an employee to a complete stop and limit maximum deceleration distance an employee travels to 3.5 ft.
- The system must have sufficient strength to withstand twice the potential impact energy of an employee free-falling a distance of 6 ft. or the free-fall distance permitted by the system, whichever is less.
- The anchorage connector must be attached to a suitable and strong attachment point.

1. **Person Fall Arrest System Rules**

1. Only Company approved lifelines, shock-absorbing lanyards and body harnesses shall be used.
2. All fall protection components shall be compatible.
3. Lifelines shall be protected against being cut or abraded. Only synthetic or wire rope shall be used for lifelines.
4. Lifelines shall be installed before working in the hoistway/wellway and shall run the full length of the hoistway/wellway. Prior to use, the fall arrest system shall be approved by a Competent Person.
5. Only one worker is permitted on a vertical lifeline, and that lifeline shall have a breaking strength greater than 5,000 lbs. after it has been attached to the anchorage point.
6. Shock-absorbing lanyards shall be anchored to the lifeline and shall be above shoulder height so that any fall shall not exceed 6 ft. When determining fall heights be sure to use

proper fall clearance distances. Lanyards shall be connected to a vertical lifeline by means of a rope grab; the lanyard shall not be attached directly to the lifeline.

7. Lifelines, harnesses, and shock-absorbing lanyards subjected to impact shall be immediately removed from service. They shall be eliminated and destroyed for employee safeguarding.
8. Tying to the hoist line is prohibited. Proper rope grab shall be used.
9. Fall protection shall be used on top of a completed elevator car when there is a fall hazard, and the car is secured from movement.
10. On a completed car secured from movement, tie off when exposed to a fall hazard. Do not position yourself where there is a fall hazard.
11. Replace personal fall protection at intervals recommended by the manufacturer.
12. Before using a body harness, shock-absorbing lanyard and lifeline, inspect them carefully each time for signs of wear or damage.

1. **Personal Fall Arrest System Inspection**

To maintain service life and high performance, personal fall arrest systems must be inspected before each use. Replace the personal fall arrest system if any defective conditions are found. An in-person inspection of personal fall arrest equipment will also be done annually at the PPE Check by another Competent Person.

1. **Webbing:** Grasp the webbing with your hands 6 in. to 8 in. apart. Bend the webbing in an inverted "U". The resulting surface tension makes damaged fibers or cuts easier to see. Follow this procedure the entire length of the webbing, inspecting both sides of each strap. Watch for frayed edges, broken fibers, pulled stitches, cuts, burns, and chemical damage.
2. **D-Rings/Back Pads:** Check D-Rings for distortion, cracks, breaks, and rough or sharp edges. The D-Ring should pivot freely. D-Ring back pads should also be inspected for damage.
3. **Attachment of Buckles:** Attachments of buckles and D-Rings should be given special attention. Note any unusual wear, frayed or cut fibers, or distortion of the buckles or D-Rings.
4. **Tongue/Grommets:** The tongue receives heavy wear from repeated buckling and unbuckling. Inspect for loose, distorted or broken grommets. Webbing shall not have additional punched holes.
5. **Tongue Buckle:** Buckle tongues shall be free of distortion in shape and motion. They should overlap the buckle frame and move freely back and forth in their socket. The roller shall turn freely on the frame. Check for distortion, sharp edges or grip failure.
6. **Friction and Mating Buckles:** Inspect the buckle for distortion. The outlet bars and center bars must be straight. Pay special attention to corners and attachment points of the center bar.

1. **Visual Indications of Damage**

The following indications refer to nylon and polyester webbing:

1. **Heat:** In excessive heat, webbing becomes brittle and has a shriveled brownish appearance. Harnesses made of these materials should not be used at temperatures above 180°F.
2. **Chemical:** Changes in color usually appearing as a brownish smear or smudge. Transverse cracks appear when bent over a mandrel. Loss of elasticity.
3. **Molten Metal or Flame:** Webbing strands fused together. Hard shiny spots appear. Hard and brittle feel.
4. **Paint and Solvents:** Paint that penetrates and dries restricts movement of the fibers. Drying agents and solvents in some paints cause chemical damage.

1. **Care and Cleaning of the Harness and Lanyard**

Basic care for fall protection safety equipment will prolong and endure the life of the equipment and contribute toward the performance of its vital safety function. Proper storage and maintenance after use is as important as cleaning the equipment of dirt, corrosives or contaminants. The storage location should be clean, dry and free of exposure to fumes, corrosive elements, and sunlight which can cause UV damage.

Nylon and Polyester: Wipe off all surfaces with a sponge dampened in plain water. Squeeze the sponge dry. Dip the sponge in a mild solution of water and commercial soap or detergent. Work up a thick lather with a vigorous back and forth motion. Then wipe the belt dry with a clean cloth. Hang freely to dry but away from excessive heat, steam or long periods of sunlight.

1. **Managing the Harness and Lanyard**

Part of wearing a fall arrest harness is the management of the harness and lanyard. Throughout the day the harness may become loose, or the lanyard may become free of its storage position creating a potential snagging hazard similar to loose clothing. At a minimum, the following activities must be followed:

1. Every time your harness is put on, confirm the harness is properly connected and adjusted so that the straps fit close to your body and the ends of the adjusting straps are secured by the retaining clips in such a manner as not to cause a snagging potential. Always inspect your harness and lanyard thoroughly prior to wearing. If any straps or buckles are suspect, contact your supervisor for replacement. (The wearing of disconnected leg straps is not permitted.)
 1. Some harness manufacturers require periodic inspections and have inspection tags on the harness. If your harness contains an inspection tag, the tag can be checked off after a thorough inspection of the harness and lanyard.
1. The lanyard must be stowed in such a way as not to have loose loops which may swing around while walking or moving. Also, the lanyard must be stowed in such a way that it does not hang away from the body when bending down or stooping and causing a snagging hazard.
2. When working in an area with exposed electrical equipment such as an elevator controller or an area with rotating equipment, or areas of restricted passage such as

passing through scaffold; always check the fall arrest system equipment to see that it is fastened close to the body before entering or passing through those areas.

3. Lifelines

Approved lifelines should be secured and padded around all corners and edges. If a typical hoistway anchorage point is not present on your job, a written JHA must be performed to consider support possibilities. It is preferred that lifelines be secured to independent overhead supports separate from false car supports (e.g., stop block cables on false cars are not to be used as lifelines).

- One lifeline per person is required.
- Two lifelines per hoistway are required.
- Lifelines may only be removed after the elevator cab has been installed.

1. Fall Protection PPE Requirements

Construction

On construction jobs, a personal fall arrest system must be worn at all times. The lanyard must be anchored at all times when there is exposure to a fall of 6 ft. or more and where guardrails are not present. When working at the edge of the building, typically for receiving material, employees must use appropriate fall protection.

Exceptions:

- The project will no longer be considered a construction site when the contractor has obtained a final occupancy permit.
- All elevators/escalators have been placed on final acceptance.
- A “handover” to the service department has been completed and construction work is 100% completed on all elevators/escalators.

Modernization

Each modernization project must be assessed using a Job Hazard Analysis. Fall protection must be utilized when a fall hazard exists. During hoistway work activity, harness should be available on the car top even with compliant guardrails in the event work activity extends beyond the guardrails.

Service, Repair and Open Order

Harness and lanyard must be worn and anchored when a fall hazard exists. This includes all work activities such as inspections, surveys and observations.

1. Fall Clearance Distance Calculation

The total fall clearance distance is the minimum vertical distance between the worker and the lower level to ensure the worker doesn't come into contact with the lower level during a fall.

To calculate the fall clearance distance, use the following formula:

Fall clearance distance = length of lanyard + deceleration distance + height of worker + safety factor

Here are some things to consider when calculating fall clearance distance:

- Deceleration distance: The distance it takes for the lanyard or SRL to activate and stop the fall. OSHA limits this distance to 3.5 feet or less.
- Safety factor: This accounts for things like an improperly fitted harness, a taller-than-average worker, or a miscalculation of distance. A safety factor of 3 feet is typically added.
- Lanyard length: Do not tie a knot in the lanyard to make it shorter, as it can impact the strength of the lanyard.
- Anchor point placement: The placement of the anchor point can affect the free fall distance calculation.
- Potential hazards: Other things that a worker can contact during a fall, like material in the hoistway, swing hazards, or pit and door equipment, can significantly reduce the total clearance height.

1. **Fall Rescue Plan**

A rescue plan is our strategy or procedure, planned in advance, to safely retrieve a person who has fallen from an elevated work surface and is suspended in a full body harness. This includes self-rescue or mechanically aided rescue. Prior to engaging in work, a rescue plan must be discussed with all workers because suspension trauma can occur. Suspension trauma is a serious medical condition that can lead to unconsciousness, injury, or death, which can occur when a worker is suspended in a harness for too long after a fall.

Self-Rescue can be aided by the use of a trauma relief pack attached to the harness that can be deployed after a fall. The worker can use the pack to attach footholds to relieve pressure from the harness on the legs. 90% of workers will be able to perform a self-rescue which should include these steps:

1. Climbing back up to the level from which he fell (from a few inches to 2-3 feet).
2. Returning to the floor or ground to be evaluated for possible medical attention.
3. Removing all components of the fall arrest system impacted by the fall event and bring equipment to management.

Assisted Self-Rescue may be required. The rescuer(s) are not to put themselves in danger. The rescue team may raise or lower the victim as needed with mechanical assistance. Once retrieved, the victim should be evaluated for the need of medical attention.

If the worker that fell is unconscious or medically disabled, 9-1-1 must be called immediately as this will require a Mechanically Aided Rescue.

Holes/Covers

A hole is defined as a void or gap, 2" or more, in the least dimension in a floor, roof, or other walking/working surface. Any hole not being utilized at any time must be covered. Covers in floors, roofs, and other walking/working surfaces shall meet the following:

- Covers located in roadways and vehicular aisles shall be able to support at least twice the maximum axle load of the largest vehicle to which the cover might be subjected.
- All other covers shall be able to support at least twice the weight of employees, equipment and materials that may be imposed on the cover at any one time.
- To prevent accidental displacement resulting from wind, equipment, or workers' activities, all covers shall be secured.
- All covers shall be color coded or bear the markings "HOLE" or "COVER".

Fall Protection on Top of a Stationary Car

Fundamental Rule: Always attach and detach your lanyard from the car top and never from the landing.

Tying off on car tops is allowed providing the elevator is locked and tagged, or control of the car has been tested and verified as described below.

If the elevator is not locked out, control of the elevator must be maintained at all times when fall protection is in place. Fall protection lanyards must never be attached/secured when the control of the car has not been established, tested and verified.

Lanyards must be connected in a manner to prevent snagging. Never attach your lanyard to the car top while standing on the landing. Never step from the car top to the landing while your lanyard is connected.

When you have reached the area where work is to be performed, stop the car and gain control to prevent inadvertent movement (e.g., inspection control and the stop switch). Other means of control can be utilized, such as broken tape switch, car or landing gate, etc. These devices must be tested and verified prior to use.

Fall Protection on a Moving Car

In general, lanyard should not be connected to a moving elevator.

Exceptions: Some elevators may have a curved or sloping car top or may be in open buildings such as scenic/observation elevators and may require fall protection while moving the elevator.

In these circumstances a written JHA must be completed and approved by the immediate supervisor. The JHA at a minimum must address the following:

- Prevention of unexpected movement of the car on normal operation (e.g., In-Car Inspection and Car Top Inspection; Machine Room Inspection and Car Top Inspection, etc.).
- Managing the lanyard so that inadvertent snagging does not occur.
- It is recommended, where possible, that cars be operated in the down direction when lanyards are connected.

Deep Pit Fall Protection

Fall protection means must be used to access/egress the pit in the following cases:

- Access to the ladder exposes a person to a fall hazard 6ft. or greater AND the ladder is farther than 29.5 in. from the interior edge of the door frame
- Access exposes a person to a fall hazard and the ladder (or handholds) extends less than 42 in. above the access landing.

Fall protection can be mitigated by improving the height with additional handholds or relocating ladders to meet requirements. If installing handholds or relocating ladders is not feasible, fall protection systems must be installed.

Definitions

Anchorage - A secure point of attachment for lifelines, lanyards or deceleration devices.

Body Harness - Straps that may be secured about the person, in a manner that distributes the fall- arrest forces over the thighs, pelvis, waist, chest, and shoulders, with a means for attaching the harness to other components of a personal fall-arrest system.

Connector - A device that is used to couple (connect) parts of a personal fall-arrest system or positioning device system together.

Deceleration Device - Any mechanism, such as rope, grab, rip stitch lanyard, specially woven lanyard, tearing or deforming lanyards, automatic self-retracting lifelines/lanyards; that serves to dissipate a substantial amount of energy during a fall-arrest, or otherwise limits the energy imposed on an employee during fall-arrest.

Deceleration Distance - The additional vertical distance a falling person travels, excluding lifeline elongation and free-fall distance, before stopping, from the point at which a deceleration device begins to operate.

Guardrail System - A barrier erected to prevent falls to lower levels.

Hole - A void or gap, two (2) inches or more, in the least dimension in a floor, roof, or other walking/ working surface.

Lanyard- A flexible rope, wire rope, or strap that has a connector at each end for connecting the body harness to a deceleration device, lifeline, or anchorage.

Leading Edge - The edge of a floor, roof, or formwork for a floor, or other walking/working surface (such as the deck), which changes location as additional floor, roof, decking, or formwork sections are placed, formed or constructed.

Lifeline A component consisting of a flexible line for connection to an anchorage at one end to hang vertically (vertical lifeline), or for connection to anchorages at both ends to stretch horizontally (horizontal lifeline) and that serves as a means for connecting other components of a personal fall- arrest system to the anchorage.

Opening - A gap or void, 30 inches or more high, and 18 inches or more wide, in a wall or partition, through which employees can fall to a lower level.

Personal Fall-Arrest System (PFAS) - A system including, but not limited to an anchorage, connectors, and a body harness used to arrest an employee in a fall from a working level. As of January 1, 1998, the use of a body belt for fall-arrest is prohibited.

Self-Retracting Lifeline/Lanyard - A deceleration device containing a drum-wound line, which can be slowly extracted from, or retracted onto the drum, under minimal tension, during normal employee movement and which, after onset of a fall, automatically locks the drum and arrests the fall.

Snap-hook - A connector, consisting of a hook-shaped member with a normally closed keeper, or similar arrangement, which may be opened to permit the hook to receive an object and, when released, automatically closes to retain the object.

Toe Board - A low, protective barrier, that prevents material and equipment from falling to lower levels, which protects personnel from falling.

Unprotected Sides and Edges - Any side or edge (except at entrances to points of access) of a walking/working surface (e.g. floor, roof, ramp, or runway), where there is no wall or guardrail system at least 39 inches high.

Enforcement

Constant awareness of and respect for fall hazards, and compliance with all safety rules are considered conditions of employment. Supervision and management as well as the Safety Manager reserve the right to issue disciplinary warnings to employees, up to and including termination, for failure to follow the guidelines of the safety policy. If an employee finds it more hazardous to follow the fall protection measures, they must contact supervision for guidance.

Recordkeeping

Training records are required to be kept for three years. Any other prevalent records regarding the Fall Protection Program should also be kept for three years. Job Hazard Analyses concerning site-specific fall protection plans shall be kept with the job file.

Training

All field employees will be trained in fall protection at hire and annually. At the time of initial training, employees will be issued fall protection PPE where the employee will show they know how to don and doff the PPE, inspect the PPE, and ensure proper fit before going to the field.

Revision Tracking Table

Date

Revision Description

Page # of Revision

Confined Space

(Reference: Policy AXX-0029)

Axxiom Elevator is committed to safe entry and work within confined spaces. This policy will establish a framework and set guidelines for safe confined space work. The objectives of the Confined Space Policy are listed below.

POLICY OBJECTIVES

1. **Safety Assurance:** The primary purpose is to prioritize the safety and well-being of our employees working in or around confined spaces. This policy outlines safety measures, procedures, and precautions to minimize risks associated with confined space entry.
2. **Regulatory Compliance:** Axxiom Elevator will comply with all applicable local, state, and federal regulations governing confined space work. This includes hazard assessment, training, permits, and emergency response planning.
3. **Risk Mitigation:** By identifying and classifying confined spaces, conducting thorough hazard assessments, and implementing control measures, this policy aims to mitigate the risks associated with confined space entry. This includes addressing atmospheric hazards, physical hazards, and other potential dangers.
4. **Consistent Procedures:** This policy establishes consistent and standardized procedures for entering confined spaces. This includes obtaining entry permits, conducting pre-entry checks, implementing isolation measures, monitoring atmospheric conditions, and developing rescue plans.

5. **Training and Competency:** This policy outlines the required training programs to ensure that individuals are competent in recognizing and addressing confined space hazards.
6. **Communication Protocols:** Clear communication is crucial during confined space entry. This policy outlines communication protocols among team members, including entry supervisors, attendants, and entrants. Effective communication is essential for coordinating entry activities and responding to emergencies.
7. **Emergency Response Planning:** This policy will address emergency response planning prior to confined space entry.
8. **Documentation and Recordkeeping:** This policy will outline proper documentation of all aspects of confined space entry, including entry permits, hazard assessments, training records, and incident reports. Documentation is essential for compliance, auditing, and continuous improvement.
9. **Continuous Improvement:** This policy encourages a culture of continuous improvement by specifying the need for regular reviews and updates. Lessons learned from incidents, changes in regulations, or improvements in safety practices are incorporated into this policy to enhance safety measures.

Roles and Responsibilities

Management: It is the responsibility of management to provide the necessary resources, training, and oversight to ensure compliance with this policy.

Supervisors: Supervisors are responsible for enforcing safety protocols, conducting regular inspections, and addressing any non-compliance promptly.

Employees and Contractors: All personnel are expected to follow established safety procedures and report any unsafe conditions or incidents promptly.

Confined Space Definition

A confined space is defined by specific criteria that involve the size, access, and hazards present within the space. The Occupational Safety and Health Administration (OSHA) provides a widely recognized definition. According to OSHA, a confined space is a space that:

1. Is Large Enough for Entry and Work:
 - The space must be large enough and configured in such a way that an employee can bodily enter and perform assigned work.
1. Has Limited or Restricted Means for Entry or Exit:
 - There must be limited or restricted means for entry or exit. This can include small openings, tight passageways, or other physical configurations that could impede the ability to enter or exit quickly.
1. Is Not Designed for Continuous Employee Occupancy:

- The space is not designed for continuous employee occupancy. This means that the primary purpose of the space is not for people to work in on a regular basis.

Additionally, a confined space may pose specific hazards, which could include:

- Atmospheric Hazards:
 - The presence of hazardous gases, lack of oxygen, or other atmospheric conditions that could be harmful to health.
- Physical Hazards:
 - Mechanical or physical conditions, such as moving parts, that could pose a threat to the safety of workers.
- Engulfment Hazards:
 - The potential for the space to contain materials that could engulf a person entering the space.

It is important to note that not all confined spaces are considered permit-required confined spaces. A permit-required confined space has characteristics that are listed below.

1. ELEVATOR PITS AS A CONFINED SPACE

An elevator pit is typically considered a confined space. Elevator pits are enclosed spaces that meet the criteria for a confined space because they are usually:

1. **Large enough for entry and work:**
 - Elevator pits are designed to be large enough for maintenance personnel to enter and perform necessary tasks.
2. **Limited or restricted means for entry or exit:**
 - Elevator pits often have limited access, with small openings or tight spaces that can impede the ability to enter or exit quickly.
3. **Not designed for continuous employee occupancy:**
 - Elevator pits are not designed for employees to occupy continuously; they are primarily intended for the operation and maintenance of the elevator equipment.

Elevator pits may also pose specific hazards that are characteristic of confined spaces, such as atmospheric hazards (lack of ventilation, presence of hazardous gases), physical hazards (moving parts, electrical equipment), and the potential for engulfment by liquids.

It is important to recognize elevator pits as confined spaces and assess them accordingly. If the elevator pit meets the criteria for a permit-required confined space, employees and Supervisors must implement safety measures appropriately.

Permit-Required Confined Space Definition

A Permit-Required Confined Space (PRCS) is a specific type of confined space that poses hazards to workers, and entry into such spaces requires a written permit specifying the

precautions and procedures to be followed. The characteristics of a permit-required confined space are defined by regulatory standards, such as those established by the Occupational Safety and Health Administration (OSHA).

To be classified as a Permit-Required Confined Space, a confined space must meet one or more of the following criteria:

1. **Contains or has the potential to contain a hazardous atmosphere:**
 - This includes the presence of gases, vapors, dust, or other substances that could be harmful to health or support combustion.
2. **Contains a material with the potential for engulfment:**
 - The space may have a substance (liquid or solid) that could engulf a person entering the space.
3. **Has an internal configuration that could trap or asphyxiate an entrant:**
 - The space is designed in a way that could trap or asphyxiate a person due to inwardly converging walls, a floor that slopes downward and tapers to a smaller cross-section, or other configurations.
4. **Contains any other recognized serious safety or health hazard:**
 - This may include physical hazards such as moving parts, electrical hazards, or other conditions that could pose a serious risk to the safety or health of workers.

Key features of Permit-Required Confined Spaces include:

Entry Permit Requirement:

- Before any worker enters a permit-required confined space, a written permit must be issued by an authorized supervisor. The permit outlines the specific precautions, control measures, and procedures to be followed during entry.

Training:

- Workers entering permit-required confined spaces must receive specialized training to understand the hazards, use protective equipment, and follow entry procedures.

Monitoring:

- Continuous monitoring of atmospheric conditions within the space is required to ensure that it remains safe for entry.

Rescue Procedures:

- Adequate rescue procedures must be in place, including both self-rescue and external rescue plans.

AXXIOM ELEVATOR EMPLOYEES ARE NOT TO ENTER PERMIT-REQUIRED CONFINED SPACES FOR ANY REASON.

If entry is required to permit-required confined spaces, a qualified contractor will be hired to perform that work. Once the contractor is on site, the Supervisor or Mechanic in Charge (MIC) will go through the entry assessment process with the contractor and remain onsite until the contractor has completed the work and exited the space.

In summary, it is essential to perform a thorough assessment of the elevator pit and its associated hazards. If conditions change or if the hazard profile evolves, reclassification may be necessary. The goal is to ensure the safety of workers entering confined spaces by accurately identifying and addressing potential hazards.

Confined Space Reclassification

Confined space reclassification refers to the process of reassessing and potentially changing the classification of a space that was initially identified and designated as a confined space. This reevaluation is typically conducted to determine whether the space still meets the criteria for being classified as a confined space or if it now presents additional hazards that require a different classification, such as a permit-required confined space. **The Safety Manager must be consulted when reclassifying Permit Required Confined Spaces.**

Key aspects of confined space reclassification include:

1. Hazard Assessment:
 - Conducting a thorough hazard assessment of the space to identify and evaluate potential hazards. This includes atmospheric hazards, physical hazards, and any other safety or health hazards that may be present. The confined space hazard assessment form is available in the Confined Space binder and is labeled as “Appendix A”.
2. Changes in Conditions:
 - Assessing whether there have been changes in conditions, equipment, or processes within the space that may impact the hazard profile. For example, the installation of new equipment or changes in the work environment could introduce new hazards.
3. Review of Initial Classification:
 - Reviewing the initial classification of the space to determine if it still meets the criteria for a confined space or if it now qualifies as a permit-required confined space.
4. Continuous Monitoring:
 - Implementing a system for continuous monitoring and periodic reassessment of the space to ensure that the initial classification remains accurate over time.
5. Documentation:
 - Maintaining documentation of the reclassification process, including the reasons for any changes in classification and the associated hazard control measures.
6. Communication:

- Communicating any changes in the confined space classification to relevant personnel, including workers, supervisors, and those involved in safety planning.

Reclassification is essential to ensure that the safety measures in place for confined spaces remain appropriate and effective. If the hazard profile changes, Axxiom Elevator may need to implement additional control measures.

Permit-Required Space Entry

As stated above, Axxiom Elevator employees (including Supervisors) are strictly prohibited from entering Permit-Required Confined Spaces. However, if entry into a PRCS is required, a contractor will be hired, and it will be the responsibility of the Supervisor or Mechanic in Charge to oversee the entry. Before entry can be made, a qualified contractor will need to be found and assessed.

Contractor Pre-qualification

Contractor pre-qualification to enter Permit-Required Confined Spaces (PRCS) involves assessing the contractor's capability to perform confined space work safely and in compliance with relevant regulations. The goal is to ensure that contractors have the necessary training, experience, equipment, and safety procedures in place. Here are key components for contractor pre-qualification for PRCS entry:

1. Confined Space Entry Program:
 - Axxiom Elevator requires contractors to have a documented confined space entry program that outlines their procedures, training, hazard assessment, and emergency response plans.
2. Training and Competency:
 - Ensure that the contractor's personnel who will be entering confined spaces are adequately trained and competent in confined space entry procedures, atmospheric monitoring, and rescue operations.
3. Safety Training Records:
 - Request documentation of the training records for all personnel involved in confined space work. This should include evidence of completion of relevant safety courses and certifications.
4. Experience and References:
 - Assess the contractor's experience in performing confined space work. Request references from previous clients or projects involving similar confined space entry activities.
5. Equipment Inspection and Maintenance:
 - Verify that the contractor has well-maintained and calibrated equipment, including gas detectors, personal protective equipment (PPE), communication devices, and rescue equipment.
6. Rescue Plan:

- Ensure that the contractor has a comprehensive confined space rescue plan in place, including procedures for self-rescue and coordination with external rescue services.
- 7. Documentation Compliance:
 - Confirm that the contractor maintains proper documentation, including confined space entry permits, hazard assessments, and any other records required by regulations.
- 8. Safety Policies and Procedures:
 - Request and review the contractor's safety policies and procedures, ensuring that they align with Axxiom Elevator's safety standards and regulatory requirements.
- 9. Insurance and Liability:
 - Verify that the contractor has appropriate insurance coverage, including liability insurance, and that they comply with any insurance requirements outlined in the contract.
- 10. Communication Protocols:
 - Establish clear communication protocols between Axxiom Elevator employees and the contractor. Ensure that both parties have contact information for key personnel involved in the confined space entry.
- 11. Compliance with Regulations:
 - Confirm that the contractor is familiar with and complies with relevant local, state, and federal regulations regarding confined space entry.
- 12. Pre-Job Planning:
 - Discuss pre-job planning with the contractor, including a joint site assessment to identify specific hazards and control measures for each confined space entry.
- 13. Review of Incidents and Near Misses:
 - Request information about any previous incidents or near misses related to confined space work and assess the contractor's lessons learned and corrective actions.
- 14. Audit and Evaluation:
 - Conduct periodic audits and evaluations of the contractor's confined space entry practices to ensure ongoing compliance and identify areas for improvement.

By thoroughly assessing these factors during the pre-qualification process, we can help ensure that contractors are well-prepared to safely conduct confined space work within our facility or on our job sites while complying with all relevant safety regulations and standards.

PRCS Contractor Oversight

Overseeing a contractor entering a Permit-Required Confined Space (PRCS) involves careful planning, coordination, and adherence to safety protocols to ensure the safety of the contractor and compliance with relevant regulations. Here are steps Axxiom Elevator personnel should take to oversee a contractor entering a PRCS:

1. Pre-Job Planning:
 - Conduct thorough pre-job planning with the contractor to discuss the scope of work, potential hazards, and the confined space entry procedure.
2. Contractor Training:

- Ensure that the contractor's personnel are trained in confined space entry procedures, hazards, and emergency response protocols.
- 3. Review Contractor's Confined Space Program:
 - Request and review the contractor's confined space entry program, including their written procedures, training records, and permits.
- 4. Communication:
 - Establish clear communication channels between your Axxiom Elevator employees and the contractor. Define roles and responsibilities and exchange necessary contact information.
- 5. Joint Site Assessment:
 - Conduct a joint site assessment to identify and evaluate confined space hazards. Share information about the confined space, including entry points, potential hazards, and emergency procedures.
- 6. Permit Issuance:
 - Axxiom Elevator will issue confined space entry permits and ensure that the contractor obtains the necessary permits before starting the work. This will be performed by the Safety Manager. (Appendix B)
- 7. Supervision and Monitoring:
 - Assign a qualified supervisor to oversee the contractor's confined space entry. This supervisor should monitor the entry process and ensure compliance with the permit conditions.
- 8. Continuous Monitoring:
 - The contractor should have their own monitoring equipment. Check-in with the contractor entry supervisor periodically to ensure that continuous air monitoring is taking place.
- 9. Emergency Response:
 - Review and confirm the emergency response procedures with the contractor, including rescue plans and communication protocols. Document this on the PRCS entry permit prior to entry.
- 10. Equipment Inspection:
 - Verify that the contractor's equipment, such as personal protective equipment (PPE), communication devices, and rescue equipment, is in good condition and meets safety standards. This includes recent calibration of air monitoring equipment.
- 11. Documentation:
 - Maintain accurate documentation, including permits, training records, and any communication related to the contractor's confined space entry.
- 12. Debriefing:
 - Conduct a debriefing session after the completion of the confined space work to discuss any issues, lessons learned, and areas for improvement.
- 13. Audit and Review:
 - Periodically audit and review the contractor's confined space entry practices to ensure ongoing compliance and identify opportunities for improvement.

Remember that overseeing a contractor in a PRCS requires collaboration and a shared commitment to safety. Regular communication, clear expectations, and a proactive approach to addressing potential hazards are essential elements of effective oversight.

Training

All employees will receive initial and periodic refresher training in confined spaces. The training will include the following:

- Definition of a Confined Space
- Characteristics of Confined Spaces
- Examples of Confined Spaces in the Workplace
- Atmospheric Hazards (e.g., oxygen deficiency, flammable gases, toxic substances)
- Physical Hazards (e.g., moving parts, engulfment risks)
- Biological Hazards
- Overview of Relevant Regulations (e.g., OSHA)
- Permit-Required Confined Spaces vs. Non-Permit Confined Spaces
- Approval process for Permit-Required Confined Space entry if needed
- Employer and Employee Responsibilities
- Entry Permit System
- Pre-Entry Checks and Assessments
- Isolation and Lockout/Tagout Procedures
- Ventilation Requirements
- Entry Permit Documentation
- Stop Work Authority reminder

Review and Update

This policy will be reviewed periodically to ensure its effectiveness and relevance. Updates will be made as needed to align with changes in regulations, technology, or work processes.

Recordkeeping

Training records must be maintained for a minimum of three years. The most recent record of training must always be kept on file, even if older than three years.

Revision Tracking Table

Date

Revision Description

Used Oil

(Reference: Policy AXX-0032)

The purpose of this used oil policy is to ensure the proper collection, storage, and disposal of used (various types) oil, preventing environmental contamination by directing it towards recycling instead of improper disposal in landfills, storm drains, or other waterways, which can harm wildlife and groundwater. Essentially, this policy aims to protect the environment by promoting responsible used oil management and resource conservation.

General Requirements

This policy will outline the following requirements:

- **Collection Procedures:** Designated collection points for used oil, including proper container types and labeling.
- **Storage Guidelines:** Proper storage methods to prevent spills and contamination including designated storage containers and secondary containment measures.
- **Transportation Requirements:** Procedures for transporting used oil to recycling facilities, including authorized transporters and manifest documentation.
- **Recycling Partner Selection:** Identifying reputable used oil recycling facilities to ensure the oil is properly processed.
- **Employee Training:** Educating employees on the importance of used oil management and proper handling procedures.

Collection Procedures

The purpose of this procedure is to describe the proper means for handling and disposing of used oil from equipment maintenance operations, process procedures, and any other activities where used oils are generated. This procedure applies to the disposal of any used oil that is collected during normal work functions at Axxiom Elevator. Used oil includes:

1. **Gasoline.** Volatile and flammable, it can be ignited by sparks and flames even at cold temperatures. Vapors can migrate to distant ignition sources and in poorly ventilated spaces, can accumulate to explosive levels. Typical gasoline contains about 150 different chemicals including benzene, toluene, and xylene.
2. **Fuel Oils.** Fuel oils such as diesel fuel are petroleum-based fluids which are somewhat volatile and flammable and can be ignited only when heated above 100°F. Vapors can travel and flash from ignition sources and can accumulate to explosive levels in poorly

ventilated areas. All fuel oils consist of complex mixtures of aliphatic and aromatic hydrocarbons such as kerosene, benzene, and styrene.

3. **Lubricating Oils.** Lubricating oils such as motor oil and hydraulic fluids are not volatile but are combustible. For lubricating oil to catch fire some other intense heat sources (i.e., other materials on fire, hot engine manifold, etc.) must be present. Mineral-based lube oils are refined from petroleum or crude oil and contain additives such as lead or metal sulfide and other polymers.
4. **Transformer Oil.** Transformer oil conducts heat away from and insulates equipment used to convert electricity from high amperage to low amperage lines. Transformer oil is a liquid by-product of the distillation of petroleum to produce gasoline.
5. **Cooking Oils and Grease.** Cooking oils and grease are not volatile, but they are combustible. With a 400°F flashpoint, another heat source must be present for cooking oils or grease to catch fire. Vegetable oils contain esters which in high enough concentrations can form solvents that are strong enough to dissolve rubber and small plastics.

Note: for all other waste chemicals, please contact the Safety Manager for further information.

1. Responsibility

Anyone handling or disposing of used oil for Axxiom Elevator must comply with this procedure.

Procedures

1. Products saturated with petroleum products require special handling and disposal by licensed transporters. If personnel are unsure if the substances to be discarded are in fact oils, they should contact the Safety manager for instructions. In addition, refer to the Safety Data Sheet (SDS) for the particular substance for relevant information regarding disposal. The SDS-GHS application provides access to an SDS database if one is not readily available.
2. During collection of used oils for disposal, some basic principles should be followed:
3. Put used oil in a clean plastic or metal container in good condition and with a tight lid.
4. If the oil is hot, avoid sudden contact with other substances because mixing may cause ignition or the receiving container to fracture due to thermal shock.
5. Do not allow used fuel and used oils to mix with any other substances because mixing unknown and dangerous chemical reactions may occur.
6. Keep used oils away from gas (propane) cylinders and gasoline.
7. Do not fill container to the top but allow a couple of inches below the rim.
8. Label the container with contents, location of extraction, and any other pertinent information.
9. If material is spilled, please contact the Safety Manager and clean up the spill using spill clean up materials. If you do not have cleanup materials readily available, at least attempt to contain the spill from going into any drains or the environment if safe to do so.
10. When arriving back at the home shop to dispose of the oil in the larger containment, ensure that what you are draining is compatible with the used oil containment. Next, fill

out the Used Oil Log which should be posted near the Used Oil Containment Area. It is your responsibility to denote what address the material is coming from, the approximate number of gallons, the date, and your name. Again, if spilling occurs, it is your responsibility to clean it up appropriately with materials provided by the Used Oil Storage Area. Ensure the Used Oil Storage Area is left clean and safe for all other employees.

11. The Warehouse Manager or designee of the Branch Manager will be responsible for maintaining a Used Oil Log binder where all completed Used Oil Logs will be stored for three years. This person will also be responsible for ensuring a fresh log is available in the Used Oil Storage Area.

1. Managing Cleanup Materials

If you have used oil on rags or other sorbent materials from cleaning up a leak or spill, you should remove as much of the free-flowing oil as possible and manage the oil as you would have before it spilled. Once the free-flowing used oil has been removed from these materials, they are not considered used oil and may be managed as solid waste as long as they do not exhibit a hazardous waste characteristic. Note, however, that materials from which used oil has been removed continue to be regulated as used oil if they are to be burned for energy recovery (regardless of the degree of removal).

Storage Guidelines

The Environmental Protection Agency (EPA) and other State agencies have management standards that govern the storage of used oil. Axxiom Elevator endeavors to adhere to the strictest standards as follows with regard to the storage of Used Oils:

- Label all containers and tanks as '**Used Oil**'. Do not label containers as waste oil, they are not the same thing.
- Keep containers and tanks in good condition. Do not allow tanks to rust, leak, or deteriorate. Fix structural defects immediately.
- Keep containment closed when not adding used oil.
- Ensure the used oil tank has some type of secondary containment. Secondary containment should be impervious, free of cracks, and chemically compatible with the oil. The secondary containment must be able to contain at least 10% of the primary container's volume, or the volume of the largest container. Spill pallets are generally used in our branch locations, as well as double-walled tanks. If you require aid picking appropriate spill containment for your location, contact the Safety Manager for help.
- Never store oil in anything other than tanks and storage containers. Used oil may also be stored in units that are permitted to store regulated hazardous waste. Tanks and containers storing oil do not need to be (Resource Conservation and Recovery Act) RCRA permitted, however, they must be labeled and in good condition. Storage of used oil in lagoons, pits, or surface impoundments are not permitted under RCRA and is prohibited.

Note: The EPA classifies used oil as a universal waste. Used oil is considered a non-RCRA hazardous waste in California.

Transportation Requirements

When transporting used oil from a jobsite to the shop, ensure the container is labeled with the originating address. Up to 55 gallons may be transported at one time, no more.

Recycling Partner Selection

When selecting a recycling partner to dispose of bulk used oil for your location, ensure the following:

- The company is fully insured.
- The company is permitted as a transfer, storage and recycling facility in your state.
- Ensure they accept what is in need of disposal.

Employee Training

Each field employee and any shop or warehouse employee that interacts with chemicals shall be trained in the proper disposal of used oil. This training will occur upon hire. Employees shall recognize environmental hazards related to oil spills and the benefits of properly used oil disposal. Training shall include, but not be limited to:

- Collection procedures
- Where the Used Oil Storage Area is and how it works
- Transportation requirements

Re-training should be provided for each employee, as necessary, so that the employee maintains the understanding and knowledge necessary for compliance with the Safety Policy.

All training records should be kept for three years.

Revision Tracking Table

Date

Revision Description

Page # of Revision

Introduction to the Bloodborne Pathogens Exposure Control Plan

(Reference: Policy AXX-0031)

Bloodborne pathogens are microorganisms in the bloodstream that are capable of causing disease. Workers in industry and construction are occasionally at risk of exposure to bloodborne pathogens such as Hepatitis B, Hepatitis C, and HIV/AIDS in the workplace. Tasks such as rendering first aid and housekeeping are examples of where these hazards could occur outside of the medical field.

Axxiom Elevator has developed this exposure control plan in order to comply with OSHA's Bloodborne Pathogens Standard, 29 CFR 1910.1030, and to eliminate or minimize employee occupational exposure to blood, bodily fluids, or other potentially infectious materials. (See 1910.1030(b) in Appendix A for a full list of Definitions pertaining to this Plan).

The latest version of this Bloodborne Pathogens Exposure Control Plan is maintained electronically and will be made available upon request.

Roles and Responsibilities

1. Program Administrator – The Safety Manager

- Administer and implement this program.
- Determine job classifications and locations in which employees may be exposed to bloodborne pathogens.
- Review the need for engineering controls.
- Ensure that provisions regarding PPE are met and maintained.
- Oversee the Hepatitis B vaccination program.
- Oversee the Bloodborne Pathogens training program.
- Review this plan annually, or whenever processes change, and incorporate any necessary changes.
- Oversee the first aid safety program, including ensuring that adequate first responders are trained or located within a reasonable amount of time and verifying that first aid supplies are maintained.

1. Management

- Provide adequate controls and equipment that, when used properly, will minimize or eliminate the risk of occupational exposure to blood or other potentially infectious materials at no cost to the employees.
- Monitor proper adherence to this plan through the performance of periodic audits.
- Ensure that medical and training records are being properly maintained.

1. Supervisors

- Ensure that employees are trained in and use proper work practices, universal precautions, personal protective equipment, and proper cleanup and disposal techniques.
- Answer and advise on employee questions including facilitating conversations with customer/property owners if bloodborne exposure issues arise.

1. **Employees**

- Comply with this plan
- Attend and cooperate in the training.
- Alert management to issues or potential problems found in the field.

Exposure Determination

All job classifications and locations in which employees may be expected to incur occupational exposure to blood or other potentially infectious materials (OPIM) have been identified and evaluated. The determination was made based on the nature of the job or collateral duties, regardless of frequency, and without regard to the use of personal protective equipment.

Job classifications/tasks with occupational exposure have been separated into two categories:

- **Exposure on a Regular Basis** – Job classifications in which employees are exposed to blood or other potentially infectious materials on a regular basis, and in which such exposures are considered to be a part of the normal course of work
- **Exposure on an Occasional Basis** - Job classifications in which employees may have an occasional exposure to blood or other potentially infectious materials, and in which such exposures occur only during certain tasks or procedures that are collateral to the normal job duties

These lists are located in Appendix B of this program. The Safety Manager is responsible for updating the lists as job classifications or work situations change.

Exposure Controls

1. **Universal Precautions**

Universal Precautions is an approach to infection control by treating all human blood and certain human body fluids as if they are known to be infected with HIV, HBV, and other bloodborne pathogens.

Axxiom Elevator requires employees to observe Universal Precautions to prevent contact with blood or other potentially infectious materials (OPIM). Under circumstances in which differentiation between body fluid types is difficult or impossible, all body fluids shall be considered potentially infectious materials and treated with appropriate precautions such as the use of:

- Specific PPE if blood or OPIM exposure is anticipated.
- Engineering and work practice controls to limit exposure.
- Call your supervisor if any unknown bodily fluids are encountered.

1. Engineering Controls

The Safety Manager is responsible for reviewing the need for new equipment and/or technologies present at the workplace annually at a minimum. At this time, there is no anticipated need for engineering controls to minimize bloodborne pathogens exposure at Axxiom Elevator.

Axxiom Elevator is required to use engineering and work practice controls to minimize or eliminate exposure to employees, where possible.

Where occupational exposure remains after the institution of these controls, personal protective equipment must also be used.

1. Sharps / Needles

At this time, Axxiom Elevator employees have no occupational exposure to sharps or needles. In the event that employees are exposed to sharps or needles, the following precautions and procedures apply:

Contaminated needles and other sharps must be placed into appropriate sharps containers immediately (or as soon as possible after use). All disposable sharps containers must be puncture resistant, labeled with a biohazard label, and leak-proof. Contaminated needles and other sharps may not be bent, recapped, removed, sheared, or purposely broken, unless otherwise stated in this plan.

Reusable sharps which are contaminated must be placed immediately after use into appropriate sharps containers. All reusable sharps containers must be puncture resistant, labeled with a biohazard label, and leak-proof.

Sharps Injury Log

A needle stick or sharps injury log (see Appendix C) is required to be maintained (for employers that keep records under 29 CFR 1904). These logs must include the following information for each incident.

1. Period of time the log covers
2. Date incident is entered on the log;
3. Date of incident;
4. Type and brand of device involved;
5. Department or area of incident; and
6. Description of incident.

The log will be retained for five years after the end of the log year.

1. Hand Washing Facilities

Hand washing facilities must be made available and readily accessible to all employees who may incur exposure to blood or other potentially infectious materials. Where hand washing facilities are not feasible, Axxiom Elevator should provide an antiseptic cleanser in conjunction with clean cloth/paper towels and/or antiseptic towelettes. When these alternatives are used, employees are to wash their hands with soap and running water as soon as feasible.

1. Work Area Restrictions

In work areas where there is a reasonable risk of exposure to blood or other potentially infectious materials, employees may not eat, drink, apply cosmetics or lip balm, smoke, or handle contact lenses. Food and beverages are not to be kept in refrigerators, freezers, shelves, cabinets, or on countertops or benchtops where blood or other potentially infectious materials may be present.

All processes and procedures are to be conducted in a manner that will minimize splashing, spraying, splattering, and generation of droplets of blood or other potentially infectious materials. Mouth pipetting or suctioning of blood or other potentially infectious materials is prohibited.

1. Contaminated Equipment

Equipment that has become contaminated with blood or other potentially infectious materials prior to servicing or shipping must be closely examined. Contaminated equipment should be decontaminated, unless decontamination is not feasible. Contaminated equipment must be tagged and labeled as such.

1. Personal Protective Equipment (PPE)

The Safety Manager shall ensure that the provisions regarding personal protective equipment described in this plan are met and maintained.

Personal protective equipment shall be chosen based on the anticipated exposure to blood or other potentially infectious materials. Protective equipment shall be considered appropriate only if it does not permit blood or other potentially infectious materials to pass through or reach an employees' clothing, skin, eyes, mouth, or other mucous membranes under normal and proper conditions of use and for the duration of time that the equipment will be used.

A list of personal protective equipment and associated tasks for Axxiom Elevator can be found in Appendix D of this plan.

PPE Use

The Safety Manager and supervisors shall ensure that employees use appropriate PPE. In cases where an employee temporarily and briefly declines to use PPE because, in the

employee's professional judgment, its use may prevent delivery of healthcare or pose an increased hazard to the safety of the worker or co-worker, then the supervisors shall investigate and document the situation to determine whether changes can be instituted to prevent such occurrences in the future.

PPE Accessibility

The Safety Manager is responsible for ensuring that the appropriate PPE in the various sizes is readily accessible at the work site or is issued at no cost to employees. Hypoallergenic gloves, glove liners, powderless gloves, or other similar alternatives shall be readily accessible to those employees who are allergic to the gloves normally provided.

PPE Cleaning, Laundering, and Disposal

All PPE shall be cleaned, laundered, repaired, replaced, or disposed of by Axxiom Elevator at no cost to the employees.

All garments penetrated by blood or other potentially infectious materials shall be removed immediately or as soon as feasible. All PPE shall be removed before leaving the work area.

When PPE is removed, it shall be placed in appropriately designated areas or containers for storage, washing, decontamination, or disposal.

Types of PPE

- Disposable gloves are not to be washed or decontaminated for re-use and are to be replaced as soon as possible when they become contaminated. Gloves that become torn or punctured (or their ability to function as a barrier is otherwise compromised) are to be replaced immediately or as soon as feasible.
- Eye & Face Protection - Masks worn in combination with eye protection devices (such as goggles or glasses with solid side shield, or chin-length face shields) are required when the occurrence of splashes, splatters, or droplets of blood or other potentially infectious materials can reasonably be anticipated to contaminate an employee's eye, nose, or mouth.
- Additional protective clothing (such as lab coats, gowns, aprons, clinic jackets, or similar outer garments) is necessary in instances when gross contamination can reasonably be expected.

Regulated Waste Disposal

Disposal of all regulated waste shall be in accordance with applicable federal, state, and local regulations.

1. Sharps

Contaminated sharps should be discarded immediately or as soon as feasible in containers that are closable, puncture resistant, leak proof on sides and bottom, and labeled or color-coded.

During use, containers for contaminated sharps must remain upright throughout use, shall be easily accessible to employees, and are to be located as close as feasible to the immediate area where sharps are used or can be reasonably anticipated to be found (including laundry areas).

The local Operations Manager is responsible for replacing sharps containers routinely and not allowing them to overfill.

When moving sharps containers from the area of use, the containers must be closed immediately prior to removal or replacement to prevent spillage or protrusion of contents during handling, storage, transport, or shipping. Sharps containers should be placed in a secondary container if leakage of the primary container is possible. The second container is to be closeable, constructed to contain all contents, and designed to prevent leakage during handling, storage, transport, or shipping. The secondary container should be labeled or color-coded to identify its contents.

Reusable containers may not be opened, emptied, or cleaned manually or in any other manner that would expose employees to the risk of puncture injury.

1. Other Regulated Waste

Other regulated waste must be placed in containers that are closeable, constructed to contain all contents, and will prevent leakage of fluids during handling, storage, transportation, or shipping.

All waste containers are to be labeled or color-coded and closed prior to removal to prevent spillage or protrusion of contents during handling, storage, transport, or shipping.

Laundry

Laundry contaminated with blood or other potentially infectious materials should be handled as little as possible. Such laundry must be placed in appropriately marked bags (biohazard labeled or color-coded bags) at the location where it was contaminated. Contaminated laundry may not be sorted or rinsed in the area of contamination.

Axxiom Elevator will use a local laundering facility to clean contaminated laundry, as necessary. The professional laundry service selected will need to utilize Body Substance Isolation and/or Universal Precautions and follow the procedures below.

- If Body Substance Isolation or Universal Precautions are used in handling of all soiled laundry (all laundry is assumed to be contaminated), then no labeling or color-coding is

necessary if all employees recognize the hazards associated with the handling of the laundry.

- If Universal Precautions are not used in the handling of all laundry, then contaminated laundry must be placed in bags or containers that are labeled or color-coded. One possible solution would be to include a requirement in the contract with the off-site laundry service that they also use the equivalent of Universal Precautions.

Hepatitis B Vaccines & Post-Exposure Evaluation & Follow-up

Axxiom Elevator will make the Hepatitis B vaccine and vaccination series available to all employees who have the potential for occupational exposure. In addition, the vaccine, vaccination series, and post-exposure follow up will be provided to employees who have experienced an exposure incident. All medical evaluations and procedures involved in the Hepatitis B vaccine and vaccination series and post-exposure follow up (including prophylaxis) will be:

- Made available at no cost to the employee;
- Made available to the employee at a reasonable time and place;
- Performed by or under the supervision of a licensed physician or other licensed healthcare professional;
- Provided in accordance with the recommendations of the United States public health service; and
- An accredited laboratory must be used to conduct all laboratory tests.

The Safety Manager is responsible for oversight of the Hepatitis B vaccination program, post exposure evaluations, and all follow-up activities.

1. Hepatitis B Vaccination

Axxiom Elevator plans to use a local hospital, or other medical facility, to administer the Hepatitis B Vaccination series. Employees may also obtain Hepatitis B vaccines, post-exposure evaluations, and follow-up through their personal physician.

Employees in job classifications which have an expected exposure to blood and/or other potentially infectious materials will be offered the Hepatitis B vaccination series after the employee has received training in occupational exposure and within 10 working days of initial assignment to job duties that involve exposure.

Employees in job classifications which have an occasional exposure to blood and/or other potentially infectious materials will be offered the Hepatitis B vaccination series no later than 10 hours after an exposure incident (November 1, 2000, OSHA Letter of Interpretation allows up to 24-hours).

All employees who decline the Hepatitis B vaccination are to sign a waiver indicating their refusal (a waiver form is provided in Appendix F), as required by OSHA. If the United States Public Health Service recommends a routine booster dose of Hepatitis B vaccine, this must also be made available free of charge to affected employees.

Exceptions to the administration of the Hepatitis B vaccination include situations where an employee has previously received the complete Hepatitis B vaccination series, antibody testing has revealed that the employee is immune, or the vaccine is contraindicated for medical reasons.

Participation in a pre-screening program is not a prerequisite for an affected employee to receive the Hepatitis B vaccination. If an employee initially declines the Hepatitis B vaccination, but later decides to accept the vaccination and is still covered under the OSHA standard, the vaccination is to be made available at that time.

1. Post-Exposure and Follow-up

Employees must report all exposure incidents to the Safety Manager immediately or as soon as possible after the occurrence. All exposure incidents should be investigated and documented by Axxiom Elevator. Following a report of an exposure incident, the exposed employee shall immediately receive a confidential post-exposure evaluation and follow up. OSHA requires the post-exposure evaluation and follow-up to include the following elements, at a minimum.

- Documentation of the route of exposure, and the circumstances under which the exposure occurred.
- Identification and documentation of the source individual, unless it can be established that identification is infeasible or prohibited by state or local law
- The source individual's blood shall be tested and documented as soon as feasible and after consent is obtained (if consent is required) in order to determine HBV and HIV infectivity. If consent cannot be obtained, the Safety Manager must establish and document that legally required consent cannot be obtained.
- When the source individual is already known to be infected with the Hepatitis B virus (HBV) or human immunodeficiency virus (HIV), testing for the source individual's known HBV or HIV status need not be repeated.
- Results of the source individual's testing are to be made available to the exposed employee, and the employee must be informed of applicable laws and regulations concerning disclosure of the identity and infectious status of the source individual.
- The exposed employee's blood is required to be collected as soon as feasible and tested after consent is obtained.
- The exposed employee is to be offered the option of having their blood tested for HBV and HIV serological status. The blood sample must be preserved for up to 90 days to allow the employee to decide if their blood should be tested for HBV and HIV serological status.
- Names of employees that contract HIV, Hepatitis, or tuberculosis are not to be recorded on the OSHA 300 log.

1. Information Provided to the Healthcare Provider

After an exposure incident occurs, the Safety Manager must provide the healthcare professional responsible for the exposed employee's Hepatitis B vaccination, as well as the healthcare provider providing the post-exposure evaluation (if different) with the following:

- A copy of 29 CFR 1910.1030, OSHA's Bloodborne Pathogens Standard, with emphasis on the confidentiality requirements contained therein;
- A written description of the exposed employee's duties as they relate to the exposure incident;
- Written documentation of the route of exposure and circumstances under which the exposure occurred;
- Results of the source individual's blood testing, if available; and
- All medical records relevant to the appropriate treatment of the employee, including vaccination status.

1. Health Professional's Written Opinion

The Safety Manager should obtain a copy of the evaluating healthcare professional's written opinion. This information must be provided to the exposed employee within 15 days of completion of the evaluation.

The healthcare professional's written opinion for HBV vaccination is to be limited to whether HBV vaccination is indicated for the employee, and if the employee has received said vaccination.

The healthcare professional's written opinion for post-exposure follow up must be limited to ONLY the following information:

- A statement that the employee has been informed of the results of the evaluation; and
- A statement that the employee has been told about any medical conditions resulting from exposure to blood or other potentially infectious materials that require further evaluation or treatment.

Other findings or diagnosis resulting from the post-exposure follow-up are to remain confidential and may not be included in the written report.

Labels and Signs

Biohazard labels must be properly affixed to containers of regulated waste, refrigerators, and freezers containing blood or other potentially infectious materials. Labels must also be affixed to any other containers used to store, transport, or ship blood or other potentially infectious materials.

Labels must be fluorescent orange or orange-red and include the universal biohazard symbol (below). Red bags or containers with the universal biohazard symbol may be substituted for

labels. However, regulated wastes must be handled in accordance with the rules and regulations of the entity with jurisdiction.

Training

Axxiom Elevator must provide training at the time of initial assignment to tasks where occupational exposure to blood or other potentially infectious materials may occur. OSHA requires that training be repeated every 12 months, or when there are any changes to tasks or procedures affecting an employee's occupational exposure. Training should be interactive and tailored to the education level and language of the affected employees. The person conducting the training must be knowledgeable in the subject matter.

Training must include:

1. A copy of 29 CFR 1910.1030, OSHA's Bloodborne Pathogens Standard;
2. A discussion of the epidemiology and symptoms of bloodborne diseases;
3. An explanation of the modes of transmission of bloodborne pathogens;
4. An explanation of Axxiom Elevator's Bloodborne Pathogens Exposure Control Plan, and how employees can obtain a copy of the plan;
5. A description and recognition of tasks that may involve exposure;
6. An explanation of the use and limitations of the methods employed by Axxiom Elevator to reduce exposure (such as engineering controls, work practices, and personal protective equipment);
7. Information about the types, use, location, removal, handling, decontamination, and disposal of personal protective equipment;
8. An explanation of the basis of selection of personal protective equipment;
9. Information about the Hepatitis B vaccination (including efficacy, safety, method of administration, and benefits), as well as an explanation that the vaccination will be provided at no charge to the employee;
10. Instruction on the appropriate actions to take and persons to contact in an emergency involving blood or other potentially infectious materials;
11. An explanation of the procedures to follow if an exposure incident occurs, including the method of reporting and medical follow up;
12. Information on the post-incident evaluation and follow up required for all exposure incidents; and
13. An explanation of signs, labels, and color-coding systems.

The Safety Manager is responsible for training oversight.

Recordkeeping and Program Review

1. Medical Records

Medical records will be maintained in accordance with 29 CFR 1910.1020 in a safe location. All records must be kept confidential and be retained for at least the duration of employment, plus 30 years.

Medical records are to include:

- Name and social security number of the employee;
- A copy of the employee's HBV vaccination status, including the dates of vaccination;
- A copy of all results of examinations, medical testing, and follow-up procedures; and
- A copy of the information provided to the healthcare professional, including a description of the employee's duties as they relate to an exposure incident, and documentation of the routes and circumstances of an exposure.

Axxiom Elevator must also ensure that all contracts for Hepatitis B vaccinations and post-exposure evaluations and follow-ups stipulate that medical providers follow all OSHA recordkeeping and retention requirements.

The Safety Manager is responsible for maintaining all medical records relating to the company bloodborne pathogens program.

1. Training Records

Axxiom Elevator must maintain training records for three years from the date of training. Records should be kept inside of the Bloodborne Pathogens Program binder or may be kept electronically. Training records are required to include:

- The dates of the training sessions;
- An outline describing the material presented;
- The names and qualifications of persons conducting the training; and
- The names and job titles of all persons attending the training sessions.

The Safety Manager is responsible for maintaining all training records relating to the company bloodborne pathogens program.

1. Availability of Records

Whenever an employee (or designated representative) requests access to a record, Axxiom Elevator must provide access to said employee's records in a reasonable time, place, and manner in accordance with 29 CFR 1910.1020(e). An employee (or designated representative) may only be given access to his or her own records.

1. Transfer of Records

If Axxiom Elevator ceases to do business and there is no successor employer to receive and retain the records for the prescribed period, the Director of the National Institute for

Occupational Safety and Health (NIOSH) must be contacted three months prior to cessation of business for instruction on final disposition of the records.

1. Evaluation and Review

This Bloodborne Exposure Control Plan will be reviewed for effectiveness at least annually by an Axxiom Elevator designate. The plan will be updated to incorporate changes to the standard or changes in the workplace, as needed.

First Aid Safety

First aid refers to medical attention that is usually administered immediately after the injury occurs and at the location where it occurred. It often consists of a one-time, short-term treatment and requires little technology or training to administer.

First aid can include the following:

- Cleaning minor cuts, scrapes, or scratches
- Treating a minor burn
- Applying bandages and dressings
- The use of non-prescription medicine
- Draining blisters
- Removing debris from the eyes
- Massage
- The use of hot/cold therapy
- Drinking fluids to relieve heat stress

The Safety Manager is responsible for administering and implementing the Axxiom Elevator workplace First Aid program. The Safety Manager is also responsible for the day-to-day supervision of this program.

1. First Responders

OSHA's Medical Services and First Aid standard requires that in the absence of an infirmary, clinic, hospital, or physician, that is reasonably accessible in terms of time and distance to the worksite, a person who has a valid certificate in first-aid training from the U.S. Bureau of Mines, the American Red Cross, or equivalent training that can be verified by documentary evidence, shall be available at the worksite to render first aid. A first responder is not located within four minutes of Axxiom Elevator's facility or every jobsite. Therefore, we have designated and trained employees in first aid & CPR.

Before starting a project, the supervisor is responsible for arranging the proper equipment for prompt transportation for the injured person to a physician or hospital or a communication system for contacting necessary ambulance service.

Axxiom Elevator will ensure that when a worksite is at an unreasonable distance from a first responder, a person who has a valid certificate in first-aid training will be available at the worksite to provide first aid.

1. First Aid Supplies

Where an employee of Axxiom Elevator may be exposed to injurious corrosive materials, suitable facilities for quick drenching or flushing of the eyes and body will be provided within the work area for immediate emergency use.

First Aid kits will be located in in company vehicles, safety bags and/or in the Conex box with other Axxiom tools/materials. The contents of each kit will be placed in a weatherproof container with individual sealed packages for each type of item.

At minimum, each first aid kit will contain the following:

ITEM

SIZE

QUANTITY

Absorbent compress

32 in²

1

Adhesive bandages

1 x 3 inches

4

Adhesive tape

5 yards

1 roll

Antiseptic

0.5g (0.14 fl. oz.)

2 packets

Burn treatment

0.5g (0.14 fl. oz.)

2 applications

Medical exam gloves

N/A

1 pair

Sterile pads

3 x 3 inches

2

Triangular bandage

40 x 40 x 56 inches

1

Revision Tracking Table

Date

Revision Description

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Hearing Conservation Program

(Reference: Policy AXX-0033)

This hearing conservation policy is essential to protect employees from occupational hearing loss by identifying and managing workplace noise levels, providing appropriate hearing protection when necessary, and educating workers on the importance of safeguarding their hearing, ultimately reducing the risk of work-related hearing damage and associated health concerns.

Responsibilities

Management

Responsible for ensuring implementation, maintenance and continuous improvement of the Hearing Conservation Program. Management is also responsible for providing hearing protection Personal Protective Equipment (PPE).

Supervisors

Supervisors are responsible for monitoring and enforcing the use of hearing protection, as well as providing support and addressing any issues that arise.

Employees

All employees are responsible for wearing provided hearing protection devices correctly and consistently, participating in training and audiometric testing when needed, and reporting concerns or discomfort related to hearing protection.

Key Elements to the Hearing Conservation Program

Several elements are in place to ensure the success of the program. They are as follows:

1. **Noise Assessment:** If field employees, or warehouse/shop employees are located permanently in an area with consistent noise levels believed to be at, near or over the established Time-Weighted Average, a noise assessment will be carried out to determine actual noise exposures so that a plan can be made to lower/eliminate the noise levels, or to establish appropriate personal protective equipment that must be worn when in those areas.
2. **Engineering Controls:** Whenever feasible, Axxiom Elevator will implement engineering controls to reduce noise levels at their source, thereby minimizing the need for hearing protection.
3. **Administrative Controls:** Axxiom Elevator will establish and enforce, when appropriate, administrative controls, such as scheduling tasks to limit noise exposure, rotating workers out of noisy areas, and designating quiet zones.
4. **Hearing Protection Devices:** Employees exposed to noise levels at or above 85 decibels (dB) will be provided with appropriate hearing protection devices (earplugs, earmuffs) and will be required to wear them consistently and correctly.

5. **Training and Education:** All employees will receive comprehensive training on the risks of noise-induced hearing loss, the proper use of hearing protection, and the procedures outlined in this policy. This training will be conducted upon hire and regularly thereafter. Hearing Conservation should be a subject addressed during Job Hazard Assessments.
6. **Audiometric Testing:** Axxiom Elevator will make audiometric testing available to all employees whose exposures equal or exceed an 8-hour time-weighted average of 85dB. More information on this can be found below.
7. **Record Keeping:** Axxiom Elevator will maintain accurate records of noise assessments, audiometric test results (kept separately in accordance with all applicable laws and regulations), training sessions, and any other relevant data related to the Hearing Conservation Program.
8. **Supervision and Monitoring:** Supervisors and safety personnel will monitor and enforce compliance with hearing protection requirements and address any concerns raised by employees.
9. **Communication and Feedback:** Axxiom Elevator encourages open communication between management and employees regarding noise-related concerns. Feedback from employees will be used to continually improve the effectiveness of the Hearing Conservation Program.

Fundamental Rule

As a rule of thumb, if you must raise your voice to speak to others at a distance of 5 ft (1.5m) or less – hearing protection should be worn.

Noise-Induced Hearing Loss

Noise-induced hearing loss (NIHL) is a type of hearing impairment that results from prolonged or repeated exposure to excessive levels of noise. It is a preventable condition that occurs when the delicate hair cells in the inner ear, which are responsible for converting sound waves into electrical signals that the brain interprets as sound, become damaged or destroyed due to the impact of loud noise.

Key characteristics of noise-induced hearing loss include:

1. **Gradual Onset:** Noise-induced hearing loss typically develops gradually over time as a result of cumulative exposure to loud noise. Initial exposure to loud noise may not cause immediate noticeable hearing changes, but repeated or sustained exposure can lead to permanent damage over time.
2. **Irreversible:** Once the hair cells in the inner ear are damaged or destroyed due to noise exposure, they do not regenerate or recover. This makes noise-induced hearing loss largely irreversible.
3. **Frequency and Pitch:** Noise-induced hearing loss often affects the ability to hear high-frequency sounds more prominently than low-frequency sounds. This can lead to

difficulties in understanding speech, particularly in noisy environments, as consonant sounds critical for speech intelligibility are usually high-frequency sounds.

4. **Sensitivity to Loud Sounds:** Individuals with noise-induced hearing loss may experience hyperacusis, which is an increased sensitivity to loud sounds, even at levels that were previously comfortable.
5. **Tinnitus:** Many people with noise-induced hearing loss also experience tinnitus, which is the perception of ringing, buzzing, hissing, or other phantom sounds in the ears. Tinnitus can be distressing and impact quality of life.
6. **Risk Factors:** Occupations or environments with prolonged exposure to high levels of noise, such as construction, manufacturing, mining, and certain entertainment industries, have a higher risk of causing noise-induced hearing loss. However, recreational activities such as attending concerts, using loud personal audio devices, or participating in motorsports can also contribute.

When Should Hearing Protection Be Worn?

Hearing protection should be worn whenever employees are exposed to hazardous levels of noise that could potentially cause noise-induced hearing loss. Here are some guidelines for when hearing protection should be worn:

1. **During Noisy Tasks:** Whenever employees are engaged in tasks or activities that generate loud noise levels, they should wear hearing protection. This includes activities such as using power tools, operating machinery, working in construction zones, and other tasks with noise levels exceeding safe limits.
2. **Continuous Noise Exposure:** If employees are exposed to continuous or intermittent noise at or above 85 decibels (dB) over an 8-hour workday (time-weighted average), they should consistently wear hearing protection.
3. **Intermittent Noise:** Even short bursts of high-intensity noise, such as hammering or sudden loud noises, can contribute to hearing damage. Hearing protection should be worn during these instances.
4. **Visiting Noisy Areas:** If employees need to enter areas with high noise levels, such as equipment rooms or construction zones, they should wear hearing protection regardless of the duration of their stay.
5. **Compliance with Regulations:** Follow occupational health and safety regulations and guidelines specific to your region. These regulations often define noise exposure limits and require the use of hearing protection when those limits are exceeded.
6. **Safety Precaution:** Err on the side of caution and wear hearing protection whenever there is uncertainty about noise levels or the potential for exposure to loud sounds.

Audiometric Testing

Axxiom Elevator will make audiometric testing available to all employees whose exposures equal or exceed an 8-hour time-weighted average of 85dB. For those employees, baseline

testing will be performed within 6 months of hire and will be retested annually to observe any standard threshold shifts (STS) in hearing. If any shifts have occurred, employees will be notified in writing (within 21 days of shift detection). All employees that have audiometric testing conducted will be notified of the results regardless of shift.

Audiometric testing is administered at no cost to employees. Employees who refuse to receive any mandatory audiometric testing will forfeit the right to work in any area where hearing protection is required.

If a threshold shift is detected, the employee will be retested. If the employee is still showing a shift, hearing protection shall be re-evaluated for that individual.

Axxiom Elevator will maintain records of noise exposure measurements for 2 years if taken, and audiometric test records for the duration of employment of the employees who are in the hearing conservation program.

Axxiom Elevator does not maintain audiometers or audiometric testing rooms/booths. All audiometric testing will be administered by personnel contracted to perform this service. Audiometric testing will be conducted by a licensed or certified audiologist, ear, nose, and throat physician (otolaryngologist), or audiology technician certified by the Council of Accreditation in Occupational Hearing Conservation (CAOHC). The interpretation of the audiograms will be performed by an audiologist.

Determining Noise Levels

When on jobsites, employees and supervisors can use the 'NIOSH SLM' application on their phones to determine the decibel level in their immediate area. Note: this is not a calibrated sound level meter, but this should give the worker an idea of what sound levels are in the area. When in doubt, wear hearing protection.

Note: If at any time noise levels are continuously high, employees should contact their supervisor.

Hearing Protection Types

Hearing protection is available in three general types:

1. Disposable Ear Plugs: Made of pliable material, one size fits all, but should only be used once.
2. Permanent Ear Plugs: Must be fitted to the individual to provide a good seal and are meant to be washed and reused.
3. Earmuffs: When properly fitted and worn correctly, these generally provide more protection than ear plugs.

The hearing protection devices that are distributed by the Company are designed for the environments in which they are used (construction and environments above 85 dBs).

Note: At times, more than one type of hearing protection may be required to bring the environmental noise levels below 85dB. Example: Ear plugs worn with earmuffs over the ear. If, after utilizing multiple types of hearing protection, noise levels are continuously high, employees should leave the area and contact their supervisor to find methods to reduce noise levels.

Axxiom Elevator will provide hearing protection to all employees at risk both upon hire and periodically as needed. If you are in need of hearing protection, notify your supervisor.

Procedure for Donning Disposable Ear Plugs

Putting on earplugs correctly is essential to ensure a proper seal and effective protection against noise. Here is a step-by-step procedure for putting on earplugs:

1 Clean Hands: Wash your hands thoroughly with soap and water before handling earplugs to prevent introducing dirt or bacteria into your ears.

2 Choose the Right Earplugs: Select earplugs that are appropriate for your ear size and the noise levels you'll be exposed to. Foam earplugs are a common choice and come in different sizes.

3 Rolling Technique: Start by rolling the earplug between your fingers to compress it into a smaller, elongated shape. Use a gentle rolling motion to compress the earplug, avoiding excessive pressure.

4 Insertion: Reach over your head with one hand and gently pull your ear up and slightly back. This helps straighten the ear canal and creates a better pathway for the earplug. With your other hand, hold the rolled-up earplug between your thumb and forefinger.

5 Insert the Earplug: Insert the compressed earplug into your ear canal gently and slowly. Aim for a snug fit without forcing it in too deep. Allow the earplug to expand slowly in your ear canal. Hold the earplug in place for a few seconds to ensure proper expansion.

6 Seal Check: Gently press the earplug into your ear canal to ensure a comfortable and secure fit. Listen for the reduction in external noise as the earplug creates a seal. You should feel a noticeable reduction in noise.

7 Repeat for the Other Ear: Repeat the entire process for the other ear to ensure both ears are adequately protected.

8 Comfort Check: Wiggle your jaw or move your head to ensure that the earplugs remain secure and comfortable. Adjust if needed to achieve a proper fit.

9. Communication Test: Test your ability to hear by speaking softly to yourself or having a conversation with someone nearby. If you can still hear clearly, the earplugs may not be inserted properly.

AirPods, Headphones and other Noise Cancelling Technology

The use of AirPods, headphones, earbuds or other Noise Cancelling Technology is prohibited for all Axxiom employees both while driving and while on a jobsite. The reasons for this are:

- When noise cancelling technology is used, the wearer may be unaware of a hazard or the notification from co-workers and others that a hazard is present.
- When noise cancelling technology is used, the wearer may be distracted by what is being listened to.

Listening to music may produce a safety hazard by masking environmental sounds that need to be heard, especially on active construction sites where attention to moving equipment, heavy machinery, vehicle traffic, and safety warning signals may be compromised.

If a phone call is needed, the employee is to pull off the road or go to a safe place on the jobsite to complete the call. If it is necessary to take a call around equipment, place the call on 'speaker' so that the rest of the environment can be readily heard.

Recordkeeping

Training records are required to be kept for three years. Noise level testing will be kept for 2 years by the Safety Manager (regardless of location performed). Audiometric Test records will be kept in the specific employee's file for the duration of employment.

Training

All field employees will be trained in hearing conservation at hire and annually.

Revision Tracking Table

Date

Revision Description

Page # of Revision

Driver & Vehicle Safety

(Reference: Policy AXX-0034)

This policy ensures all drivers of motor vehicles used on official business are qualified and drive safely. This policy applies to both Axxiom-owned and privately owned vehicles (when being used for business purposes). Its purpose is to ensure drivers have sufficient knowledge about vehicle handling, as portrayed through a safe driving record, in order to protect employees, Axxiom Elevator customers and the general public from unsafe vehicle operations. This policy does not apply to commercial motor vehicle drivers as specified in the Code of Federal Regulations (CFR) Chapter 49.

This policy applies to all locations or projects where vehicles are used by Axxiom Elevator employees while on the job as defined in CFR Chapter 49. Axxiom Elevator jobs will be evaluated to decide whether driving is a condition of employment. When driving is a condition of employment, an employment offer will be made contingent upon obtaining satisfactory proof of an acceptable driving history report. The Chief People Officer or the Safety Manager will review these documents. Applicants who do not provide the required documents will not be considered for employment or promotion for positions that involve driving.

No vehicle, while on Axxiom Elevator business, shall be operated by an improperly licensed driver or by a driver with an unacceptable driving record as defined within this policy. This ensures all employees, existing and prospective, are properly placed based on the demands of the job and the capacities of the individual.

This policy is valid and in force even if parts of this policy may become invalid due to local or state laws.

Implementation

The Chief People Officer is responsible for managing and directing the policies for driver authorization and qualification.

It is the responsibility of all managers and supervisors to administer this policy. It is the responsibility of all employees and managers involved to adhere fully to this policy. Individual responsibilities for implementing this plan are indicated below.

1. Management

1. Ensure compliance with this program
2. Conduct immediate corrective action for deficiencies found in this program
3. Conduct an annual review of the program
4. Maintain an effective driving safety policy
5. Ensure the availability of the policy and written documents associated with this program for employees or their designated representative.

6. Provide information to supervisors proactively, and as requested, regarding items of concern in MVRs making employees non-compliant with the Driving and Vehicle Safety Policy.
7. Verify driving records for all newly hired employees who are required to drive on Axxiom Elevator business.
8. Obtain a release from new employees prior to an MVR check and ensure it complies with applicable state and federal laws, and/or oversee a contract with a reputable external agency to gather and report this information
 1. Supervisors
9. Maintain a list of jobs requiring the operation of non-commercial vehicles for Axxiom Elevator business and a list of the individual employees who are holding those jobs and/or authorized to operate the vehicles.
10. Ensure each driver has met the requirements of the Driver & Vehicle Safety Policy – This information may be maintained electronically and must be available at all times.
11. Monitor the effectiveness of the program
12. Monitor employee training to ensure effectiveness
13. Keep management informed of necessary changes
14. Monitor records to ensure insurance coverage on employee-owned vehicles is current
15. Ensure Motor Vehicle Records are up to date for driver-employees
16. Provide specific driving safety training for assigned employees
17. At the time of an annual performance evaluation, discuss driving records with employees whose work duties require operating a motor vehicle – this includes verifying the current status and validity of the employee's driver's license.
18. Ensure employee-conducted vehicle inspections and records are maintained

25.1.3 Employees

1. Comply with the requirements of this program
2. Immediately report to their supervisor any suspensions, restrictions, limitations, revocations or restrictions of the driver's license or any other changes in their driving status, regardless of the actions that brought about the changes
3. Immediately report vehicle incidents and accidents to their supervisor
4. Use only those vehicles or types of vehicles for which they have been trained and are authorized to operate
5. On first day of employment and annually thereafter, provide copies of their driver's license and other needed information, to enable Axxiom Elevator to obtain their motor vehicle records
6. On first day of employment and annually thereafter, provide proof of insurance allowing for the operation of their personal motor vehicles for Company business
7. Visually inspect vehicles being used daily, and complete and provide weekly inspection reports to their immediate supervisors

General Program Information

The written Driving and Vehicle Safety Policy consists of:

- Employee training
- A list of authorized drivers
- Incident and violation reporting procedures
- Procedures for inspecting and maintaining vehicles
- Procedures for tracking certificates of insurance
- Driver motor vehicle records

Employee Training

1. Initial orientation training

All new employees shall receive driver safety orientation training from their immediate supervisor covering the elements of the Driver & Vehicle Safety Program. This training will cover:

- Driving preparation
- Defensive driving
- Driver readiness
- Sharing the road
- Emergencies and adverse conditions
 - 1. Job-specific training

Employees will receive on-the-job training from their immediate supervisor or their designee. Training will include:

- Hazards unique to their driving situations or assigned vehicles
- Defensive driving techniques to reduce hazards
- Proper use of vehicle-specific equipment
- Emergency procedures, including actions to take in the event of an accident
- Accident reporting procedures
 - 1. Periodic Refresher Training

Periodic refresher training will be conducted as part of continuing safety training by the employee's immediate supervisor.

1. Immediate on-the-spot training

Training will be conducted by supervisors, the Safety Manager and/or Branch Manager when:

- A new vehicle type is assigned to the driver
- Any employee requests additional information or exhibits a lack of understanding of safe driving requirements

Standards for Operating Vehicles

1. All persons operating a motor vehicle on behalf of Axxiom Elevator must have a valid driver's license.
2. No authorized driver shall relinquish to another individual the operation of an Axxiom Elevator vehicle unless approved by the immediate supervisor.
3. Transportation of passengers is authorized for business use of company cars, but the number of passengers cannot exceed the number of passenger seats available, and passengers may only occupy company cars for purposes authorized by this policy. To the extent that a child will be traveling in a company car, child safety seats must be installed and used as required by applicable law.
4. Towing with a company vehicle is only allowed for business purposes.
5. Drivers will receive proper instruction regarding Axxiom Elevator vehicles and equipment before being authorized to operate them. This includes the review of this policy, specific driving responsibilities of the driver, a driving performance evaluation conducted by the immediate supervisor and attending any driving safety training.
6. When private vehicles are used for Axxiom Elevator business, owners of the private vehicles will provide to their supervisor:
7. Documentation that the vehicle is operable and in sound mechanical condition by completing the vehicle inspection checklist with their supervisor – There must not be any discrepancies noted on the checklist.
8. Documentation the vehicle is insured as required under local law and as specified by Axxiom Elevator – Evidence of insurance must be provided to the immediate supervisor prior to using the vehicle for conducting business and (at least) annually thereafter.
9. Driver certification will be reviewed annually.
10. An unacceptable driving record is defined as either of the following:
11. Being convicted of a major traffic offense, including driving while under the influence of intoxicants, hit-and-run, reckless driving, fleeing or trying to elude a police officer or driving with a suspended or revoked license.
12. Three or more moving violations.
13. All vehicles shall be maintained in safe working order. If a safety feature is found to be defective, notify your supervisor and Fleet to bring the vehicle in for service.

Driver Safety

1. All authorized drivers of vehicles on Axxiom Elevator business will exercise reasonable care while operating a vehicle. Drivers are expected to obey all traffic laws and registration and licensing requirements applicable to the vehicle being operated.
2. All drivers and passengers are required to wear safety belts. It is the responsibility of the driver to ensure all occupants use seat belts while riding in the vehicle.
3. Possession use or being under the influence of alcohol, any narcotic, hallucinogen, stimulant, sedative or other controlled substance while operating an Axxiom Elevator vehicle, or a private vehicle while on Axxiom Elevator business, is prohibited.

4. A driver will be deemed to be 'under the influence' for purposes of this policy if they operate a vehicle within 4 hours of having consumed alcoholic beverages, when mental faculties are impaired to any perceptible degree or when the driver's blood alcohol content exceeds 0.02 percent.
5. If the use of legally prescribed controlled substances (or withdrawal symptoms from them) adversely affect a driver's physical or mental faculties to any perceptible degree or if the driver tests positive for controlled substances by screening and confirmation tests, the driver will be deemed to be 'under the influence' for the purpose of this policy.
6. Employees using any prescription medication which limits or affects their ability to operate a motor vehicle while on Axxiom Elevator business must report the use of the medication to their supervisor prior to operation of the vehicle. The supervisor, in collaboration with another manager, will determine whether the driver can safely operate the vehicle based upon a written recommendation of the driver's physician.
7. Axxiom Elevator vehicles are not to be used to transport controlled substances or alcoholic beverages, including open containers of alcoholic beverages, under any circumstances.
8. Eating while operating Axxiom Elevator vehicles is prohibited. Non-alcoholic beverages may be consumed while driving; however, drivers must exercise caution.
9. All cargo must be adequately secured before driving away. Utilize straps and other secure means to tie-down loads. If the load on a vehicle extends beyond the vehicle's sides or rear by four inches or more, a solid red or fluorescent orange flag or cloth must be displayed on the extreme end of the load. This flag must be 18 inches square.
10. The following applies to cell phone use by an employee operating a vehicle as part of their job duties or while operating a personal vehicle on Axxiom Elevator business:
11. Employees must use and comply with a hands-free device (including dialing) to include Apple Car Play, Google and similar devices.
12. Axxiom Elevator employees may not use a hand-held cell phone while operating a vehicle, whether the vehicle is in motion or stopped at a traffic light. This includes, but is not limited to, answering or making phone calls, engaging in phone conversations and reading responses to emails, instant messages or text messages.
13. If Axxiom Elevator employees need to use their phones, they must pull over safely to the side of the road or another safe location.
14. Additionally, employees are required to:
15. Inform customers, associates and business partners of this policy as an explanation of why calls may not be returned immediately.

Traffic Violations or Accidents

1. Notify your immediate supervisor and the Safety Manager as soon as possible if you have obtained a traffic violation in the course of your duties while operating a company vehicle or are involved in any vehicular accident.
2. Follow State laws pertaining to filing an accident report should an accident occur while operating an Axxiom Elevator vehicle.

3. As soon as possible, the immediate supervisor will complete the First Report of Incident (F.R.O.I.) form and e-mail the completed form and any additional pictures or witness statements to the Chief People Officer and the Safety Manager. If anyone is requiring hospitalization (either the Axxiom employee or another person), call the Safety Manager immediately.
4. All vehicle accidents will be reviewed to determine where they are chargeable or non-chargeable. A chargeable accident is defined as:
5. An accident which results from the driver's negligence, in which the driver failed to do everything he or she reasonably could have done to prevent it.
6. An accident that is cited as the driver's fault by a law enforcement officer.
7. If the accident is non-chargeable, but the employee had been involved in another Axxiom Elevator vehicle accident within the past 12 months, the supervisor will review the circumstances of the accidents and take appropriate action.
8. If the accident is chargeable, the supervisor in collaboration with the Human Resources Department and the Safety Department will determine appropriate corrective action. Documentation of the action taken will be provided to the employee, employee representation (if applicable), and be placed in the employee's personnel file.

Corrective Action

1. Management will be consulted prior to any disciplinary or corrective action.
2. Management may obtain a current driving history on each operator involved in a chargeable accident. This information will be reviewed and considered in making recommendations regarding the continuation of driving privileges while on Axxiom Elevator business.
3. The following are violation corrective actions. Nothing in this policy precludes termination on the first offense where the circumstances warrant it.
4. Citations: Any driver conducting Axxiom Elevator business who is convicted of a moving violation may be disciplined. Any driver who receives three moving violations in a one-year period will be reviewed by Management who may recommend appropriate action including suspension of driving privileges for Axxiom Elevator business. In cases where driving is a job requirement, this may result in an inability to do assigned work and, therefore, termination.
5. Accidents: Any operator involved in a chargeable accident while on Axxiom Elevator business should be appropriately disciplined.
6. Driving under the influence: Any driver convicted of driving under the influence of an intoxicating substance shall not operate a vehicle on behalf of Axxiom Elevator until the individual is in possession of a valid license of the appropriate class, the case has been reviewed and driving duties are approved and authorized by the Chief People Officer.

Motor Vehicle Records (MVR)

1. MVRs are supplied by individual states where employees reside or by private firms that provide such records as a service. An MVR provides details on the following items: driver's license status or license suspension; license points and violation codes; traffic tickets, violations and fines; auto accidents; license classification and endorsements (if applicable).
2. Once obtained, MVRs will be kept in an employee's personnel file or electronically.
3. Prior to beginning work, and annually thereafter, the MVR must be obtained and accepted by management.
4. Human Resources must ensure an MVR is obtained and then discussed with employees during the personnel evaluation process.

Accident Reporting and Emergencies

1. In case of an emergency, follow the driver emergency actions covered in the training. If in doubt, call 9-1-1.
2. If involved in a vehicular accident, drivers are to follow the procedures outlined below and those contained in the accident report kit stored in the vehicle glove compartment.
3. Render aid or assistance to the injured.
4. Do not admit fault and do not discuss the accident with anyone except Axxiom Elevator managers and supervisors, the Safety Manager, or law enforcement authorities.
5. Notify the nearest law enforcement agency immediately if an accident involves a fatality, injury, or property damage.
6. If the accident involves another party, please use the information contained in the accident report kit. Exchange insurance and identification information for all involved parties. Take pictures, noting damage, license plates and any other information relevant to the scene.
7. Immediately notify your supervisor as soon as safe to do so.
8. With your supervisor, complete the First Report of Incident (FROI) report while details are still fresh in your mind.
9. First Report of Incident (FROI) forms are available from the Safety Manager, within Dayforce, or within the Driving & Vehicle Safety Binder at all local offices.

Insurance Coverage for Employee-Owned Vehicles

1. Management will maintain records to ensure insurance coverage on employee-owned vehicles is current.
2. Employees will provide proof of insurance annually (or near expiration date of original insurance) to allow for the operation of their personal motor vehicle for organization business.

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Crystalline Silica Policy

(Reference: Policy AXX-0037)

Silica is a term which refers broadly to the mineral compound silicon dioxide (SiO₂) and can be in the form of crystalline or amorphous silica. Crystalline silica is significantly more hazardous to employees than amorphous silica, and the term 'silica' refers specifically to crystalline silica.

Crystalline silica is the basic component of sand, quartz, and granite rock. In addition to causing the disabling and irreversible lung disease known as silicosis, crystalline silica has been classified as a human carcinogen by the International Agency for Research on Cancer (IARC). Crystalline silica is an important industrial material, and occupational exposure occurs across a broad range of industries.

More than 2 million U.S. workers are exposed to crystalline silica. OSHA estimates that more than 840,000 of these workers are exposed to silica levels that exceed the new permissible exposure limit (PEL). Processes associated historically with high rates of silicosis include sandblasting, sand-casting foundry operations, mining, tunneling, and granite cutting. Workers may be exposed to crystalline silica in a variety of industries including, but not limited to construction, mining, foundry work, stone cutting, ceramics or pottery work, glass manufacturing, shipyards, or manufacturing.

At this time, Axxiom Elevator has no regular exposure to respirable crystalline silica, nor do we anticipate that to change. This program was created and implemented to protect our workers against secondary exposure and the minimal potential for any added job tasks in the future that could bring exposure to respirable crystalline silica.

In the event that a task is added or changed in which exposure to respirable crystalline silica becomes a potential, supplemental training and administrative, engineering and personal protective equipment controls will be implemented to comply with OSHA and MSHA's Crystalline Silica standards.

Health Effects

Occupational exposure to crystalline silica dust has long been known to produce pulmonary silicosis, a disabling, nonreversible, incurable lung disease caused by the inhalation of respirable crystalline silica particles. Additionally, there is evidence that exposure to crystalline

silica-containing dusts causes or is associated with the following conditions: lung cancer, tuberculosis, chronic obstructive pulmonary disease (including emphysema and bronchitis), autoimmune diseases or immunologic disorders, chronic renal disease, and subclinical renal changes. Smoking adds to the damage. Silicosis was considered to be the most serious occupational hazard during the 1930's when federal, state, and professional efforts started to focus attention on prevention strategies.

Once the microscopic particles are deposited in the lungs, they cause tissue damage and lead to the formation of fibrosis (scar tissue). The scarring decreases the lungs' ability to extract oxygen from the air. Smoking adds to the damage. There are three types of silicosis:

- **Chronic silicosis:** Usually occurs after 10 or more years of exposure to crystalline silica at low levels. This is the most common type of silicosis. Symptoms may not appear in the early stages of chronic silicosis. In fact, chronic silicosis may go undetected for 15 to 20 years after exposure. Eventually, silicosis can result in respiratory failure and death.
- **Accelerated silicosis:** Results from exposure to higher levels of crystalline silica and occurs 5 to 10 years after exposure. Inflammation, scarring, and symptoms progress faster in accelerated silicosis than in chronic silicosis.
- **Acute silicosis:** Can occur after only weeks or months of exposure to very high levels of crystalline silica. Death occurs within months. The lungs become very inflamed and may fill with fluid, causing shortness of breath and low blood oxygen levels. Acute silicosis may develop after short periods of overexposure.

As silicosis progresses, symptoms may include shortness of breath, severe cough, weakness, occasional bluish skin at the ear lobes or lips, fatigue, or loss of appetite. Due to the presence of silica in the lungs, the body's ability to fight infections may be weakened and other illnesses (such as tuberculosis) may result and can cause fever, weight loss, night sweats, chest pains, or respiratory failure. These symptoms can become worse over time, leading to death.

General Requirements

Axxiom Elevator has made a commitment to prevent silicosis at the worksite by taking the necessary steps to reduce employee exposure. It is our goal to comply with OSHA and MSHA's Crystalline Silica standards and to ensure that no employee is exposed to an airborne concentration of respirable silica in excess of OSHA and MSHA's permissible exposure limit (PEL) of 50 µg/m³ calculated as an 8-hour time weighted average (TWA).

This plan will be available for examination and copying, upon request, to each employee covered by the governing Authority Having Jurisdiction's (AHJ's) Crystalline Silica standard, their designated representatives, the Assistant Secretary and the Director. The plan will be reviewed and evaluated for effectiveness periodically and updated as necessary.

Control Methods

In order to reduce our workers' exposure to silica, tools will be operated and maintained in accordance with the manufacturer's instructions in order to minimize dust emissions. The following tasks involve exposure to respirable crystalline silica. Axxiom Elevator plans to utilize the control methods outlined below in order to reduce exposure if ever necessary:

- **Rig mounted core saws or drills** – An integrated water delivery system that supplies water to the cutting surface.
- **Vehicle-mounted drilling rigs for rock and concrete** – A dust collection system with close capture hood or shroud around drill bit with a low-flow water spray to wet the dust at the discharge point from the dust collector OR operate from within an enclosed cab and use water for dust suppression on drill bit.
- **Handheld and standmounted drills (including impact and rotary hammer drills)** – A commercially available shroud or cowling with a dust collection system that provides air flow recommended by the tool manufacturer, or greater, and having a filter with 99% or greater efficiency and filter-cleaning mechanism. A HEPA filtered vacuum will be used when cleaning holes.
- **Jackhammers and handheld powered chipping tools** – Water delivery system with continuously stream or spray at the point of impact. Respiratory protection with an APF of 10 is required when used indoors, in an enclosed area, or outdoors for more than 4 hours. OR a tool equipped with a shroud and dust collection system will be used along with a filter with 99% or greater efficiency and a filter changing mechanism. Respiratory protection with an APF of 10 is required when used indoors, in an enclosed area, or outdoors for more than 4 hours.

When control methods are not feasible or practical, Axxiom Elevator will conduct an exposure assessment for each employee who is or may reasonably be expected to be exposed to respirable crystalline silica at or above the action level of 25 µg/m³ in accordance with OSHA's prescribed methods, detailed in 29 CFR 1926.1153(d).

When it is not feasible to fully and properly comply with the engineering controls, work practice controls, and respiratory protection described in OSHA's exposure Methods table outlined in 29 CFR 1926.1153(c)(1), the following measures will be implemented:

- Ensure that no employee is exposed to an airborne concentration of respirable silica in excess of 50µg/m³ (micrograms of silica per cubic meter of air), calculated as an 8-hour TWA.
- Assess the exposure of each employee who is or may be expected to be exposed to respirable silica at or above the action level of 25 µg/m³.
- Use engineering and work practice controls to reduce and maintain employee exposure to respirable silica to or below the PEL, unless such controls are not feasible.
- Reduce employee exposure to the lowest feasible level and shall supplement them with the use of respiratory protection.
- In any instance - SWA

Housekeeping

Housekeeping practices that have the potential to expose workers to silica such as dry sweeping, dry brushing and use of compressed air on surfaces that could contribute to employee exposure of respirable silica shall be replaced with methods that minimize the likelihood of exposure where feasible. Some alternative measures consist of wet sweeping, HEPA-filtered vacuuming, and ventilation systems that effectively capture dust clouds.

Access

Access to worksites will be restricted when deemed a necessary precaution in order to minimize the number of employees exposed to respirable crystalline silica through exposure generated by Axxiom Elevator or other employees or personnel. Access will be restricted through the use of signage, barricades, structural barriers such as walls, or other means determined by the jobsite supervisor.

The regulated work area will be limited to access by authorized persons, their representatives designated to conduct monitoring or observation, or any person authorized by OSHA. Signs posted at all entrances will read:

Medical Surveillance

If employees will be required to use a respirator for 20 or more days per year due to silica exposure as outlined in OSHA's Crystalline Silica standard, medical surveillance will be offered at no cost to the employee.

Axxiom Elevator will ensure that medical surveillance complies with the following:

- All medical examinations and procedures required are performed by a physician or other licensed health care professional (PLHCP).
- An initial (baseline) medical examination is made available within 30 days of initial assignment unless the employee has received a medical examination in the last three years that meets OSHA's exam requirements.
- Medical examinations shall be available at least every three years.

Exams will include at least the following items:

- A medical and occupational history to collect data on worker exposure.
- Physical examination with special emphasis on the respiratory system.
- Chest X-rays
- Pulmonary function testing
- Testing for latent tuberculosis

A form that may be used to document if an employee declines participation in the medical surveillance program is located within the Silica Safety Program binder.

Worker Responsibilities

Signs warning workers of hazardous work areas will be posted. These signs will contain the hazards and specify any personal protective equipment that is required. Workers are responsible for the following:

- Using engineering controls properly
- Following safe work practices
- Participating in air monitoring and training programs
- Using the appropriate type of respirator when necessary
- Washing hands and face before eating, drinking, etc.
- Eating, drinking, etc. only in areas where there is no crystalline silica dust;
- If possible, wear disposable or washable work clothes and change into clean clothing before leaving the worksite.

Training

Workers will be trained about jobs that present a silica hazard, crystalline silica's health effects, engineering and work practice controls to reduce exposures, the importance of maintenance and good housekeeping, the proper type and fitting of respirators (according to Axxiom Elevators written Respiratory Protection Program), Hazard Communications, and the proper use and care of personal protective equipment at least annually.

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